AOS-CX 10.18.xxxx IP Services Guide (5420, 6200 Switch series)

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Table of contents

About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Subtopics

Applicable products

This document applies to the following products:

- HPE Aruba Networking 5420 Switch Series (S0U67A, S0U55A, S0U63A, S0U64A, S0U65A, S0U75A, S0U72A, S0U78A, S0U58A, S0U73A, S0U74A, S0U71A, S0U76A, S0U70A, S0U77A, S0U60A, S0U61A, S0U62A, S0U66A, S0U68A)
- HPE Aruba Networking 6200 Switch Series (JL724A, JL725A, JL726A, JL727A, JL728A, R8Q67A, R8Q68A, R8Q69A, R8Q70A, R8Q71A, R8V08A, R8V09A, R8V10A, R8V11A, R8V12A, R8Q72A, JL724B, JL725B, JL726B, JL727B, JL728B, S0M81A, S0M82A, S0M83A, S0M84A, S0M85A, S0M86A, S0M87A, S0M88A, S0M89A, S0M90A, S0G13A, S0G14A, S0G15A, S0G16A, S0G17A)

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Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in Support and Other Resources.

Command syntax notation conventions

Convention	Usage
`example - text`	Identifies commands and their options and operands , code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets (**[ ]**).
`example - text`	In code and screen examples, indicates text entered by a user.
Any of the following: • <example - text> • <example - text> • example - text • example - text	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: • For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (**< >**). Substitute the text—including the enclosing angle brackets—with an actual value. • For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
|	Vertical bar. A logical **OR** that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[ ]	Brackets. Indicates that the enclosed item or items are optional.
… or ...	Ellipsis: • In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. • In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if)#
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100)#
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>)#
```

Where <VLAN-ID> is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: **member/slot/port.**

On the HPE Aruba Networking 6200 Switch Series

- member: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 8. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.

- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 in slot 1 on member 1.

On the HPE Aruba Networking 6400 and 5420 Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Specifies physical location of a module in the switch chassis.
  - Management modules are on the front of the switch in slots 1/1 and 1/2.
  - Line modules are on the front of the switch starting in slot 1/3.
- port: Physical number of a port on a line module.

For example, the logical interface **1/3/4** in software is associated with physical port 4 in slot 3 on member 1.

IRDP

ICMP Router Discovery Protocol (IRDP), an extension of the ICMP, is independent of any routing protocol. It allows hosts to discover the IP addresses of neighboring routers that can act as default gateways to reach devices on other IP networks.

> **NOTE**
> On the switches covered by this guide, IRDP is configured on a VLAN interface.

IRDP operation

IRDP uses the following types of ICMP messages:

- Router advertisement (RA): Sent by a router to advertise IP addresses (including the primary and secondary IP addresses) and preference.
- Router solicitation (RS): Sent by a host to request the IP addresses of routers on the subnet.

An interface with IRDP enabled periodically broadcasts or multicasts an RA message to advertise its IP addresses. A receiving host adds the IP addresses to its routing table, and selects the IP address with the highest preference as the default gateway.

When a host attached to the subnet starts up, the host multicasts an RS message to request immediate advertisements. If the host does not receive any advertisements, it retransmits the RS several times. If the host does not discover the IP addresses of neighboring routers because of network problems, the host can still discover them from periodic RAs.

IRDP allows hosts to discover neighboring routers, but it does not suggest the best route to a destination. If a host sends a packet to a router that is not the best next hop, the host will receive an ICMP redirect message from the router.

IP address preference

Every IP address advertised in RAs has a preference value. A larger preference value represents a higher preference. The IP address with the highest preference is selected as the default gateway address.

You can specify the preference for IP addresses to be advertised on a router interface.

An address with the minimum preference value (-2147483648) will not be used as a default gateway address.

Lifetime of an IP address

An RA contains a lifetime field that specifies the lifetime of advertised IP addresses. If the host does not receive a new RA for an IP address within the address lifetime, the host removes the route entry.

All the IP addresses advertised by an interface have the same lifetime.

Advertising interval

A router interface with IRDP enabled sends out RAs randomly between the minimum and maximum advertising intervals. This mechanism prevents the local link from being overloaded by a large number of RAs sent simultaneously from routers.

As a best practice, shorten the advertising interval on a link that suffers high packet loss rates

Destination address of RA

An RA uses either of the following destination IP addresses:

- Broadcast address 255.255.255.255.
- Multicast address 224.0.0.1, which identifies all hosts on the local link.

By default, the destination IP address of an RA is the multicast address. If all listening hosts in a local area network support IP multicast, specify 224.0.0.1 as the destination IP address.

Proxy-advertised IP addresses

By default, an interface advertises its primary and secondary IP addresses. You can specify IP addresses of other gateways for an interface to proxy-advertise.

VRF support

In IP-based computer networks, virtual routing and forwarding (VRF) is a technology that allows multiple instances of a routing table to co-exist within the same router at the same time. Because the routing instances are independent, the same or overlapping IP addresses can be used without conflicting with each other.

IRDP is VRF aware. As the router advertisements and solicit processing occurs on the interface, packet is through the interface and corresponding VRF.

VSX synchronization

IRDP supports VSX synchronization. For more information on using VSX, see the Virtual Switching Extension (VSX) Guide for your switch and software version

Subtopics

[Configuring IRDP](#)
[IRDP commands](#)

Configuring IRDP

Prerequisites

A layer 3 interface.

Procedure

1. Enable IRDP on an interface with the command `ip irdp`.
2. Set the maximum hold time with the command `ip irdp holdtime`.
3. Set the maximum router advertisement interval with the command `ip irdp maxadvertinterval`.
4. Set the minimum router advertisement interval with the command `ip irdp minadvertinterval`.
5. Set the IRDP preference level with the command `ip irdp preference`.
6. Review IRDP configuration settings with the command `show ip irdp`.

Example

This example creates the following configuration:

- Enables IRDP on the layer 3 VLAN interface 2 with packet type set to broadcast.
- Sets the hold time to 5000 seconds.
- Sets the advertisement interval to 30 seconds.
- Sets the minimum advertisement interval to 25 seconds.
- Sets the IRDP preference level to 25.

```
switch(config)#
              interface vlan 2
```

```
switch(config-if-vlan)# ip irdp broadcast
switch(config-if-vlan)# ip irdp holdtime 5000
switch(config-if-vlan)# ip irdp maxadvertinterval 30
switch(config-if-vlan)# ip irdp minadvertinterval 25
switch(config-if-vlan)# ip irdp preference 25
```

IRDP commands

Select a command from the list in the left navigation menu.

Subtopics

- diag-dump irdp basic
- ip irdp
- ip irdp holdtime
- ip irdp maxadvertinterval
- ip irdp minadvertinterval
- ip irdp preference
- show ip irdp

`diag-dump irdp basic`

Syntax

```
diag-dump irdp basic
```

Description

Displays diagnostic information for IRDP.

Example

```
switch# diag-dump irdp basic
=============================================================================
[Start] Feature irdp Time : Thu Jun  8 09:50:28 2017
=============================================================================
-----------------------------------------------------------------------------
[Start] Daemon hpe-rdiscd
-----------------------------------------------------------------------------
Interface: 1/1/1 (state : Up)
rdisc ipv4 (enabled: 0, max:600, min:450, hold:1800, pref:0, isBcast:0)
```

```
Router IPs - 192.168.1.2,
Interface: 1/1/2 (state : Up)
rdisc ipv4 (enabled: 0, max:600, min:450, hold:1800, pref:0, isBcast:0)
Router IPs - 192.168.2.2,
-------------------------------------------------------------------------
[End] Daemon hpe-rdiscd
-------------------------------------------------------------------------
=========================================================================
[End] Feature irdp
=========================================================================
Diagnostic dump captured for feature irdp
switch# diag-dump irdp basic
=========================================================================
[Start] Feature irdp Time : Thu Jan  7 04:46:25 2021
=========================================================================
-------------------------------------------------------------------------
[Start] Daemon hpe-rdiscd
-------------------------------------------------------------------------
Interface: vlan2 (state : Down)
rdisc ipv4 (enabled: 1, max:600, min:450, hold:1800, pref:0, isBcast:0)
No advertisable IPv4 addresses on the interface
Interface: vlan1 (state : Down)
rdisc ipv4 (enabled: 0, max:600, min:450, hold:1800, pref:0, isBcast:0)
No advertisable IPv4 addresses on the interface
-------------------------------------------------------------------------
[End] Daemon hpe-rdiscd
-------------------------------------------------------------------------
=========================================================================
[End] Feature irdp
=========================================================================
Diagnostic-dump captured for feature irdp
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`ip irdp`

Syntax

```
ip irdp [broadcast | multicast]
no ip irdp
```

Description

Enables IRDP on an interface and specifies the packet type that is used to send advertisements. By default, the packet type is set to `multicast`. IRDP is only supported on layer 3 interfaces.

The **no** form of this command disables IRDP on an interface.

Parameter	Description
`broadcast`	Advertisements are sent as broadcast packets to IP address 25 5.255.255.255.
`multicast`	Advertisements are sent as multicast packets to the multicast g roup with IP address 24.0.0.1. Default.

Examples

Enabling IRDP on interface vlan 2 with packet type set to the default value (multicast).

```
switch(config)#
                interface vlan 2
switch(config-if-vlan)# ip irdp
```

Enabling IRDP on interface 1/1/1 with packet type set to broadcast.

```
switch(config)#
                interface vlan 2
```

```
switch(config-if-vlan)# ip irdp broadcast
```

Disabling IRDP.

```
switch(config)#
            interface vlan 2
switch(config-if-vlan)# no ip irdp
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip irdp holdtime`

Syntax

```
ip irdp holdtime <TIME>
no ip irdp holdtime <TIME>
```

Description

Specifies the maximum amount of time the host will consider an advertisement to be valid until a newer advertisement arrives. When a new advertisement arrives, hold time is reset. Hold time must be greater than or equal to the maximum advertisement interval. Therefore, if the hold time for an advertisement expires, the host can reasonably conclude that the router interface that sent the advertisement is no longer available. The default hold time is three times the maximum advertisement interval.

The **no** form of this command removes the specified maximum amount of time the host will consider an advertisement to be valid until a newer advertisement arrives and update it to the default value.

Parameter	Description
<TIME>	Specifies the lifetime of router advertisements sent from this in terface. Range: 4 to 9000 seconds. Default: 1800 seconds.

Example

Setting the hold time for VLAN interface 2 to 5000 seconds:

```
switch(config)#
            interface vlan 2
switch(config-if-vlan)# ip irdp holdtime 5000
```

Removing the the hold time for VLAN interface 2 to 5000 seconds:

```
switch(config)#
            interface vlan 2
switch(config-if-vlan)# no ip irdp holdtime 5000
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip irdp maxadvertinterval`

Syntax

```
ip irdp maxadvertinterval <TIME>
no ip irdp maxadvertinterval <TIME>
```

Description

Specifies the maximum router advertisement interval.

The **no** form of this command removes the specified maximum router advertisement interval and reverts to the default value.

Parameter	Description
`<TIME>`	Specifies the maximum time allowed between the sending of u nsolicited router advertisements. Range: 4 to 1800 seconds. De fault: 600 seconds.

Example

Setting the advertisement interval for VLAN interface 2 to 30 seconds:

```
switch(config)#
          interface vlan 2
switch(config-if-vlan)# ip irdp maxadvertinterval 30
```

Removing the advertisement interval for VLAN interface 2 to 30 seconds:

```
switch(config)#
          interface vlan 2
switch(config-if-vlan)# no ip irdp maxadvertinterval 30
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip irdp minadvertinterval`

Syntax

```
ip irdp minadvertinterval <TIME>
no ip irdp minadvertinterval <TIME>
```

Description

Specifies the minimum amount of time the switch waits between sending router advertisements. By default, this value is automatically set by the switch to be 75% of the value configured for maximum router advertisement interval. Use this command to override the automatically configured value.

The **no** form of this command removes the specified minimum amount of time the switch waits between sending router advertisements and reverts to the default value.

Parameter	Description
<TIME>	Specifies the minimum time allowed between the sending of u nsolicited router advertisements. Range: 3 to 1800 seconds. De fault: 450 seconds (75% of the default value for maximum route r advertisement interval).

Example

Setting the minimum advertisement interval for VLAN interface 2 to 25 seconds:

```
switch(config)#
            interface vlan 2
switch(config-if-vlan)# ip irdp minadvertinterval 25
```

Removing the minimum advertisement interval for VLAN interface 2 to 25 seconds:

```
switch(config)#
            interface vlan 2
```

```
switch(config-if-vlan)# no ip irdp minadvertinterval 25
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip irdp preference`

Syntax

```
ip irdp preference <LEVEL>
no ip irdp preference <LEVEL>
```

Description

Specifies the IRDP preference level. If a host receives multiple router advertisement messages from different routers, the host selects the router that sent the message with the highest preference as the default gateway.

The **no** form of this command removes the specified IRDP preference level and reverts to the default value.

Parameter	Description
<LEVEL>	Specifies the IRDP preference level. Range: - 2147483648 to 2147483647. Default: 0.

Example

Setting the IRDP preference level for VLAN interface 2 to 25.

```
switch(config)#
            interface vlan 2
switch(config-if-vlan)# ip irdp preference 25
```

Removing the IRDP preference level for VLAN interface 2 to 25.

```
switch(config)#
            interface vlan 2
switch(config-if-vlan)# no ip irdp preference 25
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`show ip irdp`

Syntax

```
show ip irdp
```

Description

Displays IRDP configuration settings.

Example

```
switch# show ip irdp
ICMP Router Discovery Protocol
```

```
 Interface Status    Advertising Minimum  Maximum  Holdtime Preference
                     Address     Interval Interval
 --------- -------- ----------- -------- -------- -------- -----------
 1/1/1     Enabled  multicast   6        8        10       10
 1/1/2     Disabled multicast   450      600      1800     0
 1/1/3     Enabled  broadcast   450      600      1800     115
switch# sh ip irdp
 ICMP Router Discovery Protocol
 Interface        Status    Advertising Minimum  Maximum  Holdtime Preference
                            Address     Interval Interval
 ---------------- -------- ----------- -------- -------- -------- -----------
 vlan1            Disabled multicast   450      600      1800     0
 bridge_normal    Disabled multicast   450      600      1800     0
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

IPv6 Router Advertisement

IPV6 RA provides a method for local IPV6 hosts to automatically configure their own IP address (and other settings such as a preferred DNS server) based on information advertised by switches/routers operating on the network.

IPv6 flags

Behavior of IPv6 hosts to IPv6 RA messages is controlled by the managed address configuration flag (M flag), and other stateful configuration flag (O flag).

M flag	O flag	Description
0	0	Indicates that no information is available via DHCPv6.
0	1	Indicates that other configuration information is available via DHCPv6. Examples of such information are DNS - related information or information on other servers within the network.
1	0	Indicates that addresses are available via Dynamic Host Configuration Protocol (DHCPv6).
1	1	If the M flag is set, the O flag is redundant and can be ignored because DHCPv6 will return all available configuration information.

Subtopics

Configuring IPv6 RA

Syntax

Procedure

1. Enable transmission of IPv6 router advertisements with the command `no ipv6 nd suppress-ra`.
2. Optionally, configure IPv6 unicast address prefixes with the command `ipv6 nd prefix`.
3. Optionally, configure support for DNS name resolution with the commands `ipv6 nd ra dns server` and `ipv6 nd ra dns search-list`.
4. For most deployments, the default values for the following features do not need to be changed. If your deployment requires different settings, change the default values with the indicated command:

IPv6 RA setting	Default value	Command to change it
Number of neighbor solicitations to be sent when performing DAD.	1	ipv6 nd dad attempts
Number of neighbor entries in th e ND cache.	131072	`ipv6 nd cache-limit`
Hop limit to be sent in the RA m essages.	64	`ipv6 nd hop-limit`
MTU value to be sent in the RA messages.	1500 bytes	`ipv6 nd mtu`
Neighbor solicitation interval	1000 milliseconds	`ipv6 nd ns-interval`
Lifetime of a default router.	1800 seconds	`ipv6 nd ra lifetime`
Retrieval of an IPv6 address by d evices.	Disabled	`ipv6 nd ra managed-config-flag`
Maximum interval between trans missions of IPv6 RAs.	600 seconds	`ipv6 nd ra max-interval`
Minimum interval between trans missions of IPv6 RAs.	200 seconds	`ipv6 nd ra min-interval`
Time that an interface considers a device to be reachable.	0 milliseconds (no limit)	`ipv6 nd ra reachable-time`
Retry period between ND solici tations.	0 (Use locally configured NS - in terval)	`ipv6 nd ra retrans-timer`
Default routing preference for an interface.	Medium	`ipv6 nd router-preference`

5. Review IPv6 RA configuration settings with the commands `show ipv6 nd interface`, `show ipv6 nd interface prefix`, `show ipv6 nd ra dns server`, and `show ipv6 nd ra dns search-list`.

Example

This example creates the following configuration:

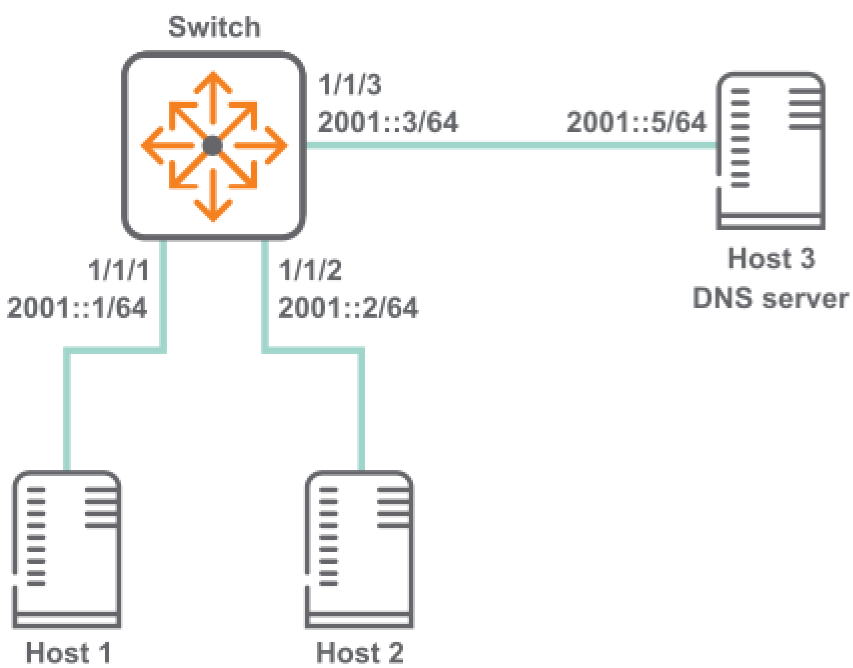
- Enables IPV6 RA on interface 1/1/3 .
- Sets the recursive DNS server address to `4001::1` with a lifetime of `400` seconds.
- Sets the minimum interval between transmissions to `3` seconds.
- Sets the maximum interval between transmissions to `13` seconds.
- Sets the lifetime of a default router to `1900` seconds.

```
switch(config)# interface 1/1/3
switch(config-if)# no ipv6 nd suppress-ra
switch(config-if)# ipv6 nd ra dns server 4001::1 lifetime 400
switch(config-if)# ipv6 nd ra min-interval 3
switch(config-if)# ipv6 nd ra max-interval 13
switch(config-if)# ipv6 nd ra lifetime 1900
switch(config-if)# end
switch# show ipv6 nd interface 1/1/3
Interface 1/1/3 is up
  Admin state is up
  IPv6 address:
    2006::1/64 [VALID]
  IPv6 link-local address: fe80::98f2:b321:368:6dc6/64 [VALID]
  ICMPv6 active timers:
      Last Router-Advertisement sent: 0 Secs
      Next Router-Advertisement sent in: 13 Secs
  Router-Advertisement parameters:
      Periodic interval: 3 to 13 secs
      Router Preference: medium
      Send "Managed Address Configuration" flag: false
      Send "Other Stateful Configuration" flag: false
      Send "Current Hop Limit" field: 64
      Send "MTU" option value: 1500
      Send "Router Lifetime" field: 1900
      Send "Reachable Time" field: 0
      Send "Retrans Timer" field: 0
      Suppress RA: false
      Suppress MTU in RA: true
  ICMPv6 error message parameters:
      Send redirects: false
  ICMPv6 DAD parameters:
      Current DAD attempt: 1
```

```
switch# show ipv6 nd ra dns server
Recursive DNS Server List on: 1/1/3
      Suppress DNS Server List: No
      DNS Server 1: 2001::1    lifetime 400
```

IPv6 RA scenario

In this scenario, two host computers are auto-configured with IP addresses using IPv6 RA. In addition, the switch provides the hosts with an address of a recursive DNS server. The physical topology of the network looks like this:



Procedure

1. Configure the interfaces with IPv6 addresses.
   switch# **config** switch(config)# **interface 1/1/1** switch(config-if)# **ipv6 address 2001::1/64** switch(config)# **interface 1/1/2** switch(config-if)# **ipv6 address 3001::1/64** switch(config)# **interface 1/1/3** switch(config-if)# **ipv6 address 4001::1/64**

2. Enable transmission of all IPv6 RA messages.
   switch(config-if)# **no ipv6 nd suppress-ra**

IGMP snooping commands

Select a command from the list in the left navigation menu.

Subtopics

`ip igmp snooping (config mode)`

Syntax

```
ip igmp snooping
   drop-unknown vlan-shared|vlan-exclusive
   fastlearn <PORT-LIST>
```

Description

Configures drop-unknown and fastlearn modes on the ports. While IGMP snooping is enabled, the traffic will be forwarded only to ports that made an IGMP request for the multicast. Drop unknown filters ensure that packets are not forwarded to ports that did not make a request for the traffic stream. This could either be a filter across all VLANs (**vlan-shared**) or per VLAN (**vlan-exclusive**). The default is **vlan-shared**. Fast learn enables the port to learn group information when receiving a topology change notification. By default, fast learn is not enabled on ports.

Parameter	Description
`drop-unknown`	Drop unknown filters ensure that packets are not forwarded to ports that did not make a request for the traffic stream.
`vlan-shared`	Enables a shared VLAN filter on the switch. Default is **vlan - shared.**

Parameter	Description
`vlan-exclusive`	Enables an exclusive drop unknown filter per VLAN.
`fastlearn <PORT-LIST>`	Enable fast learn on ports. This parameter specifies a list of one or more ports to be configured as fast learn ports. You can spec ify a single port, a comma - separated list of ports or a range o f ports such as 1/1/1 - 1/1/3. You may also enter an L2 LAG (1 - 128)
`no ...`	Negates any configured parameter.

Example

Configuring fast learn ports:

```
switch(config)# ip igmp snooping fastlearn 1/1/3
switch(config)# ip igmp snooping fastlearn 1/1/1-1/1/2
switch(config)# ip igmp snooping fastlearn 1/1/5,1/1/6
```

Configuring a shared VLAN filter on the switch:

```
switch(config)# ip igmp snooping drop-unknown vlan-shared
```

Configuring a exclusive drop unknown filter per VLAN:

```
switch(config)# ip igmp snooping drop-unknown vlan-exclusive
```

Disabling drop unknown on the switch:

```
switch(config)# no ip igmp snooping drop-unknown
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip igmp snooping (interface mode)`

Syntax

```
ip igmp snooping
   auto vlan <VLAN-LIST>
   blocked vlan <VLAN-LIST>
   fastleave vlan <VLAN-LIST>
   forced-fastleave vlan <VLAN-LIST>
   forward vlan <VLAN-LIST>
   no ...
```

Description

Configure IP IGMP snooping for the VLAN on the interface. When IGMP snooping is enabled, the L2 snooping switch forwards multicast packets of known multicast groups to only the receivers. When IGMP snooping is not enabled, the snooping switch floods multicast packets to all hosts on the VLAN.

Parameter	Description
`auto vlan` *<VLAN-LIST>*	Instruct the device to monitor incoming multicast traffic on the specified ports on a VLAN or VLAN range. This is the default behavior. Enter the number of a single VLAN or a series of num bers for a range of VLANs, separated by commas (10, 20, 30, 4 0), dashes (10 - 40), or both (10 - 40,60).
`blocked vlan` *<VLAN-LIST>*	Configures the specified ports in blocked mode for the specifie d VLAN list. In blocked mode, joins and traffic are always blocke d on this port. Enter the number of a single VLAN or a series of numbers for a range of VLANs, separated by commas (10, 20, 3 0, 40), dashes (10 - 40), or both (10 - 40,60).
`fastleave vlan` *<VLAN-LIST>*	IGMP fastleave is configured for ports on a per - VLAN basis. U pon receiving a Leave Group message, the querier sends an IG MP Group - Specific Query message out of the interface to ens ure that no other receivers are connected to the interface. If rec eivers are directly attached to the switch, it is inefficient to send the membership query as the receiver wanting to leave is the o nly connected host. When a fastleave - enabled switch port is connected to a single host and receives a leave, the switch does not wait for the queri er status update interval, but instead immediately removes the I GMP client from its IGMP table and ceases transmitting multica st traffic to the client. (If the switch detects multiple end nodes on the port, Fastleave does not activate regardless of whether o

Parameter	Description
	ne or more of these end nodes are IGMP clients.) This processin g speeds up the overall leave process and also eliminates the C PU overhead of having to generate an IGMP Group - Specific Q uery message. This parameter specifies a list of VLANs on which the port sho uld be configured as a fastleave port. Specifies the number of a single VLAN or a series of numbers for a range of VLANs, sepa rated by commas (10, 20, 30, 40), dashes (10 - 40), or both (1 0 - 40,60).
`forced-fastleave vlan <VLAN-LIST>`	With forced fastleave enabled, IGMP speeds up the process of b locking unnecessary multicast traffic to a switch port that is con nected to multiple end nodes. When a port having multiple end nodes receives a leave group request from one end node for a g iven multicast group, forced fastleave activates and waits for a s econd to receive a join request from any other member of the s ame group on that port. If the port does not receive a join requ est for that group within the forced fastleave interval, the switc h then blocks any further traffic to that group on that port. This parameter specifies a list of VLANs on which the port shou ld be configured as a forced fastleave port. Specifies the numbe r of a single VLAN or a series of numbers for a range of VLANs, separated by commas (10, 20, 30, 40), dashes (10 - 40), or bot h (10 - 40,60). This command is available in **config - if** mode.
`forward vlan <VLAN-LIST>`	Configures the specified ports in forward mode in the given VL AN list. In forward mode, traffic is always forwarded on this por t, irrespective of joins. Specify a list of VLANs on which the port should be configured as a forward port. Specifies the number of a single VLAN or a series of numbers for a range of VLANs, sep arated by commas (10, 20, 30, 40), dashes (10 - 40), or both (10 - 40,60). This command is available in **config - if** mode.
`no ...`	Negates any configured parameter.

Example

Configure auto ports for VLAN on the interface:

```
switch# configure terminal
switch(config)# int 1/1/1
switch(config-if)# no shut
switch(config-if)# no routing
```

```
switch(config-if)# vlan trunk allowed 10-20
switch(config-if)# ip igmp snooping auto vlan 10
switch(config-if)# ip igmp snooping auto vlan 10-20
```

Configuring fastleave ports for the VLAN on the interface:

```
switch# configure terminal
switch(config)# int 1/1/1
switch(config-if)# no shut
switch(config-if)# no routing
switch(config-if)# vlan trunk allowed 10-20
switch(config-if)# ip igmp snooping fastleave vlan 10
switch(config-if)# ip igmp snooping fastleave vlan 10-20
```

Configuring blocked ports for the VLAN on the interface:

```
switch# configure terminal
switch(config)# int 1/1/1
switch(config-if)# no shut
switch(config-if)# no routing
switch(config-if)# vlan trunk allowed 10-20
switch(config-if)# ip igmp snooping blocked vlan 10
switch(config-if)# ip igmp snooping blocked vlan 10-20
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ip igmp snooping (vlan mode)`

Syntax

```
ip igmp snooping
   apply access-list <ACL-NAME>
   enable|disable
   no ...
   static-group <MULTICAST-IP-ADDRESS>
   version <2-3> (vlan interface mode)
```

Description

These commands enable or disable IP IGMP snooping on the VLAN, create IGMP snooping static multicast groups, set the IGMP snooping version and configurethe ACL on a particular interface.

> **NOTE**
> Disabling and enabling IGMP snooping on a VLAN causes IGMP querier re-election.

Parameter	Description
`access-list`	Associates an ACL with the IGMP.
`enable|disable`	Enables or disables IGMP snooping on the VLAN. Starting with AOS - CX 10.16, IGMP snooping is enabled by default.
`no ...`	Negates any configured parameter.
`static-group` *<MULTICAST-IP-ADDRESS>*	This parameter configures an IGMP snooping static multicast g roup. Specify the IGMP static multicast group IP address in A. B.C.D format. You can configure a maximum of 32 IGMP snoopin g static
version <2-3>	Configures the IGMP snooping version on the VLAN. Select **2** fo r IGMPv2 (RFC2236). Select **3** for IGMPv3 (RFC3376).

Usage

- Existing classifier commands are used to configure the ACL.
- In case an IGMPv3 packet with multiple group addresses is received, the switch only processes the permitted group addresses based on the ACL rule set. The packet is forwarded to querier and PIM router even though one of the groups present in the packet is blocked by the ACL. This avoids the delay in learning of the permitted groups. Since the access switch configured with ACL blocks the traffic for the

groups which are denied, forwarding of joins has no impact. If all the groups in the packet are denied by the ACL rule, the packet is not forwarded to the querier and PIM router. Existing joins will timeout.

- In case of IGMPv2, if there is no match or if there is a deny rule match, the packet is dropped.

> **NOTE**
> If the access list is configured for both L2 VLAN and L3 VLAN, the L3 VLAN configuration will be applied.

Example

Disable IGMP snooping on a VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# ip igmp snooping disable
```

Reenable IGMP snooping on a VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# ip igmp snooping enable
```

Configuring an IGMP snooping static group:

```
switch(config)# vlan 2
switch(config-vlan)# ip igmp snooping static-group 239.1.1.1
switch(config-vlan)# no ip igmp snooping static-group 239.1.1.1
```

Configuring IGMP snooping version on the VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# ip igmp snooping version 2
```

Command History

Release	Modification
10.16	Starting with this release, IGMP snooping is enabled on VLANs by default, inclu ding the Management VLAN (vlan 1) and VLANs with a associated VNI.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-vlan-<VLAN-ID>`	Administrators or local user group members with execution righ ts for this command.

`ip igmp snooping all-vlans`

Syntax

```
ip igmp snooping all-vlans
no ip igmp snooping all-vlans
```

Description

Enable IGMP snooping on all VLANs. This configuration setting enables IGMP snooping on all existing VLANs, as well as on any VLANs that are created after the **ip igmp snooping all-vlans** command is issued. When IGMP snooping is enabled globally, IGMP snooping can not be configured on individual VLANs within the VLAN context of the command-line interface.

The **no** form of this command disables global IGMP snooping, allows IGMP snooping to remain enabled at the VLAN level, and allows the configuration of VLAN-level IGMP snooping settings.

Example

Configure IGMP snooping globally:

```
switch(config)# ip igmp snooping all-vlans
```

Disable global IGMP snooping:

```
switch(config)# no ip igmp snooping all-vlans
```

Command History

Release	Modification
10.16.1000	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip igmp snooping apply access list`

Syntax

```
ip igmp snooping apply access list <ACL-NAME>
no ip igmp snooping apply access list <ACL-NAME>
```

Description

Configures the access list (ACL) in a particular interface to filter IGMP join or leave packets based on rules set in a particular access list name. The **no** form of this command removes the configuration.

Parameter	Description
<ACL-NAME>	Specifies the access list name.

Usage

Existing classifier commands are used to configure ACL. In case of IGMPv3 packets with multiple group addresses received, only permitted group addresses based on the ACL rule set are proccessed. The packet is forwarded to querier and PIM router even though one of the groups present in the packet is blocked by ACL to avoid the delay in learning of the permitted groups because the access switch configured with the ACL blocks the traffic for the groups which are denied forwarding of joins have no impact. If all of the groups in a packet are denied by the ACL rule packet, it is not forwarded to the querier and PIM router. If the ACE has the source address configured, the source address in the IGMPv3 report is matched against the ACL and corresponding action is taken. Existing joins timeout.

With IGMPv2, if there is no match or if there is a deny rule match, the packet is dropped.

NOTE

If the access list is configured for both L2 VLAN and L3 VLAN, then the L3 VLAN configuration is applied.

Example

Configure the access list:

```
switch(config)# vlan 2
switch(config-vlan-2)# ip igmp snooping apply access-list mygroup
```

Remove the access list:

```
switch(config)# vlan 2
switch(config-vlan-2)# no ip igmp snooping apply access-list mygroup
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
All platform	`config-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip igmp snooping filter unknown mcast`

Syntax

```
ip igmp snooping filter-unknown-mcast
no ip igmp snooping filter-unknown-mcast
```

Description

Enables the avoidance of initial flooding of unknown multicast traffic on IGMP-snooping-enabled VLANs.

The no form of this command returns to the default behavior of initial flooding of unknown multicast traffic.

Usage

In the default behavior, the unknown multicast traffic is flooded until the IP Multicast Flow programming is done on the hardware. This is known as initial flooding of unknown multicast. Use this command to filter unknown multicast instead of flooding.

> **NOTE**
>
> Initial flooding of multicast traffic is observed for a few seconds after the device comes up from a reboot. This issue is only seen when the multicast source connected device is rebooted. Once the device is up after a reboot, it takes a few seconds for the CPU Rx rule to be programmed during the timeframe that the initial flooding is observed. This is an expected behavior.

Example

Configure the unknown multicast to steal globally on IGMP snooping enabled VLANs.

```
switch# configure terminal
switch(config)# ip igmp snooping filter-unknown-mcast
```

Removing the configuration of the unknown multicast to steal globally on IGMP snooping enabled VLANs.

```
switch# configure terminal
switch(config)# no ip igmp snooping filter-unknown-mcast
```

Command History

Release	Modification
10.11	Command introduced on the 6200, 6300, 6400, 8100, and 8360.

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`ip igmp snooping hardware-packet-forwarding`

Syntax

```
ip igmp snooping hardware-packet-forwarding [enable | disable]
```

Description

This command configures the hardware packet forwarding feature for IGMP control packets on IGMP snooping enabled VLANs. When this feature is enabled, IGMP control packets are forwarded in the hardware to the filter additive ports instead of software reforwarding, so that all the switches in a L2 cascaded network will learn the control packets at the same time. This feature is intended to reduce the join/leave propagation delay in L2 cascaded networks.

Parameter	Description
enable	Enable hardware - packet - forwarding on the VLAN. Required
	Disable hardware - packet - forwarding on the VLAN. Required

Parameter	Description
disable	

NOTE

- This feature is mutually exclusive with Report suppression of both Join and Leave. Also configured IGMP and MLD ACLs will not be honored when this feature is enabled. It is recommended not to enable this feature, if there is an intention to block or rate-limit the IGMP and MLD packets in the control path.
- In deployments with VSX, ISL ports are added as filter additive ports and IGMP control packets are forwarded to the ISL port also from the hardware. Hardware forwarding of packets should ideally honor the spanning tree settings for the ports on which the packets are forwarded.
- The feature is not supported on the VLANs associated with a VNI. Also, the feature should not be enabled on a VLAN if static join or static group is configured on that VLAN. It is not recommended to enable the feature in a non-cascaded L2 network as it can cause some extra processing overhead in the control plane.

Examples

Configuring hardware packet forwarding for IGMP control packets on VLAN 2:

```
switch# configure terminal
switch(config)# vlan 2
switch(config)# ip igmp snooping enable
switch(config-vlan)# ip igmp snooping hardware-packet-forwarding enable
switch# show running-config
Current configuration:
vlan 2
  ip igmp snooping enable
  ip igmp snooping hardware-packet-forwarding enable
switch(config)# vlan 2
switch(config-vlan)# ip igmp snooping hardware-packet-forwarding enable
Enabling Hardware Packet Forwarding is not allowed without Igmp Snooping
enabled on VLAN 2
```

Disabling packet forwarding for IGMP control packets:

```
switch# configure terminal
switch(config)# vlan 2
```

```
switch(config-vlan)# ip igmp snooping hardware-packet-forwarding disable
switch(config-vlan)#
```

NOTE

- IGMP hardware packet forwarding is supported only on IGMP snooping enabled VLANs.
- Attempting to enable this feature without IGMP snooping enabled will result in an error.
- If IGMP snooping is disabled on a VLAN, this feature will also be automatically disabled.

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip igmp snooping preprogram-starg-flow`

Syntax

```
ip igmp snooping preprogram-starg-flow {enable | disable}
```

Description

Configures preprogramming of the starg (*,G) flow feature on the IGMP snooping enabled VLAN.

Parameter	Description
`enable`	Enables preprogramming starg flows on the VLAN.
`disable`	Disables preprogramming starg flows on the VLAN.

Usage

When this feature is enabled, a summarized multicast bridge entry is programmed into the hardware table when an IGMPv2 or IGMPv3 join is received on the IGMP snooping enabled VLAN. This enables multicast flow to be programmed in the hardware before the data packet arrives for a multicast flow. If an unknown packet is received for a multicast flow, having this feature enabled triggers programming of starg entry in the hardware on selected platforms, which is helpful in optimizing hardware resource utilization and PIM registration in deployments where a L2 device is connected along the PIM registration path.

This feature is currently supported for IGMPv2 and IGMPv3 joins, so IGMPv3 joins that are sent for a specific source are treated as IGMPv2 joins and summarized entry is programmed in the corresponding hardware.

Preprogramming of (*,G) flows is supported only on the IGMP snooping enabled VLANs. If IGMP snooping is disabled on a VLAN, this feature is auto-disabled.

This feature is currently supported for IGMPv2 and IGMPv3 joins, as a result, summarized multicast flow is programmed in advance when an IGMPv2 join or IGMPv3 join for a specific group is received. For IGMPv3 deployments, traffic from all sources for a specific multicast group is sent to all clients, regardless of whether they send IGMPv2 or IGMPv3 joins for this group. Keeping this feature disabled is recommended on VLANs where traffic from the specific source is only expected for the IGMPv3 clients.

On the HPE Aruba Networking 6200, 6300, 6400, 8100, and 8360 Switch series, a single (*,G) entry is programmed in advance for each join received. Data driven programming of SG entries does not occur when traffic is received from a specific source for this group. A single (*,G) entry is used to forward the traffic to the clients for all of the active joins in the feature enabled VLANs.

When an unknown multicast packet is received on a VLAN where this feature is enabled, it triggers programming of a (*,G) entry in the hardware instead of the SG.

It is highly recommended to not enable this feature on devices where PIM or L3 multicast routing is enabled as it can lead to issues like permanent traffic loss.

Configuring this feature on devices where there are multiple sources sending traffic for the same group address is recommended.

This feature is mutually exclusive with the IGMP snooping static group feature.

Example

Enable preprogramming multicast (*,G) flows:

```
switch(config)# vlan 2
switch(config-vlan-2)# ip igmp snooping preprogram-starg-flow enable
```

Disable preprogramming multicast (*,G) flows:

```
switch(config)# vlan 2
switch(config-vlan-2)# ip igmp snooping preprogram-starg-flow disable
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip igmp snooping querier-address`

Syntax

```
ip igmp snooping querier-address <QUERIER-IP-ADDRESS>
no ip igmp snooping querier-address <QUERIER-IP-ADDRESS>
```

Description

Configures the querier IP address for the IGMP-MLD proxy feature in EVPN.

The **no** form of this command removes the configuration.

Parameter	Description
`<QUERIER-IP-ADDRESS>`	Specifies the proxy querier IP address.

Usage

Fabrics running the IGMP-MLD Proxy feature must perform distributed querier functionality on every VTEP where the VLAN is extended. Typically, a valid IP address must be configured on the SVI for IGMP querier functionality. However, in centralized routing scenarios within the fabric, most VTEPs only have Layer 2 VLANs. In such cases, IGMP querier functionality needs to be enabled on those VTEPs. This configuration provides a valid querier IP address for those Layer 2 VLANs.

This command is used in conjunction with the IGMP-MLD proxy feature of EVPN.

> **NOTE**
> This is an optional command applicable to VTEPs that have Layer 2 VLANs (without an IP address) and requires IGMP-MLD proxy querier functionality.

If a Layer 3 SVI with an IP address is configured for the VLAN, it takes precedence. In the absence of a valid IP address or SVI, the given IP address is used for querier operations. This configuration is global and applies to all tenants' Layer 2 VLANs where the IGMP-MLD proxy feature is enabled. Fabrics running the **igmp-mld proxy** feature must perform distributed querier functionality on every VTEP where the VLAN is extended. Typically, a valid IP address must be configured on the SVI for IGMP querier functionality needs to be enabled on those VTEPs. This configuration provides a valid querier IP address for those Layer 2 VLANs. This configuration is applicable only when **igmp-mld-proxy** configuration is configured under EVPN.

In the event of underlay tunnel fluctations, the querier state reinitializes and transitions back to the querier state only after the default querier wait of 260 seconds.

This configuration is not applicable to VLANs that are not mapped to a VNI.

> **NOTE**
> The specified IP address does not need to be assigned to any interface.

Example

Configure the querier IP address for the IGMP-MLD proxy feature in EVPN.

```
switch(config)# ip igmp snooping querier-address 10.10.10.1
```

Remove the configuration of the querier IP address for the IGMP-MLD proxy feature in EVPN:

```
switch(config)# no ip igmp snooping querier-address 10.10.10.1
```

Command History

Release	Modification
10.16	Command introduced.

Command Information

Platforms	Command context	Authority
All platform	`config`	Administrators or local user group members with execution righ ts for this command.

`ip igmp snooping report-suppression`

Syntax

```
ip igmp snooping report-suppression enable|disable
```

Description

Enable IGMP report suppression to minimize unnecessary multicast control traffic by aggregating and suppressing duplicate reports for the same group from multiple clients within a VLAN. IGMP Report Suppression only works with IGMPv2 reports, not IGMPv3. However, if IGMPv3 is explicitly configured, this feature still will suppress reports for IGMPv2 join and leave messages.

Usage

This feature is enabled by default.

Example

Enabling IGMP report suppression:

```
switch(config)# vlan 100
switch(config-vlan)# ip igmp snooping enable
switch(config-vlan)# ip igmp snooping report-suppression enable
```

Disabling IGMP report suppression:

```
switch(config-vlan)# ip igmp snooping report-suppression disable
```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip igmp snooping static group`

Syntax

```
ip igmp snooping static group <GROUP-NAME>
no ip igmp snooping static group <GROUP-NAME>
```

Description

Configures static multicast group. The **no** form of this command removes the configuration.

Parameter	Description
<GROUP-NAME>	Specifies the group name.

Example

Configure static multicast group on group 239.1.1.1:

```
switch(config)# vlan 2
switch(config-vlan-2)# ip igmp snooping static-group 239.1.1.1
```

Remove static multicast group on group 239.1.1.1:

```
switch(config)# vlan 2
switch(config-vlan-2)# no ip igmp snooping static-group 239.1.1.1
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip igmp snooping vxlan replicate-all`

Syntax

```
ip igmp snooping vxlan replicate-all
no ip igmp snooping vxlan replicate-all
```

Description

This command enables AOS-CX leaf devices (access-layer switches that connect directly to servers, storage, and other endpoints) to unconditionally forward multicast traffic to all VTEPs that do not support the exchange of IGMP reports over the fabric. By default, when IGMP snooping is enabled in AOS-CX, multicast traffic is selectively forwarded only to VTEPs connected to have interested hosts. This is achieved by exhanging IGMP reports over VXLAN tunnels where each VTEP processes the remote reports and uses the membership information to forward multicast traffic efficiently, conserving fabric bandwidth. However, in a multi-vendor deployment, some VTEPs may not support the exchange of IGMP reports, which can result in the switch failing to learn the complete group membership, and can cause subsequent multicast traffic loss. This command ensures that multicast traffic sourced from the leaft switch is always replicated to all VTEPs, regardless of report exchange capability, thereby maintaining service continuity in mixed-vendor environments.

Usage

When this configuration is applied, multicast traffic sourced from hosts attached to the local VTEP is unconditionally replicated to third-party VTEPs. This ensures that receivers connected to those remote VTEPs can receive the multicast streams. It is expected that the remote VTEP also replicates its locally sourced multicast traffic back to the local VTEP. Based on the local IGMP reports, the traffic will then be forwarded to the corresponding host ports. Therefore, before you issue the **ip igmp snooping vxlan**

replicate-all commend, ensure that an IGMP querier is configured in the same VTEP to maintain group memberships.

> **NOTE**
>
> While this command supports multicast traffic exchange across VTEPs in deployments where the source and receiver VLANS reside within the same subnet, it does not enable inter-VLAN multicast routing. Routing multicast traffic between VLANs still requires valid group membership reports from the peer VTEPs, and in the absence of such reports, traffic will not be routed from the source VLAN to the receiver VLAN. The configuration created by this command is not applicable when operating in standards-based IGMP-MLD Proxy mode and is supported only in the AOS-CX native IGMP/MLD mode over VXLANs.

Example

```
switch(config)# ip igmp snooping vxlan replicate-all
switch(config)# no ip igmp snooping vxlan replicate-all
```

Release	Modification
10.17.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`show ip igmp snooping`

Syntax

```
show ip igmp snooping
   counters
   detail
   groups [vlan <vlan-id>]
   no ...
```

```
   packet-exceptions
   static-groups
   statistics
   vlan <vlan-id> [group {<ip-addr> [client_details]}|{port <IF-NAME>}|
{vtep-peer <A.B.C.D>}]
   vsx-peer
```

Description

Shows IGMP snooping configuration information and status for all VLANs. Specify a VLAN ID or a VLAN and a group to display details for only that VLAN or VLAN group.

Parameter	Description
`counters`	Shows IGMP query packets transmitted (Tx), received (Rx), and error packet counters.
`detail`	Shows IGMP Snooping details for all VLANs, including joined po rts or VXLAN tunnel endpoints (VTEPs) for each group in the V LAN.
`groups`	Shows IGMP snooping groups information. Include the optional **vlan <vlan - id>** parameter to display information for groups o n a specific VLAN.
`no ...`	Negates any configured parameter.
`packet-exceptions`	Troubleshoot issues in L2 multicast bridge entries for data pac kets forwarded to the CPU.
`static-groups`	Shows MLD snooping static group details, including the number of static groups joined.
`statistics`	Shows MLD snooping statistics.
`vlan <vlan-id>`	Shows IGMP snooping protocol information and number of diffe rent groups joined for the VLAN.
`group`	Shows IGMP snooping group information for the specified VLA N, including the number of different groups joined for the VLA N. Identify the group by IP address or interface name.
`<ip-addr> [client-details]`	Shows IGMP snooping group address information. Include the o ptional **client details** parameter to display IGMP snooping clien t details.
`port <IF-NAME>`	Shows IGMP snooping group information for the interface name in **member/slot/port** format.

Parameter	Description
`vtep-peer <A.B.C.D>`	Shows IGMP snooping info for the specified VTEP.

Examples

Showing IGMP snooping configuration and status:

```
switch# show ip igmp snooping
IGMP Snooping Protocol Info
Total VLANs with IGMP enabled                 : 1
IGMP Drop Unknown Multicast                   : Global
Configured IGMP Proxy Querier address         :10.10.10.1
VLAN ID : 1
VLAN Name : DEFAULT_VLAN_1
IGMP Snooping is not enabled
VLAN ID : 2
VLAN Name : VLAN2
IGMP Configured Version : 3
IGMP Operating Version : 3
IGMP preprogram-starg-flow is operational
Querier Address [this switch] : 20.1.1.1
Querier Port :
Querier UpTime :0m 21s
Querier Expiration Time :0m 2s
```

Include the **detail** parameter for additional information on joined ports or VTEPs, as shown in the example below:

```
switch# show ip igmp snooping detail
IGMP Snooping Protocol Info
Total VLANs with IGMP enabled                 : 1
Current count of multicast groups joined      : 4
IGMP Drop Unknown Multicast                   : Global
VLAN ID : 100
VLAN Name : VLAN100
IGMP Configured Version : 3
IGMP Operating Version : 3
IGMP preprogram-starg-flow is not operational
Querier Address [this switch] : 15.1.1.1
Querier Port :
Querier UpTime :9m 32s
Querier Expiration Time :0m 10s
Router Detected Port(s) :
Active Group Address   Tracking  Vers Mode Uptime    Expires    Ports/Vteps
--------------------- ---------- ---- ---- --------- ----------
```

```
------------------------------
225.1.1.1               Filter    3     EXC   1m 2s      0s
200.1.1.1,200.1.1.2
                                                                   1/6/22
225.1.1.2               Filter    3     EXC   1m 2s      0s
200.1.1.1,200.1.1.2
                                                                   1/6/22
226.1.1.1               Filter    3     EXC   1m 4s      0s          200.1.1.3
226.1.1.2               Filter    3     EXC   1m 4s      0s          200.1.1.3
```

NOTE

With standard based IGMP-MLD Proxy mode, the membership reports from remote VTEPs does not have an expiry timer associated and shall be shown as 0 seconds.

Command History

Release	Modification
10.16	Updated example to include new IGMP proxy querier address.
10.13	Programming starg flow is now supported.
10.10	The **packet - exceptions** parameter is introduced.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

sFlow

sFlow is a technology for monitoring traffic in switched or routed networks. The sFlow monitoring system is comprised of:

- An sFlow Agent that runs on a network device, such as a switch. The agent uses sampling techniques to capture information about the data traffic flowing through the device and forwards this information to an sFlow collector.
- An sFlow Collector that receives monitoring information from sFlow agents. The collector stores this information so that a network administrator can analyze it to understand network data flow patterns.

> **NOTE**
> The sFlow UDP datagrams sent to a collector are not encrypted, therefore any sensitive information contained in an sFlow sample is exposed.

Subtopics

sFlow agent
Configuring the sFlow agent
sFlow scenario
sFlow scenario 2
sFlow agent commands

sFlow agent

The sFlow agent on the switch provides ingress sampling of all forwarded layer 2 and layer 3 traffic on LAG and Ethernet ports. High-availability is supported (packet sampling continues to work after switch-over).

The sFlow agent can communicate with up to three sFlow collectors at the same time. The agent communicates with collectors on the default VRF and non-default VRF.

Although you can configure very high sampling rates, the switch may drop samples if it cannot handle the rate of sampled packets. High sampling rates may also cause high CPU usage resulting in control plane performance issues.

A single sFlow datagram sent to a collector contains multiple flow and counter samples. The total number of samples an sFlow datagram can contain varies depending on the settings for header size and maximum datagram size.

Default settings

- sFlow is disabled on all interfaces.
- Collector port: UDP port 6343.
- sflow sampling mode: Ingress
- Sampling rate: 4096.
- Polling interval: 30 seconds.

- Header size: 128 bytes.
- Max datagram size: 1400 bytes.

Supported features

- Global sampling rate
- Global sampling mode (Ingress, Egress, and Both)
- Interface counters polling
- Agent IP configuration for IPv4 and IPv6
- Header size configuration
- Max datagram size configuration
- Ingress sampling for all forwarded traffic (L2, L3)
- Egress sampling for all forwarded traffic (L2, L3)
- Enable/Disable sFlow per interface
- Support for three remote collectors
- A collector can be defined on the default and non-default VRF
- Sampling on Ethernet and LAG interfaces
- High availability support (sampling continues to work after switch-over)
- Source IP support (setting source IP for sFlow datagrams sent to a remote collector)

Limitations

- Sampling rate cannot be set per interface (global only)
- sFlow is not configurable through SNMP.

`Configuring the sFlow agent`

Procedure

1. Configure one or more sFlow collectors with the command `sflow collector`. This determines where the sFlow agent sends sFlow information.
2. Enable the sFlow agent on all interfaces, or on a specific interface, with the command `sflow`.

3. Define the address of the sFlow agent with the command `sflow agent-ip`.
4. By default, the source IP address for sFlow datagrams is set to the IP address of the outgoing switch interface on which the sFlow client is communicating with a collector. Since the switch can have multiple routing interfaces, datagrams can potentially be sent on different paths at different times, resulting in different source IP addresses for the same client. To resolve this issue, define a single source IP address. For details, see Single source IP address in the Fundamentals Guide.
5. For most deployments, the default values for the following settings do not need to be changed. If your deployment requires different settings, change the default values with the indicated commands:

sFlow setting	Default value	Command to change it
Rate at which packets are sampled.	1 in every 4096 packets	`sflow sampling`
Rate at which the switch sends data to an sFlow collector.	30 seconds	`sflow polling`
Size of the sFlow header.	128 bytes	`sflow header-size`
Maximum size of an sFlow datagram.	1400 bytes	`sflow max-datagram-size`

6. Review sFlow configuration settings with the command `show sflow`.

Example

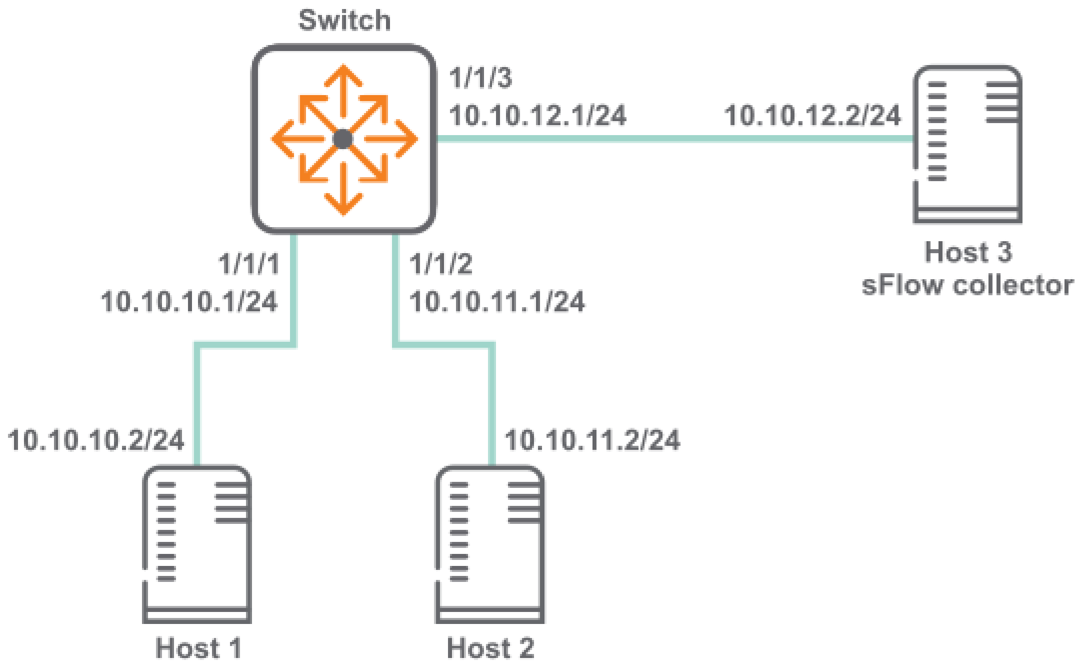
This example creates the following configuration:

- Configures an sFlow collector with the IP address **10.10.20.209** .
- Enables the sFlow agent on all interfaces.
- Defines the sFlow agent IP address to be **10.10.1.5**.

```
switch(config)# sflow collector 10.10.20.209
switch(config)# sflow
switch(config)# sflow agent-ip 10.0.0.1
```

sFlow scenario

In this scenario, two hosts send sFlow traffic through a switch to an sFlow collector. The physical topology of the network looks like this:



Procedure

1. Enable sFlow globally.
   switch# **config** switch(config)# **sflow**

2. Set the sFlow agent IP address to **10.10.12.1**.
   switch(config)# **sflow agent-ip 10.10.12.1**

3. Set the sFlow collector IP address to **10.10.12.2**.
   switch(config)# **sflow collector 18.2.2.2**

4. Configure sFLow sampling rate and polling interval.
   switch(config)# **sflow sampling 5000** switch(config)# **sflow polling 20**

5. Configure interface **1/1/1** with IP address **10.10.10.1/24.**
   switch(config)# **interface 1/1/1** switch(config-if)# **no shutdown** switch(config-if)# **ip address 10.10.10.1/24** switch(config)# **quit**

6. Configure interface **1/1/2** with IP address **10.10.11.1/24.**
   switch(config)# **interface 1/1/2** switch(config-if)# **no shutdown** switch(config-if)# **ip address 10.10.11.1/24** switch(config)# **quit**

7. Configure interface **1/1/3** with IP address **10.10.12.1/24.**
   switch(config)# **interface 1/1/3** switch(config-if)# **no shutdown** switch(config-if)# **ip address 10.10.12.1/24** switch(config)# **quit**

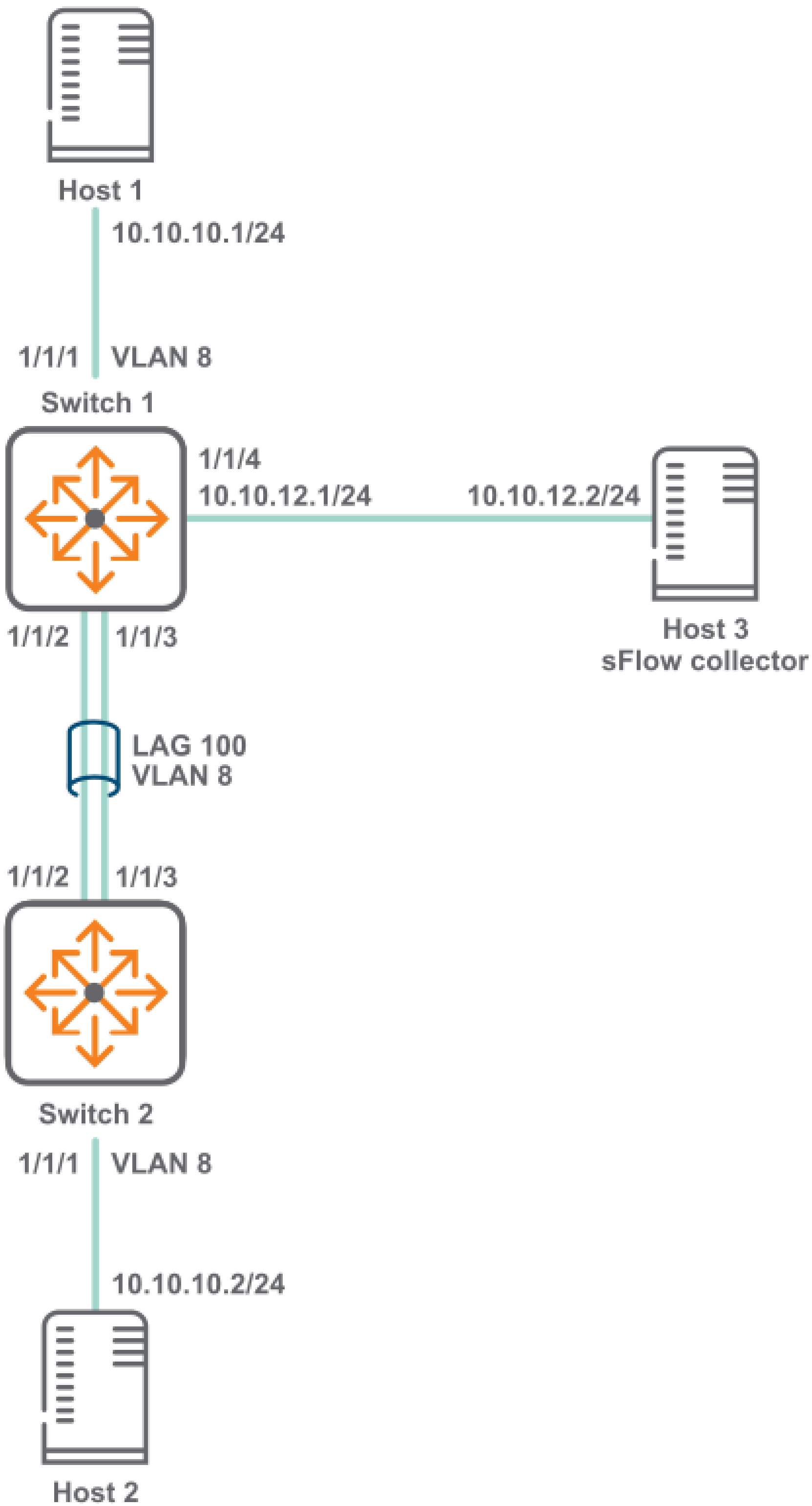
8. Verify sFlow configuration

```
switch# show sflow
sFlow Global Configuration
-----------------------------------------
sFlow                         enabled
Collector IP/Port/Vrf         10.10.10.2/6343/default
Agent Address                 10.0.0.1
Sampling Rate                 1024
Polling Interval              30
Header Size                   128
Max Datagram Size             1400
sFlow Status
-----------------------------------------
Running - Yes
sFlow enabled on Interfaces:
-----------------------------------------
lag100
sFlow Statistics
-----------------------------------------
Number of Samples             200
```

```
switch# show sflow
sFlow Global Configuration
-----------------------------------------
sFlow                         enabled
Collector IP/Port/Vrf         10.10.10.2/6343/default
Agent Address                 10.0.0.1
Sampling Rate                 1024
Polling Interval              30
Header Size                   128
Max Datagram Size             1400
Sampling Mode                 both
sFlow Status
-----------------------------------------
Running - Yes
sFlow enabled on Interfaces:
-----------------------------------------
lag100
sFlow Statistics
-----------------------------------------
Number of Ingress Samples     200
Number of Egress Samples      0
```

sFlow scenario 2

In this scenario, two hosts connected to different switches send sFlow traffic to a collector. A LAG is used to connect the two switches. The physical topology of the network looks like this:



Procedure

1. Configure switch 1.

   a. Enable sFlow globally.

   ```
   switch# config
   switch(config)# sflow
   ```

   b. Set the sFlow agent IP address to **10.10.12.1**.

   ```
   switch(config)# sflow agent-ip 10.10.12.1
   ```

   c. Set the sFlow collector IP address to **10.10.12.2**.

   ```
   switch(config)# sflow collector 10.10.12.2
   ```

   d. Configure sFLow sampling rate and polling interval.

   ```
   switch(config)# sflow sampling 5000
   switch(config)# sflow polling 10
   ```

   e. Create VLAN **8**.

   ```
   switch(config)# vlan 8
   switch(config-vlan-8)# no shutdown
   switch(config)# exit
   ```

   f. Define LAG **100** and assign VLAN **vlan 8** to it.

   ```
   switch(config)# interface lag 100
   switch(config-lag-if)# no shutdown
   ```

   ```
   switch(config-lag-if)# vlan access 8
   switch(config-lag-if)# lacp mode active
   ```

   g. Configure interface **1/1/1** .

   ```
   switch(config)# interface 1/1/1

   switch(config-if)# no shutdown
   switch(config-lag-if)# no routing
   switch(config-if)# vlan access 8
   ```

   h. Configure interface **1/1/2** and **1/1/3** as members of LAG **100.**

   ```
   switch# (config)#interface 1/1/2
   ```

```
switch(config-if)# no shutdown
switch(config-if)# lag 100
switch(config-if)# exit
switch(config)# interface 1/1/3

switch(config-if)# no shutdown
switch(config-if)# lag 100
switch(config-if)# exit
```

i. Configure interface **1/1/4** with IP address **10.10.12.1/24**.

```
switch# (config)#interface 1/1/4

switch(config-if)# no shutdown
switch(config-if)# ip address 10.10.12.1/24
switch(config-if)# quit
```

j. Verify sFlow configuration.

```
switch# show sflow
sFlow Global Configuration
-----------------------------------------
sFlow                         enabled
Collector IP/Port/Vrf         10.10.10.2/6343/default
Agent Address                 10.0.0.1
Sampling Rate                 1024
Polling Interval              30
Header Size                   128
Max Datagram Size             1400
sFlow Status
-----------------------------------------
Running - Yes
sFlow enabled on Interfaces:
-----------------------------------------
lag100
sFlow Statistics
-----------------------------------------
Number of Samples             200
```

```
switch# show sflow
sFlow Global Configuration
-----------------------------------------
sFlow                         enabled
Collector IP/Port/Vrf         10.10.10.2/6343/default
Agent Address                 10.0.0.1
```

```
Sampling Rate                 1024
Polling Interval              30
Header Size                   128
Max Datagram Size             1400
Sampling Mode                 both
sFlow Status
----------------------------------------
Running - Yes
sFlow enabled on Interfaces:
----------------------------------------
lag100
sFlow Statistics
----------------------------------------
Number of Ingress Samples     200
Number of Egress Samples      0
```

2. Configure switch 2.

    a. Create VLAN **8**.

    ```
    switch(config)# vlan 8
    switch(config-vlan-8)# no shutdown
    switch(config)# exit
    ```

    b. Define LAG **100** and assign VLAN **vlan 8** to it.

    ```
    switch(config)# interface lag 100
    switch(config-lag-if)# no shutdown
    ```

    ```
    switch(config-lag-if)# vlan access 8
    switch(config-lag-if)# lacp mode active
    ```

    c. Configure interface **1/1/1**.

    ```
    switch(config)# interface 1/1/1
    switch(config-if)# no shutdown
    ```

    ```
    switch(config-if)# vlan access 8
    ```

    d. Configure interface **1/1/2** and **1/1/3** as members of LAG **100.**

    ```
    switch#  (config)#interface 1/1/2
    switch(config-if)# no shutdown
    switch(config-if)# lag 100
    switch(config-if)# exit
    switch(config)-if# interface 1/1/3
    ```

```
switch(config-if)# no shutdown
switch(config-if)# lag 100
switch(config-if)# exit
```

sFlow agent commands

Select a command from the list in the left navigation menu.

Subtopics

- **clear sflow statistics**
- **sflow**
- **sflow agent-ip**
- **sflow collector**
- **sflow disable**
- **sflow header-size**
- **sflow max-datagram-size**
- **sflow mode**
- **sflow polling**
- **sflow sampling**
- **show sflow**

`clear sflow statistics`

Syntax

```
clear sflow statistics {global | interface <INTERFACE-NAME>}
```

Description

This command clears the sFlow sample statistics counter to 0 either globally or for a specific interface.

Parameter	Description
`global`	Specifies all interfaces on the switch.
`interface <INTERFACE-NAME>`	Specifies the name of an interface on the switch.

Examples

Clearing the global sFlow sample statistics counter to 0 globally:

```
switch(config)# clear sflow statistics global
```

Clearing the global sFlow sample statistics counter to 0 for interface 1/1/1:

```
switch(config)# clear sflow statistics interface 1/1/1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`sflow`

Syntax

```
sflow
no sflow
```

Description

Enables the sFlow agent.

- In the **config** context, this command enables the sFlow agent globally on all interfaces.
- In an **config-if** context, this command enables the sFlow agent on a specific interface. sFlow cannot be enabled on a member of a LAG, only on the LAG.

The sFlow agent is disabled by default.

The **no** form of this command disables the sFlow agent and deletes all sFlow configuration settings, either globally, or for a specific interface.

Examples

Enabling sFlow globally on all interfaces:

```
switch(config)# sflow
```

Disabling sFlow globally on all interfaces:

```
switch(config)# no sflow
```

Enabling sFlow on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# sflow
```

Disabling sFlow on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no sflow
```

Enabling sFlow on interface **lag100**:

```
switch(config)# interface lag100
switch(config-if)# sflow
```

Disabling sFlow on interface **lag100**:

```
switch(config)# interface lag100
switch(config-if)# no sflow
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config` `config-if`	Administrators or local user group members with execution righ ts for this command.

`sflow agent-ip`

Syntax

```
sflow agent-ip <IP-ADDR>
no sflow agent-ip [<IP-ADDR>]
```

Description

Defines the IP address of the sFlow agent to use in sFlow datagrams. This address must be defined for sFlow to function. HPE recommends that the address:

- can uniquely identify the switch
- is reachable by the sFlow collector
- does not change with time

The **no** form of this command deletes the IP address of the sFlow agent. This causes sFlow to stop working and no datagrams will be sent to the sFlow collector.

Parameter	Description
`<IP-ADDR>`	Specifies an IP address in IPv4 format (**x.x.x.x**), where **x** is a de cimal number from 0 to 255, or IPv6 format (**xxxx:xxxx:xxxx:x xxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number f rom 0 to F. The agent address is used to identify the switch in a ll sFlow datagrams sent to sFlow collectors. It is usually set to a n IP address on the switch that is reachable from an sFlow colle ctor.

Examples

Setting the agent address to **10.10.10.100**:

```
switch(config)# sflow agent-ip 10.0.0.100
```

Setting the agent address to **2001:0db8:85a3:0000:0000:8a2e:0370:7334**:

```
switch(config)# sflow agent-ip 2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Removing the address configuration from the switch, which results in sFlow being disabled:

```
switch(config)# no sflow agent-ip
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`sflow collector`

Syntax

```
sflow collector <IP-ADDR> [port <PORT>] [vrf <VRF>]
no sflow collector <IP-ADDR> [port <PORT>] [vrf <VRF>]
```

Description

Defines a collector to which the sFlow agent sends data. Up to three collectors can be defined. At least one collector should be defined, and it must be reachable from the switch for sFlow to work.

Parameter	Description
`collector <IP-ADDR>`	Specifies the IP address of a collector in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255, or IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexad ecimal number from 0 to F.
`port <PORT>`	Specifies the UDP port on which to send information to the sFlo w collector. Range: 0 to 65536. Default: 6343.
`vrf <VRF>`	Specifies the VRF on which to send information to the sFlow col lector. The VRF must be defined on the switch. If no VRF is spec ified, the default VRF (**default**) is used.

Example

Defining a collector with IP address **10.10.10.100** on UDP port **6400**:

```
switch(config)# sflow collector 10.0.0.1 port 6400
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`sflow disable`

Syntax

```
sflow disable
```

Description

Disables the sFlow agent, but retains any existing sFlow configuration settings. The settings become active if the sFlow agent is re-enabled.

Example

Disabling sFlow support:

```
switch(config)# sflow disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`sflow header-size`

Syntax

```
sflow header-size <SIZE>
no sflow header-size [<SIZE>]
```

Description

Sets the sFlow header size in bytes.

The **no** form of this command sets the header size to the default value of 128.

Parameter	Description
`header-size <SIZE>`	Specifies the sFlow header size in bytes. Range: 64 to 256. Defa ult: 128.

Examples

Setting the header size to **64** bytes:

```
switch(config)# sflow header-size 64
```

Setting the header size to the default value of **128** bytes:

```
switch(config)# no sflow header-size
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`sflow max-datagram-size`

Syntax

```
sflow max-datagram-size <SIZE>
no sflow max-datagram-size [<SIZE>]
```

Description

Sets the maximum number of bytes that are sent in one sFlow datagram.

The **no** form of this command sets maximum number of bytes to the default value of 1400.

Parameter	Description
`max-datagram-size <SIZE>`	Specifies the maximum datagram size in bytes. Range: 1 to 900 0. Default: 1400.

Examples

Setting the datagram size to **1000** bytes:

```
switch(config)# sflow max-datagram-size 1000
```

Setting the header size to the default value of **1400** bytes:

```
switch(config)# no sflow max-datagram-size
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`sflow mode`

Syntax

```
sflow mode {ingress | egress | both}
no sflow mode {ingress | egress | both}
```

Description

Sets the sFlow sampling mode. The default mode is ingress.

The no form of the command sets the sampling mode to ingress. Executing the no form of the command with the ingress option will have no impact as ingress is the default mode.

Parameter	Description
`ingress`	Samples only ingress traffic.
`egress`	Samples only egress traffic.
`both`	Samples both ingress and egress traffic.

Examples

Setting the sFlow mode to only sample egress traffic:

```
switch# configure terminal
switch(config)# sflow mode egress
```

Setting the sFlow mode to only sample ingress traffic:

```
switch# configure terminal
switch(config)# sflow mode ingress
```

Setting the sFlow mode to sample both sample ingress and egress traffic:

```
switch# configure terminal
switch(config)# sflow mode both
```

Resetting the sFlow sampling mode to the default of ingress from previously configured mode of egress:

```
switch# configure terminal
switch(config)# no sflow mode egress
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`sflow polling`

Syntax

```
sflow polling <INTERVAL>
no sflow polling [<INTERVAL>]
```

Description

Defines the global polling interval for sFlow in seconds.

The **no** form of this command sets the polling interval to the default value of 30 seconds.

Parameter	Description
<INTERVAL>	Specifies the polling interval in seconds. Range: 10 to 3600. Default: 30.

Examples

Setting the polling interval to 10:

```
switch(config)# sflow polling 10
```

Setting the polling interval to the default value.

```
switch(config)# no sflow polling
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

`sflow sampling`

Syntax

```
sflow sampling <RATE>
no sflow sampling [<RATE>]
```

Description

Defines the global sampling rate for sFlow in number of packets. The default sampling rate is 4096, which means that one in every 4096 packets is sampled. A warning message is displayed when the sampling rate is set to less than 4096 and proceeds only after user confirmation.

The **no** form of this command sets the sampling rate to the default value of 4096.

Parameter	Description
`sampling <RATE>`	Specifies the sampling rate. Range: 1 to 1000000000. Default: 4096.

Examples

Setting the sampling rate to **5000**:

```
switch(config)# sflow sampling 5000
```

Setting the sampling rate to the default:

```
switch(config)# no sflow sampling
```

Setting the sampling rate to **1000**:

```
switch(config)# sflow sampling 1000
Setting the sFlow sampling rate lower than 4096 is not recommended and might
affect system performance.
Do you want to continue [y/n]? y
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

show sflow

Syntax

```
show sflow [interface <INTERFACE-NAME>]
```

Description

Shows sFlow configuration settings and statistics for all interfaces, or for a specific interface. It also displays the current status of sFlow on the device and reports any errors that require attention.

> **NOTE**
>
> If sFlow is enabled on the interfaces associated with a lag interface, then the interfaces will not be shown as separate entries under `sFlow enabled on Interface` in the output. Only the associated lag interface will have an entry in the column.

Parameter	Description
`interface <INTERFACE-NAME>`	Specifies the name of an interface on the switch.

Examples

Showing sFlow information for all interfaces:

```
switch# show sflow
sFlow Global Configuration
-----------------------------------------
sFlow                         enabled
Collector IP/Port/Vrf         10.0.0.2/6343/default
                              10.0.0.3/6400/default
Agent Address                 10.0.0.1
Sampling Rate                 1024
Polling Interval              30
Header Size                   128
Max Datagram Size             1400
Sampling Mode                 ingress
sFlow Status
-----------------------------------------
Running - Yes
sFlow enabled on Interfaces:
-----------------------------------------
```

```
1/1/2
1/1/3
lag100
sFlow Statistics
-----------------------------------------
Number of Ingress Samples     200
Number of Egress Samples      120
```

```
switch# show sflow
sFlow Global Configuration
-----------------------------------------
sFlow                         enabled
Collector IP/Port/Vrf         10.10.10.2/6343/default
Agent Address                 10.0.0.1
Sampling Rate                 1024
Polling Interval              30
Header Size                   128
Max Datagram Size             1400
sFlow Status
-----------------------------------------
Running - Yes
sFlow enabled on Interfaces:
-----------------------------------------
lag100
sFlow Statistics
-----------------------------------------
Number of Samples             200
```

Showing sFlow information for interface **1/1/1**:

```
switch# show sflow interface 1/1/1
sFlow configuration - Interface 1/1/1
-----------------------------------------
sFlow                         enabled
Sampling Rate                 1024
Sampling Mode                 both
Number of Ingress Samples     81
Number of Egress Samples      20
sFlow Sampling Status         success
```

```
switch# show sflow interface 1/1/1
sFlow configuration - Interface 1/1/1
-----------------------------------------
sFlow                         enabled
Sampling Rate                 1024
Number of Samples             30
sFlow Sampling Status         success
```

Showing sFlow information for interface **lag 10**:

```
switch# show sflow interface lag 10
sFlow Configuration - Interface lag10
-----------------------------------------
sFlow                         enabled
Sampling Rate                 4096
Sampling Mode                 both
Number of Ingress Samples     0
Number of Egress Samples      0
sFlow Sampling Status         error
Sampling Status on LAG members
------------------------------------
Intf 1/1/2                     no agent
Intf 1/1/3                     no agent
```

```
switch# show sflow interface lag 10
sFlow Configuration - Interface lag10
-----------------------------------------
sFlow                         enabled
Sampling Rate                 4096
Number of Samples             0
sFlow Sampling Status         error
Sampling Status on LAG members
------------------------------------
Intf 1/1/2                     no agent
Intf 1/1/3                     no agent
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

DHCP

The Dynamic Host Configuration Protocol (DHCP) enables the automatic assignment of IP addresses and other configuration settings to network devices.

The DHCP relay agent acts an intermediary, forwarding DHCP requests/response between DHCP clients/ servers on different networks. This enables DHCP clients to use the services of DHCP servers that are not on the same subnet on which they are located.

```
switch(config)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-if)# vlan access 100
switch(config-if)# exit
switch(config-if)# interface vlan 100
switch(config-if-vlan)# <DHCP feature CLI>
```

Subtopics

DHCP client

Subtopics

Protocol and feature details

DHCP Client is supported on IPv4 and IPv6 (for IPV6 it is supported only on management VRF).

The IP DHCP assignment with HPE Aruba Networking AOS-CX switch is to allow the following:

- IP switch assignment.
- Managing the initial IP access to the switch.

- Prepare for Zero Touch Provisioning (ZTP) with HPE Aruba Networking Central.

> **NOTE**
>
> Zero Touch Provisioning (ZTP) is only supported for IPV4 in HPE Aruba Networking Central

- Prepare for Zero Touch Provisioning (ZTP) by using DHCP options along with a TFTP server.

Supported platform and standards

The following table list the supported VLANs and ports for the AOS-CX Switch.

Table 1. Supported VLAN and Ports for DHCP

Platform	VLAN 1
5400	Yes
6200	Yes

Configuration task list

To run the DHCP Client do the following:

- Enter the interface VLAN or Management VRF context
- Enable DHCP

Considerations and best practices

The IP DHCP can only be configured on one VLAN per switch.

If an OOB Management port is available, you can configure the DHCP service simultaneously for the VLAN IN Band and the OOB Management port. When both the IN Band data port and the OOB Management port receive a DHCP address, then the DHCP address received on the OOBM port is preferred over the IN Band data port for ZTP.

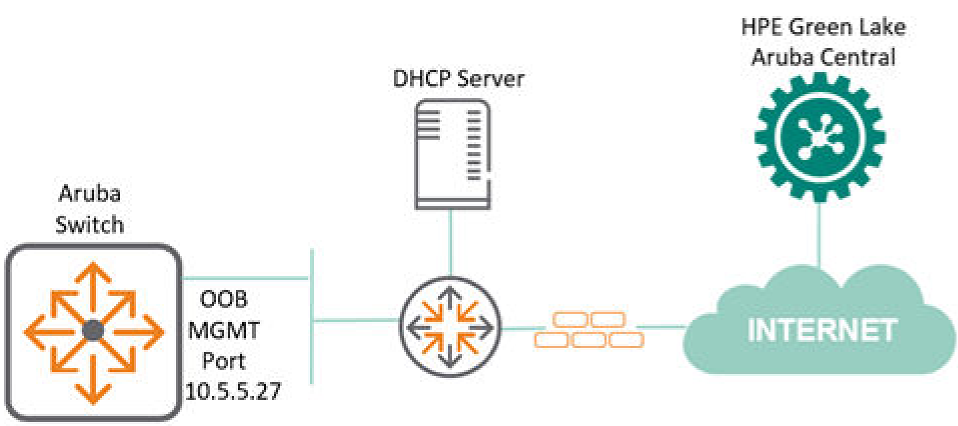
Use case

About this task

IP DHCP assignment with HPE Aruba Networking AOS-CX switches is to allow IP switch assignment and Zero Touch Provisioning using HPE Aruba Networking Central, as shown below.

In the following DHCP client scenario the OOB Management port is in use. The IN Band data ports can also be used for Zero Touch Provisioning (ZTP).

For more information related to the supported ports for the different switches, see Table 1, Supported VLAN and Ports for DHCP.



Procedure

1. Switch configuration from factory on OOB Management port.

```
switch#
!
interface mgmt
    no shutdown
    ip dhcp
!
```

2. OOB Management port gets assigned with an IPv4 address 10.5.5.27 from DHCP server.

```
switch# show interface mgmt
  Address Mode                    : dhcp
  Admin State                     : up
  Link State                      : up
  Mac Address                     : 88:3a:30:9a:39:01
  IPv4 address/subnet-mask        : 10.5.5.27/24
  Default gateway IPv4            : 10.5.5.254
  IPv6 address/prefix             :
2a00:23c5:ac8a:3e01:8a3a:30ff:fe9a:3901/64
  IPv6 link local address/prefix: fe80::8a3a:30ff:fe9a:3901/64
  Default gateway IPv6            : fe80::f286:20ff:fe0b:fe5
  Primary Nameserver              : 10.5.19.69
  Secondary Nameserver            : 10.5.19.79
  Tertiary Nameserver             : 8.8.4.4
```

Results

The following output shows a switch connected to HPE Aruba Networking Central allocated with an IP address of 10.5.5.27 on its OOB MGMT port by a DHCP server. Based on its designated configuration in the public cloud, the switch can get its configuration parameters directly from HPE Aruba Networking Central.

```
switch# show hpe_anw-central
  Central admin state                      :  enabled
  Central location                         :  device-prod2-
d2.central.arubanetworks.com
  VRF for connection                       :  mgmt
  Central connection status                :  connected
  Central source                           :  activate
  Central source connection status         :  connected
  Central source last connected on         :  Wed Sep 22 18:39:05 UTC 2021
  System time synchronized from Activate   :  True
  Activate Server URL                      :  devices-v2.arubanetworks.com
  CLI location                             :  N/A
  CLI VRF                                  :  N/A
```

```
Source IP                                    :  10.5.5.27
Source IP Overridden                         :  False
Central support mode                         :  disabled
```

DHCP client commands

Select a command from the list in the left navigation menu.

Subtopics

`ip dhcp`

Syntax

```
ip dhcp
no ip dhcp
```

Description

Enables the DHCP client on the management interface or any interface VLAN to automatically obtain an IP address from a DHCP server on the network. By default, the DHCP client is enabled on the management interface and VLAN 1.

The **no** form of the command disables DHCP mode and is supported only on interface VLANs; it is not supported on the management interface.

Examples

Enabling the DHCP client on the management interface:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# ip dhcp
switch(config-if-mgmt)# no shutdown
```

Enabling the DHCP client on the interface vlan 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip dhcp
switch(config-if-vlan)# no shutdown
```

Disabling the DHCP client on the interface vlan 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip dhcp
```

```
switch(config)# interface vlan 4
switch(config-if-vlan)# vrf attach red
switch(config-if-vlan)# ip dhcp
```

If the interface is not enabled, you can enable it by entering the `no shutdown` command.

> **NOTE**
> DHCP client can be enabled on one or more user VLANs.

General Limitations

1. ZTP (Zero Touch Provisioning) options from a DHCP server are only supported on VLAN 1. These options will be ignored if received on any other VLAN.
2. CoP and HTTPS-proxy DHCP options should be received on only one DHCP client-enabled VLAN. If multiple VLANs receive the same option, the selection of the VLAN from which the options are applied is non-deterministic. Therefore, configuring multiple VLANs to receive these options is not recommended.
3. It is recommended to limit the number of DHCP client VLANs in a system to 32. This total includes both DHCPv4 and DHCPv6 client instances.
4. The ip dhcp option broadcast-flag configuration must be enabled to successfully obtain DHCP IP addresses across multiple VLANs when the default gateway route is present.

Command History

Release	Modification
10.17	Support enabled on one or more user VLANs.
10.16.1000	Support added for the 8100, 832x, 8360, 8400, 9300, 9300S, and 10000 Switch Series.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-if-mgmt` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ipv6 dhcp`

Syntax

```
ipv6 dhcp
no ipv6 dhcp
```

Description

Configure to enable DHCPv6 mode on a VLAN. In DHCPv6 Mode, the VLAN will get a DHCP-Server allocated IPv6 address along with DNS and NTP attributes.

The **no** form of the command disables DHCP mode and is supported only on interface VLANs; it is not supported on the management interface.

> **NOTE**
>
> ZTPv6 attributes are included in the request packet, but they are not utilized by ZTP on the data port. Currently, ZTPv6 attributes are only supported on the management port. Also, IPv6 addresses assigned through DHCPv6 only support a /64 prefix.

Examples

Enabling the DHCPv6 client on the interface vlan 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 dhcp
```

Enabling the DHCPv6 on two VLANs:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 dhcp
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 dhcp
```

Enabling the DHCPv6 client on the interface vlan 4 under non-default VRF:

```
switch(config)# interface vlan 4
switch(config-if-vlan)# vrf attach red
```

```
switch(config-if-vlan)# ipv6 dhcp
```

Disabling the DHCPv6 client on the interface vlan 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 dhcp
```

If the interface is not enabled, you can enable it by entering the `no shutdown` command.

General Limitations

1. ZTP (Zero Touch Provisioning) options from a DHCP server are only supported on VLAN 1. These options will be ignored if received on any other VLAN.
2. CoP and HTTPS-proxy DHCP options should be received on only one DHCP client-enabled VLAN. If multiple VLANs receive the same option, the selection of the VLAN from which the options are applied is non-deterministic. Therefore, configuring multiple VLANs to receive these options is not recommended.
3. It is recommended to limit the number of DHCP client VLANs in a system to 32. This total includes both DHCPv4 and DHCPv6 client instances.
4. The ip dhcp option broadcast-flag configuration must be enabled to successfully obtain DHCP IP addresses across multiple VLANs when the default gateway route is present.

Command History

Release	Modification
10.17	Support added for DHCPv6 on multiple VLANs.
10.16.1000	Support added for the 8100, 832x, 8360, 8400, 9300, 9300S, and 10000 Switc h Series.
10.15	Command was introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config-if-mgmt` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip dhcp option`

Syntax

```
ip dhcp option [host-name | broadcast-flag]
no ip dhcp option[host-name | broadcast-flag]
```

Description

This command enables the DHCP client host name and broadcast flag globally.

If the **ip dhcp option broadcast-flag** command is enabled, then the DHCP offer and ack packets in the DHCP requests will be treated as broadcast packets. These packets will not be forwarded due to the presence of a default static route.

The **no** form of this command globally disables the host name and DHCP client broadcast flag options.

> **NOTE**
> The **ip dhcp option broadcast-flag** command should be configured before configuring the **ip dhcp** command.

Example

Enabling the DHCP client broadcast flag globally:

```
switch(config)# ip dhcp option
broadcast-flag  DHCP Client broadcast-flag option (Default:disabled)
host-name       DHCP Client hostname option
switch(config)# ip dhcp option
% Command incomplete.
switch(config)# interface vlan 1
switch(config-if-vlan)# ip dhcp
switch(config-if-vlan)# ip dhcp op
% There is no matched command.
switch(config-if-vlan)# ip dhcp op
Invalid input: ip
switch(config-if-vlan)#
```

Enabling the DHCP client host name globally:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip dhcp option host-name
```

Disabling the DHCP client broadcast flag globally:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip dhcp option broadcast-flag
```

Disabling the DHCP client host name globally:

```
switch(config)# ip dhcp option
broadcast-flag  DHCP Client broadcast-flag option (Default:disabled)
host-name       DHCP Client hostname option
switch(config)# ip dhcp option
% Command incomplete.
switch(config)# interface vlan 1
switch(config-if-vlan)# ip dhcp
switch(config-if-vlan)# no ip dhcp option host-name
% There is no matched command.
switch(config-if-vlan)# ip dhcp op
Invalid input: ip
switch(config-if-vlan)#
```

Command History

Release	Modification
10.16.1000	Support added for the 8100, 832x, 8360, 8400, 9300, 9300S, and 10000 Switc h Series.
10.13.1000	Command Introduced

Command Information

Platforms	Command context	Authority
All platforms	`config-if-mgmt` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip dhcp preferred-vlan`

Syntax

```
ip dhcp preferred-vlan <VLAN-ID> [vrf <VRF-NAME>]
no ip dhcp preferred-vlan vlan <VLAN-ID> [vrf <VRF-NAME>]
```

Description

Specifies the preferred VLAN for DHCP client options, including DNS, NTP, and Default Gateway. The preferred VLAN is the VLAN used to select DHCP client options within a specified VRF when multiple

VLANs have DHCP client enabled in the same VRF. If no VRF is specified, the configuration applies to the default VRF. In cases where the preferred VLAN is not explicitly configured, the lowest VLAN with DHCP client enabled will be automatically selected.

The **no** form of the command removes the preferred VLAN configuration and restores the default behavior.

Parameter	Description
<VLAN-ID>	VLAN ID to set as preferred VLAN. VLAN identifier range: 1 to 4094.
<VRF-NAME>	Specifies the VRF in which the preferred VLAN should be confi gured.

General Considerations

1. If the preferred VLAN configuration status is not success, the DHCP client-enabled VLAN with the lowest VLAN ID will be used for DHCP attribute selection.

   Use the command `show ip dhcp preferred-vlan` to check the configuration status.

2. The preferred or lowest VLAN will be inactive in the following scenarios:

   a. When the DHCP interface is in a shutdown state and corresponds to the preferred or lowest VLAN ID.

   b. When a static IP is configured on an interface that also has DHCP enabled — the static IP takes precedence over DHCP.

   c. When DHCP attributes are not received from the server on the preferred or lowest VLAN.

Examples

Selecting VLAN 1 as the preferred VLAN:

```
switch(config)# ip dhcp preferred-vlan 1
```

Selecting VLAN 10 as the preferred VLAN with VRF name:

```
switch(config)# ip dhcp preferred-vlan 10 vrf red
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`show ip dhcp`

Syntax

```
show ip dhcp
```

Description

Displays DHCP IPv4 information on the ports.

Examples

Displaying the DHCP IPv4 information on the ports:

```
switch# show ip dhcp
DHCP Options: Broadcast-flag, Hostname
INTERFACE-NAME   ADDRESS           DEFAULT_GATEWAY   DOMAIN_NAME   VRF
DNS-SERVERS
----------------------------------------------------------------------------
-----------------------------
vlan1            10.254.239.10/27                    domain.com    default
50.0.0.2, 50.0.0.3, 50.0.0.4
```

Command History

Release	Modification
10.16.1000	Support added for the 8100, 832x, 8360, 8400, 9300, 9300S, and 10000 Switc h Series.
10.13.1000	The output parameters, **Broadcast - flag** and **Hostname** were introduced.
10.09 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show ipv6 dhcp`

Syntax

```
show ipv6 dhcp
```

Description

This command displays DHCP IPv6 information on the ports.

Examples

Displaying the DHCP IPv6 information on the ports:

```
6410(config)# show ipv6 dhcp
VRF default
--------------
Interface vlan1
---------------------
IPv6 address     : 3013::1050/64
Domain search    : paramv6.org, test1.in, test2.in, test3.in, test4.in,
test5.in
DNS servers      : 3013::100, 3013::101, 3013::102
NTP servers      : 3013::201, 3013::202
```

Command History

Release	Modification
10.16.1000	Support added for the 8100, 832x, 8360, 8400, 9300, 9300S, and 10000 Switc h Series.
10.15	Command was introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config-if-mgmt` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`show ip dhcp preferred-vlan`

Syntax

```
show ip dhcp preferred-vlan [{vrf <VRF-NAME> | all-vrfs}]
```

Description

Displays the preferred DHCP client VLANs configured on the system.

Parameter	Description
`vrf <VRF-NAME>`	Show information for a specified VRF.
`all-vrf`	Show information for all VRFs.

Examples

Displaying the DHCP information for the preferred VLAN and VRF red:

```
switch# show ip dhcp preferred-vlan vrf red
 VRF                Preferred-VLAN      Config-status
-------------------------------------------------------------
 red                10                  L3 interface not found
```

Displaying the DHCP information for the all VRFs:

```
switch# show ip dhcp preferred-vlan all-vrfs
 VRF                Preferred-VLAN      Config-status
-------------------------------------------------------------
 default            1                   Success
 red                9                   VRF Mismatch
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

DHCP relay agent

The function of the DHCP relay agent is to forward the DHCP messages to other subnets so that the DHCP server does not have to be on the same subnet as the DHCP clients. The DHCP relay agent transfers DHCP messages from the DHCP clients located on a subnet without a DHCP server, to other subnets. It also relays answers from DHCP servers to DHCP clients.

Subtopics

Supported platform and standards

The following table list the supported platforms for DHCPv4 relay and DHCPv6 relay.

Table 1. Support platforms for DHCPv4 relay and DHCPv6 relay

Platform	DHCPv4 Relay
5420	Yes
6200	Yes

Table 2. Scale

Platform	DHCP v4/v6 enabled SVI
5420	512
6200	128

DHCP relay behaviors are as follows:

- DHCPv6 relay is disabled by default.
- DHCPv4 Smart relay is disabled by default.
- DHCP relay hop count is enabled by default.
- In DHCPv4 relay the option 82 policy is replace by default.
- The MAC address of the switch is used as the option 82 remote ID by default.
- In DHCPv6 relay the option 79 policy is disabled by default.
- In DHCPv4 relay the source-interface is disabled by default.

Protocol and feature details

Supported interfaces

The DHCP relay agent is supported on layer 3 VLAN interfaces, and LAG interfaces. DHCP relay is not supported on the management interface.

DHCP server interoperation

Both DHCP relay and DHCP server can be configured on the same VRF.

DHCP Relay VxLAN support

DHCPv4 Relay is supported on VXLAN with IPv4 underlay .

> **NOTE**
> DHCPv6 Relay requires the egress interface to be configured for the IPv6 multicast helper address. DHCPv6 relay multicast helper configuration is not allowed over VxLAN tunnels.

DHCPv4 relay agent

Hop count in DHCP requests

When a DHCP client broadcasts request, the DHCP relay agent in the switch receives the packets and forwards them to the DHCP server as unicast requests. During this process, the DHCP relay agent increments the hop count before forwarding DHCP packets to the server. The DHCP server, in turn, includes the hop count in the DHCP header in the response sent back to a DHCP client.

DHCP relay option 82

Option 82 is called the relay agent information option. When a DHCP relay agent forwards client-originated DHCP packets to a DHCP server, the option 82 field is inserted/replaced, or the packet with this option is dropped. Servers recognizing the relay agent information option may use the information to implement policies for the assignment of IP addresses and other parameters. The relay agent relays the server-to-client replies to the client.

If a second relay agent is configured to add its own option 82 information, it can encapsulate option 82 information in messages from a first relay agent. The DHCP server uses the option 82 information from both relay agents to decide the IP address for the client.

> **NOTE**
> A DHCP relay option 82 source-interface needs to be enabled when a source interface is used for an Intra-VRF deployment. In this deployment type, the client and DHCP server are in same VRF, but the relay will use the source interface.

Configuring a BOOTP/DHCP relay gateway

The DHCP relay agent selects the lowest-numbered IP address on the interface to use for DHCP messages. The DHCP server then uses this IP address when it assigns client addresses. However, this IP address may not be the same subnet as the one on which the client needs the DHCP service. This feature provides a way to configure a gateway address for the DHCP relay agent to use for relayed DHCP requests, rather than the DHCP relay agent automatically assigning the lowest-numbered IP address.

> **NOTE**
>
> On configuring a bootp-gateway the dhcp-smart-relay is disabled. This can occur with dhcp-relay as well.

DHCP smart relay

The DHCP Smart Relay feature first attempts to use the primary IP address from the client-connected interfaces as a gateway IP address (giaddr) and as an IP address for pool selection. If the DHCP server does not reply to the DHCP discover messages with the primary IP address, the feature attempts to use secondary IP addresses in sequential order from the lowest to the highest. The DHCP Smart Relay forwards three client discover messages with every IP address configured on the client interface until it receives a response from the server. If the list of IP addresses from the client-connected interface is exhausted or the client times out, the DHCP Smart Relay uses the primary IP address. If the DHCP Smart Relay is not configured, the giaddr is the lowest IP address on the interface.

The DHCP Smart Relay maintains the client cache which includes the information about the client, giaddr, retry count for giaddr, port, the total number of processed discovery messages the timestamp of the discover packets. This client cache is rebuilt upon high availability and VSF switchover, VSX-MM (redundancy) switchover, and when the switch reboots.

> **NOTE**
>
> DHCP Smart Relay is only supported for IPv4.

Subtopics

Configuring the DHCPv4 relay agent

Procedure

1. The DHCPv4 relay agent is enabled by default. If it was previously disabled, enable it with the command `dhcp-relay`.
2. Configure one or more IP helper addresses with the command `ip helper-address`. This determines where the DHCPv4 relay agent forwards DHCP requests. IP helper addresses can be configured on layer 3 interfaces, layer 3 VLAN interfaces, and LAG interfaces.

3. If you want to modify the content of forwarded DHCP packets or drop DHCP packets, configure option 82 support with the command `dhcp-relay option 82`.
4. Define the gateway address that the DHCPv4 relay agent will use with the command `ip bootp-gateway`.
5. If required, enable the hop count increment feature with the command `dhcp-relay hop-count-increment`.
6. Review DHCPv4 relay agent configuration settings with the commands `show dhcp-relay`, `show ip helper-address`, and `show dhcp-relay bootp-gateway`.

Example

This example creates the following configuration:

- Enables the DHCPv4 relay agent.
- Enables interface **1/1/1** and assigns an IPv4 address to it. (By default, all interfaces are layer 3 and disabled.)
- Defines an IP helper address of **10.10.20.209** on the interface.
- Enables DHCP option 82 support and replaces all option 82 information with the values from the switch with the switch MAC address as the remote ID.

```
switch(config)# dhcp-relay
switch(config)# interface 1/1/1
switch(config-if)# vlan access 100
switch(config-if)# exit
switch(config)# interface vlan 100
switch(config-if-vlan)# ip address 198.51.100.1/24
switch(config-if-vlan)# ip helper-address 10.10.20.209
switch(config-if-vlan)# exit
switch(config)# dhcp-relay option 82 replace mac
switch# show dhcp-relay
 DHCP Relay Agent                  : enabled
 DHCP Request Hop Count Increment : enabled
 Option 82                         : disabled
 Source-Interface                  : disabled
 Response Validation               : disabled
 Option 82 Handle Policy           : replace
 Remote ID                         : mac
 DHCP Relay Statistics:
  Valid Requests Dropped Requests Valid Responses Dropped Responses
  -------------- ---------------- --------------- -----------------
  60             10               60              10
```

```
DHCP Relay Option 82 Statistics:
 Valid Requests Dropped Requests Valid Responses Dropped Responses
 -------------- ---------------- --------------- -----------------
 50             8                50              8
```

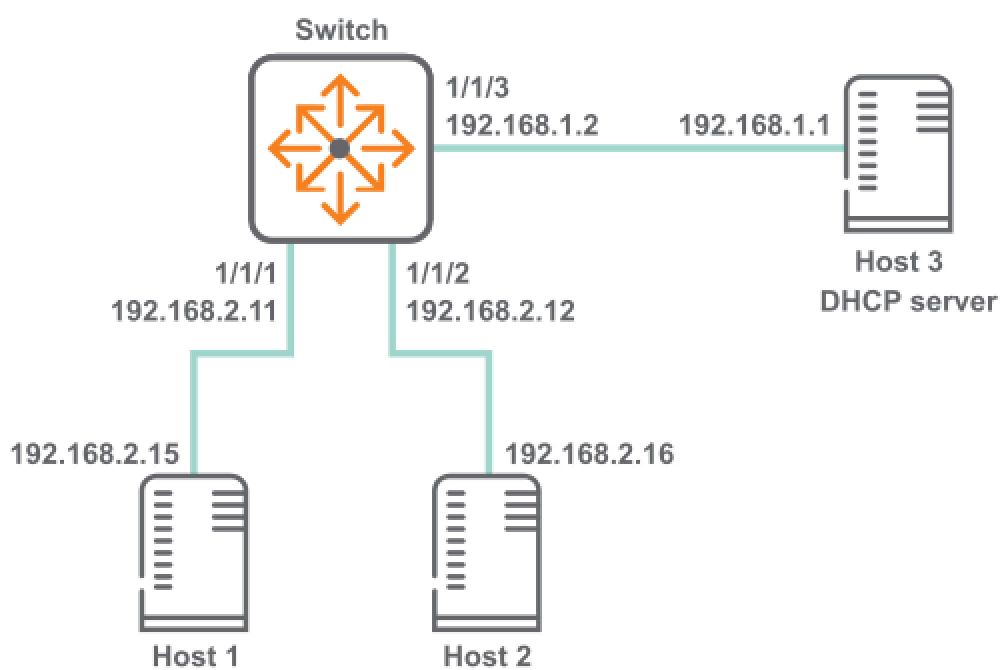
Use Case

This section explain the use cases for DHCPv4 relay agent.

Subtopics

DHCPv4 relay scenario 1

In this scenario, DHCP relay on the switch enables two hosts to obtain their IP addresses from a DHCP server on a different subnet. The physical topology of the network looks like this:



Procedure

1. DHCP relay is enabled by default. If it was previously disabled, enable it.

```
switch# config
switch(config)# dhcp-relay
```

2. Define an IPv4 helper address on interface vlan 100

```
switch(config)# interface 1/1/1
switch(config-if)# vlan access 100
switch(config-if)# exit
switch(config)# interface 1/1/2
switch(config-if)# vlan access 100
switch(config-if)# exit
switch(config)# interface vlan 100
switch(config-if-vlan)# ip address 192.168.2.11/24
switch(config-if-vlan)# ip helper-address 192.168.1.1
```

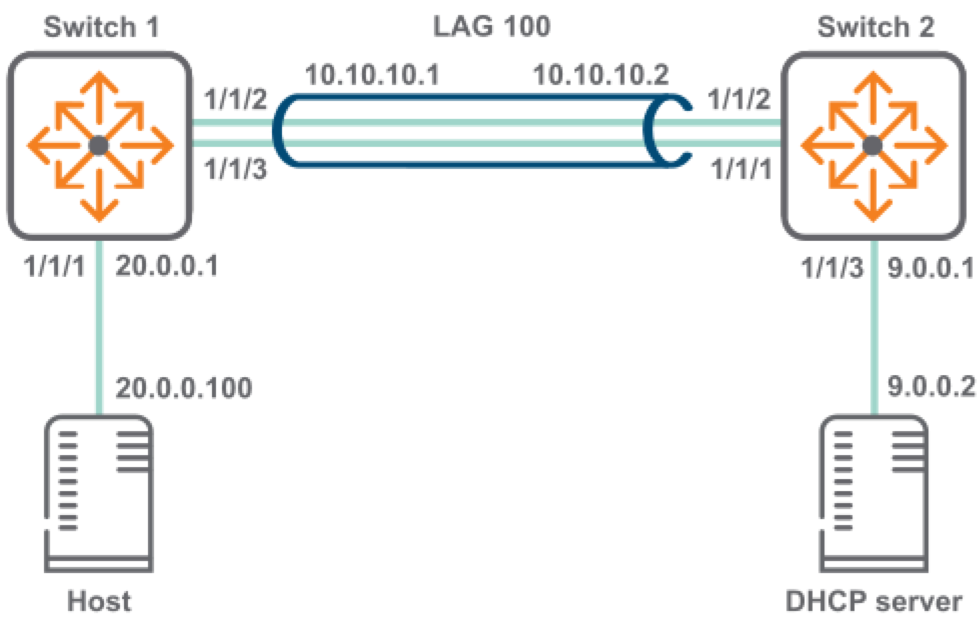
3. Verify DHCP relay configuration.

```
switch# show dhcp-relay
 DHCP Relay Agent                   : enabled
 DHCP Request Hop Count Increment : enabled
 Option 82                          : disabled
 Source-Interface                   : disabled
 Response Validation                : disabled
 Option 82 Handle Policy            : replace
 Remote ID                          : mac
 DHCP Relay Statistics:
  Valid Requests Dropped Requests Valid Responses Dropped Responses
  -------------- ---------------- --------------- -----------------
  60             10               60              10
 DHCP Relay Option 82 Statistics:
  Valid Requests Dropped Requests Valid Responses Dropped Responses
  -------------- ---------------- --------------- -----------------
  50             8                50              8
```

```
switch(config)# show ip helper-address
IP Helper Addresses
Interface: vlan100
  IP Helper Address                   VRF
  -----------------                  -----------------
  192.168.1.1                         default
```

DHCPv4 relay scenario 2

In this scenario, host on switch 1 reaches the DHCP server on switch two via a LAG. The physical topology of the network looks like this:



Procedure

1. On switch 1:

   Create LAG 100 and assign an IP address to it.

   ```
   switch# config
   switch(config)# interface lag 100
   switch(config-lag-if)# no shutdown
   switch(config-lag-if)# vlan access 200
   switch(config-lag-if)# lacp mode active
   switch(config-lag-if)# exit
   switch(config)# interface vlan 200
   switch(config-if-vlan)# ip address 10.0.10.1/24
   ```

   Assign an IP address to interface 1/1/1 and an IP helper address to reach the DHCP server.

   Assign an IP address to interface vlan 100 and an IP helper address to reach the DHCP server.

   ```
   switch(config)# interface 1/1/1
   switch(config-if)# no shutdown
   switch(config-if)# vlan access 100
   switch(config-if)# exit
   switch(config)# interface vlan 100
   switch(config-if-vlan)# ip address 20.0.0.1/8
   ```

```
switch(config-if-vlan)# ip helper-address 9.0.0.2
```

Assign interfaces 1/1/2 and 1/1/3 to LAG 100.

```
switch(config-if)# interface 1/1/2
switch(config-if)# lag 100
switch(config-if)# interface 1/1/3
switch(config-if)# lag 100
```

Create a route between 10.0.10.2 and 9.0.0.0.

```
switch(config)# ip route 9.0.0.0/24 10.0.10.2
```

2. On switch 2:

   Create LAG 100 and assign an IP address to it.

```
switch# config
switch(config)# interface lag 100
switch(config-lag-if)# no shutdown
switch(config-lag-if)# vlan access 200
switch(config-lag-if)# lacp mode active
switch(config-lag-if)# exit
switch(config)# interface vlan 200
switch(config-if-vlan)# ip address 10.0.10.2/24
```

Assign an IP address to interface vlan 300.

```
switch(config)# interface 1/1/3
switch(config-if)# no shutdown
switch(config-if)# vlan access 300
switch(config-if)# exit
switch(config)# interface vlan 300
switch(config-if-vlan)# ip address 9.0.0.1/24
```

Assign an IP address to interface 1/1/3.

Assign interfaces 1/1/1 and 1/1/2 to LAG 100.

```
switch(config-if)# interface 1/1/1
switch(config-if)# lag 100
switch(config-if)# interface 1/1/2
switch(config-if)# lag 100
```

Create a route between 20.0.0.0 and 10.0.10.1.

```
switch(config)# ip route 20.0.0.0/8 10.0.10.1
```

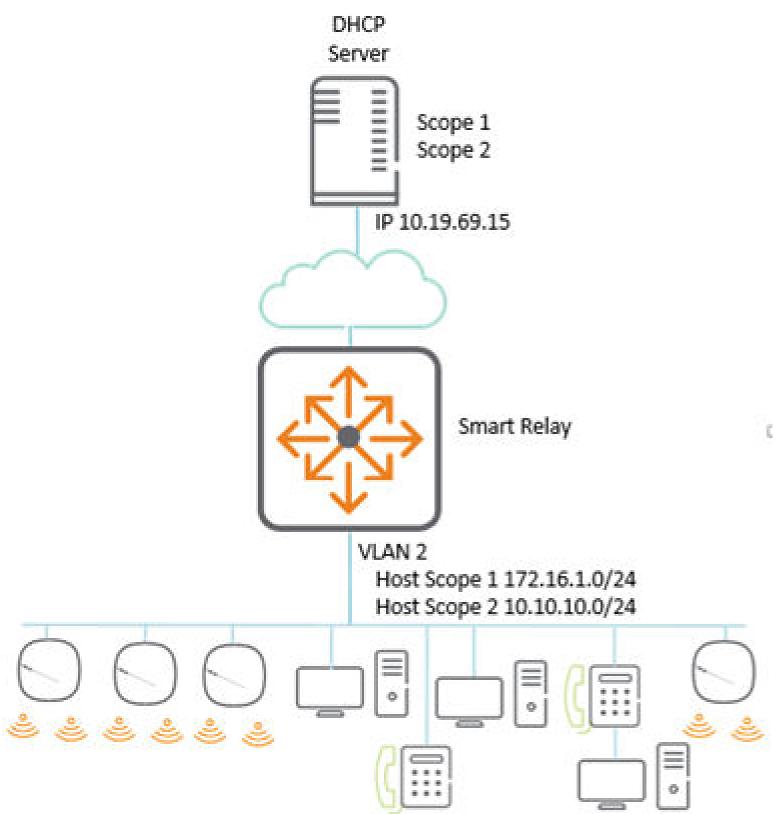
DHCPv4 relay scenario 3

About this task

In the following scenario of the DHCP smart relay, the primary address is used as the GIADDR; if the server is unavailable, then the secondary address is used. From AOS-CX 10.10.1000 and later releases, the GIADRR use the secondary address in an ascending order when there is more than one secondary address.

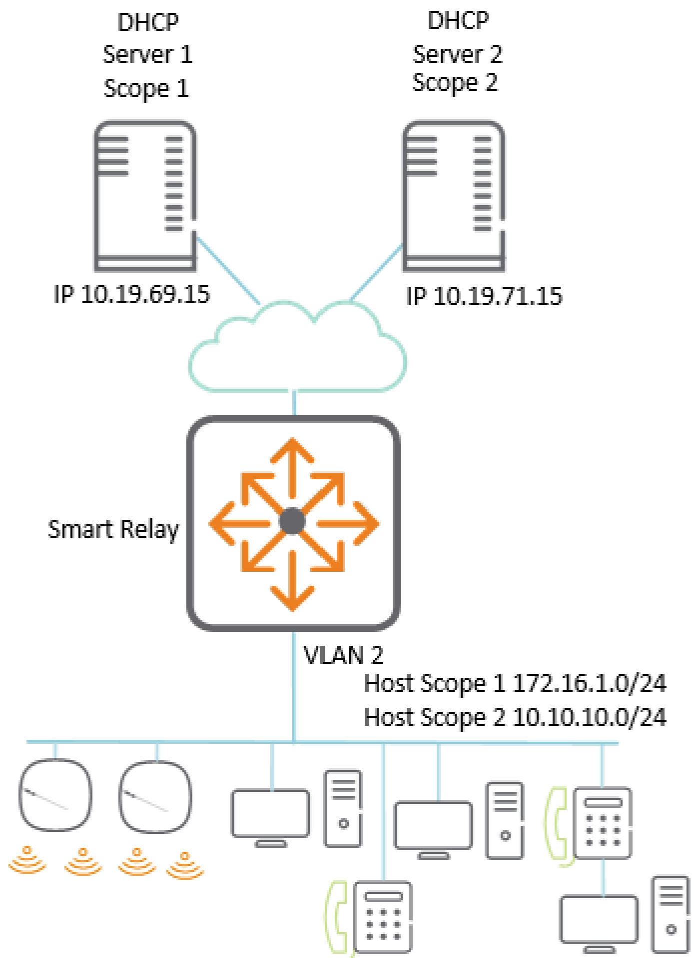
Procedure

1. If DHCP scope exhaustion happens at a site, an additional scope can be added to increase the IP address capacity while keeping the existing IP scope and subnet. The physical topology of the network looks like



this:

2. If the DHCP server is resilient or scope is required, then whole scopes can be placed on separate servers. If a server or scope becomes unavailable, then for address assignment, the next available scope is



attempted.

Results

In both the above topology, the access switch is set up the same way. The only difference is in the server which is used for helper addresses.

```
!
dhcp-smart-relay                                 <-- Enable DHCP smart relay
!
interface vlan 2
    ip address 172.16.1.254/24                   <-- Add the local IPS which
will act as GIADDR
    ip address  10.10.610.254/24 secondary
    ip helper-address 10.19.69.15                <-- Add the IP helpers to
reach the required DHCP servers
    ip helper-address 10.19.71.15
```

DHCPv4 relay commands

Select a command from the list in the left navigation menu.

Subtopics

`dhcp-relay`

Syntax

```
dhcp-relay
no dhcp-relay
```

Description

Enables DHCP relay support. DHCP relay is enabled by default. DHCP relay is not supported on the management interface.

The **no** form of this command disables DHCP relay (and DHCP relay option 82) support.

Examples

This example enables DHCP relay support.

```
switch(config)# dhcp-relay
```

This example removes DHCP relay support.

```
switch(config)# no dhcp-relay
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`dhcp-relay hop-count-increment`

Syntax

```
dhcp-relay hop-count-increment
no dhcp-relay hop-count-increment
```

Description

Enables the DHCP relay hop count increment feature, which causes the DHCP relay agent to increment the hop count in all relayed DHCP packets. Hop count is enabled by default.

The **no** form of this command disables the hop count increment feature.

Examples

Enabling the hop count increment feature.

```
switch(config)# dhcp-relay hop-count-increment
```

Disabling the hop count increment feature.

```
switch(config)# no dhcp-relay hop-count-increment
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`dhcp-relay option 82`

Syntax

```
dhcp-relay option 82 [replace {ip | mac | validate} | drop {ip | mac |
validate} | keep {ip | mac} |
    source-interface | circuit-id port-id | validate {drop {ip | mac} |
replace {ip | mac}}]
dhcp-relay option 82 [replace {ip | mac | validate} | drop {ip | mac |
validate} | keep {ip | mac} |
    source-interface | circuit-id port-id | validate {drop {ip | mac} |
replace {ip | mac}}]
```

Description

Configures the behavior of DHCP relay option 82. A DHCP relay agent can receive a message from another DHCP relay agent having option 82. The relay information from the previous relay agent is replaced by default.

The **no** form of this command disables the DHCP relay option 82 configurations. Option 82 is disabled when DHCP relay is disabled globally. When DHCP relay is re-enabled, option 82 also needs to be re-enabled using the **dhcp-relay option 82** command.

> **NOTE**
> DHCP Relay is supported over VXLAN with both IPv4 and IPv6 underlay.

Parameter	Description
`replace`	Replace the existing option 82 field in an inbound client DHCP packet with the information from the switch. The remote ID and circuit ID information from the first relay agent is lost. Default.
`drop`	Drop any inbound client DHCP packet that contains option 82 i nformation.
`keep`	Keep the existing option 82 field in an inbound client DHCP pac ket. The remote ID and circuit ID information from the first relay agent is preserved.
`source-interface`	Configures the DHCP relay to use a configured source IP addres s for inter - VRF server reachability. Set the source IP address w ith the command `ip source-interface`.
`circuit-id port-id`	Configures the circuit ID and remote ID options. Client connecte d port information to be used in Option 82 circuit ID and rem ote ID. **NOTE** When enabled, DHCP Relay drops packets from single - homed clients connected to VSX Secon dary, unless the `dhcp-relay vsx activ e-active` command is enabled. To avoid thi s, connect single - homed clients to the VSX Pri mary or enable `dhcp-relay vsx activ e-active`.
`validate`	Validate option 82 information in DHCP server responses and d rop invalid responses.
`ip`	Use the IP address of the interface on which the client DHCP pa cket entered the switch as the option 82 remote ID.
`mac`	Use the MAC address of the switch as the option 82 remote ID. Default.

Example

This example enables DHCP option 82 support and replaces all option 82 information with the values from the switch, with the switch MAC address as the remote ID.

```
switch(config)# dhcp-relay option 82 replace mac
```

Command History

Release	Modification
10.17	The **circuit - id port - id** parameter was introduced.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`dhcp-relay vsx active-active`

Syntax

```
dhcp-relay vsx active-active
no dhcp-relay vsx active-active
```

Description

Enables DHCP relay active-active mode in VSX. When enabled, DHCP relay is active on both VSX Primary and Secondary peers. DHCP client requests are processed on the same VSX peer that receives them. A VSX peer that receives DHCP client request packets from its peer silently ignores them.

By default, this feature is disabled, and DHCP request packets are forwarded to the configured helper addresses only by the VSX primary.

The **no** form of this command disables DHCP relay active-active mode, and DHCP request packets are forwarded to configured helper addresses only by the VSX primary.

Example

Enabling relay active-active mode in VSX.

```
switch(config)# dhcp-relay vsx active-active
```

Disabling relay active-active mode in VSX.

```
switch(config)# no dhcp-relay vsx active-active
```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
5420	`config`	Administrators or local user group members with execution righ ts for this command.

`dhcp-smart-relay`

Syntax

```
dhcp-smart-relay
no dhcp-smart-relay
```

Description

Enables DHCP Smart Relay on the device and on all the interfaces where IP helper addresses are configured. Disabled by default at the device level. Not supported on the management interface.

The **no** form of this command disables DHCP Smart Relay.

> **NOTE**
> Prior to enabling DHCP Smart Relay, enable IP helper address configuration and configure secondary IP addresses on the interface.

Examples

Enabling DHCP Smart Relay:

```
switch(config)# dhcp-smart-relay
```

Disabling DHCP Smart Relay support:

```
switch(config)# no dhcp-smart-relay
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`diag-dump dhcp-relay basic`

Syntax

```
diag-dump dhcp-relay basic
```

Description

Dumps DHCP relay configurations for all interfaces.

Examples

This example enables DHCP relay support.

```
switch# diag-dump dhcp-relay basic
=========================================================================
[Start] Feature dhcp-relay Time : Sun Apr 26 06:38:10 2020
=========================================================================
-------------------------------------------------------------------------
[Start] Daemon hpe-relay
-------------------------------------------------------------------------
DHCP Relay : 1
DHCP Relay hop-count-increment : 1
DHCP Relay Option82 : 1
DHCP Relay Option82 validate : 0
DHCP Relay Option82 policy : keep
DHCP Relay Option82 remote-id : mac
DHCP Relay Option82 Source Intf : Disable
```

```
DHCP Smart Relay : Enable
System Mac [f4:03:43:80:27:00]
VRF :BLUE, Source Ip:200.0.0.10


vsx: Present
isl_status: Up
remote_idl : 1
VSX Active-Active Mode : Enable

Interface vlan2: 1
  Client Packet Statistics:
  Valid         Dropped      O82_Valid    O82_Dropped   vsx_drops
  -----         -------      ---------    -----------   ---------
  0             0            0            0             0
  Server Packet Statistics:
  Valid         Dropped      O82_Valid    O82_Dropped   Invalid_IP_Drops
To_Dsnoop
  -----         -------      ---------    -----------   ----------------
---------
  0             0            0            0             0                   0
client request dropped packets with extn option 82 = 0
client request valid packets with extn option 82 = 0
server request dropped packets with extn option 82 = 0
server request valid packets with extn option 82 = 0
Port 67 - 200.0.0.100,2
source vrf-BLUE.
Interface vlan3: 1
  Client Packet Statistics:
  Valid         Dropped      O82_Valid    O82_Dropped   vsx_drops
  -----         -------      ---------    -----------   ---------
  0             0            0            0             0
  Server Packet Statistics:
  Valid         Dropped      O82_Valid    O82_Dropped   Invalid_IP_Drops
To_Dsnoop
  -----         -------      ---------    -----------   ----------------
---------
  0             0            0            0             0                   0
client request dropped packets with extn option 82 = 0
client request valid packets with extn option 82 = 0
server request dropped packets with extn option 82 = 0
server request valid packets with extn option 82 = 0
Port 67 - 200.0.0.100,2
```

```
source vrf-BLUE.
DHCP Smart Relay Client Cache:
Total Number of entries: 2
--------------------------------------------------------------------------
Client-MAC          PortIndex   Timestamp   RetryCount  DiscCount  GWIP
--------------------------------------------------------------------------
00:50:56:bd:6a:7a  20          1636105218  1           4          30.0.0.1
00:50:56:bd:71:17  20          1636105214  1           4          30.0.0.1
-------------------------------------------------------------------------
[End] Daemon hpe-relay
-------------------------------------------------------------------------
=========================================================================
[End] Feature dhcp-relay
=========================================================================
Diagnostic-dump captured for feature dhcp-relay
switch# diag-dump dhcp-relay basic
=========================================================================
[Start] Feature dhcp-relay Time : Sun Apr 26 06:38:10 2020
=========================================================================
-------------------------------------------------------------------------
[Start] Daemon hpe-relay
-------------------------------------------------------------------------
DHCP Relay : 1
DHCP Relay hop-count-increment : 1
DHCP Relay Option82 : 1
DHCP Relay Option82 validate : 0
DHCP Relay Option82 policy : keep
DHCP Relay Option82 remote-id : mac
DHCP Relay Option82 Source Intf : Disable
System Mac [f4:03:43:80:27:00]
VRF :BLUE, Source Ip:200.0.0.10
vsx: Not Present
Interface vlan2: 1
Client Packet Statistics:
Valid Dropped O82_Valid O82_Dropped vsx_drops
----- ------- --------- ----------- ---------
0 0 0 0 0
Server Packet Statistics:
Valid Dropped O82_Valid O82_Dropped Invalid_IP_Drops To_Dsnoop
----- ------- --------- ----------- ---------------- ---------
0 0 0 0 0 0
client request dropped packets with extn option 82 = 0
client request valid packets with extn option 82 = 0
```

```
server request dropped packets with extn option 82 = 0
server request valid packets with extn option 82 = 0
Port 67 - 200.0.0.100,2
source vrf-BLUE.
Interface vlan3: 1
Client Packet Statistics:
Valid Dropped O82_Valid O82_Dropped vsx_drops
----- ------- --------- ----------- ---------
0 0 0 0 0
Server Packet Statistics:
Valid Dropped O82_Valid O82_Dropped Invalid_IP_Drops To_Dsnoop
----- ------- --------- ----------- ---------------- ---------
0 0 0 0 0 0
client request dropped packets with extn option 82 = 0
client request valid packets with extn option 82 = 0
server request dropped packets with extn option 82 = 0
server request valid packets with extn option 82 = 0
Port 67 - 200.0.0.100,2
source vrf-BLUE.
[End] Daemon hpe-relay
--------------------------------------------------------------------------
==========================================================================
[End] Feature dhcp-relay
==========================================================================
Diagnostic-dump captured for feature dhcp-relay
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`ip bootp-gateway`

Syntax

```
ip bootp-gateway <IPV4-ADDR>
no ip bootp-gateway <IPV4-ADDR>
```

Description

Configures a gateway address for the DHCP relay agent to use for DHCP requests. By default DHCP relay agent picks the lowest-numbered IP address on the interface.

The **no** form of this command removes the gateway address.

Parameter	Description
`<IPV4-ADDR>`	Specifies the IP address of the gateway in IPv4 format (**x.x.x.x**), where **x** is a is a decimal number from 0 to 255.

Examples

Setting the IP address of the gateway for interface vlan 100 to **10.10.10.10**:

```
switch(config)# interface vlan100
switch(config-if)# ip bootp-gateway 10.10.10.10
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ip helper-address`

Syntax

```
ip helper-address <IPV4-ADDR> [vrf <VRF-NAME>]
no ip helper-address <IPV4-ADDR> [vrf <VRF-NAME>]
```

Description

Defines the address of a remote DHCP server or DHCP relay agent. Up to eight addresses can be defined. The DHCP relay agent forwards DHCP client requests to all defined servers.

If IP helper adddress is defined with VRF argument then this command requires you define a source IP address for DHCP relay with the command `ip source-interface`. The configured source IP on the VRF is used to forward DHCP packets to the server.

A helper address cannot be defined on the OOBM interface.

The **no** form of this command removes an IP helper address.

Parameter	Description
`helper-address <IPV4-ADDR>`	Specifies the helper IP address in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: **default**.

Examples

```
switch(config)# interface 1/1/1
switch(config-if)# ip helper-address 10.10.10.209
```

```
switch(config-if)# no ip helper-address 10.10.10.209
```

Defining the IP helper address 10.10.10.209 on interface vlan 100:

```
switch(config)# interface vlan100
switch(config-if-vlan)# ip helper-address 10.10.10.209
```

Removing the IP helper address 10.10.10.209 on interface vlan 100:

```
switch(config-if-vlan)# no ip helper-address 10.10.10.209
switch(config)# interface 1/1/2
switch(config-if)# ip helper-address 10.10.10.209 vrf myvrf
switch(config-if)# no ip helper-address 10.10.10.209 vrf myvrf
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`show dhcp-relay`

Syntax

```
show dhcp-relay
```

Description

Shows DHCP relay configuration settings.

Example

Showing DHCP relay settings:

```
switch(config)# show dhcp-relay
DHCP Relay Agent                  : enabled
DHCP Smart Relay                  : enabled
DHCP Request Hop Count Increment : enabled
Option 82                         : enabled
Source-Interface                  : disabled
Response Validation               : disabled
Option 82 Handle Policy           : replace
Remote ID                         : mac
DHCP Relay Statistics:
  Valid Requests Dropped Requests Valid Responses Dropped Responses
  -------------- ---------------- ---------------- ------------------
  60             10               60               10
```

```
DHCP Relay Option 82 Statistics:
  Valid Requests Dropped Requests Valid Responses Dropped Responses
  -------------- ---------------- --------------- -----------------
  50               8                50                8
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show dhcp-relay bootp-gateway`

Syntax

```
show dhcp-relay bootp-gateway [interface <INTERFACE-NAME>]
```

Description

Shows the bootp gateway defined for all interfaces or a specific interface.

Parameter	Description
`<INTERFACE-NAME>`	Specifies an interface. Format: `member/slot/port`.

Examples

Showing the designated bootp gateway for all interfaces:

```
switch(config)# show dhcp-relay bootp-gateway
BOOTP Gateway Entries
Interface             Source IP
-------------------- ---------------
vlan10                1.1.1.1
vlan20                2.2.2.2
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show ip helper-address`

Syntax

```
show ip helper-address [interface <INTERFACE-ID>]
```

Description

Shows the IP helper addresses defined for all interfaces or a specific interface.

Parameter	Description
`interface <INTERFACE-ID>`	Specifies an interface. Format: `member/slot/port.`

Examples

Showing the IP helper addresses for all interfaces:

```
switch(config)# show ip helper-address
IP Helper Addresses
Interface: vlan11
  IP Helper Address                    VRF
  -----------------                   -----------------
  10.10.10.209                         default
Interface: vlan20
  IP Helper Address                    VRF
  -----------------                   -----------------
  3.3.3.3                              default
  20.20.20.20                          default
```

Showing the IP helper addresses for interface vlan20:

```
switch(config)# show ip helper-address interface vlan20
IP Helper Addresses
Interface: vlan20
  IP Helper Address                    VRF
  -----------------                   -----------------
  3.3.3.3                              default
  20.20.20.20                          default
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

DHCPv6 relay agent

DHCPv6 relay VRF support

DHCPv6 relay is VRF aware. DHCPv6 client requests received on an interface are forwarded to the configured servers via that interface's VRF. DHCPv6 server responses received on an interface are forwarded to the client which is reachable via that interface's VRF.

> **NOTE**
>
> DHCPv6 relay does not have inter-VRF support.

Limitations

DHCPv6 Relay requires an egress interface to be configured for the IPv6 multicast helper address. DHCPv6-Relay multicast helper configuration is not allowed over VxLAN tunnels. The following configuration is required in order to use source interface for DHCPv6 relay.

Configuring the IPv6 source-interface interface to use for DHCPv6 relay:

```
switch(config)# ipv6 source-interface dhcp_relay interface loopback3 vrf
test1
```

Enabling DHCPv6 relay to use the configured source interface:

```
switch(config)# dhcpv6-relay source-interface
```

Subtopics

Configuring the DHCPv6 relay agent
Use Case
DHCP relay (IPv6) commands

Configuring the DHCPv6 relay agent

Procedure

1. Enable the DHCPv6 relay agent with the command `dhcpv6-relay`.
2. Configure one or more IP helper addresses with the command `ipv6 helper-address`. This determines where the DHCPv6 agent forward DHCP requests.
3. If you want to enable DHCP option 79 support to forward client link-layer addresses, use the command `dhcpv6-relay option 79`.
4. Review DHCPv6 relay agent configuration settings with the commands `show dhcpv6-relay` and `show ipv6 helper-address`.

Example

This example creates the following configuration:

- Enables the DHCPv6 relay agent.
- Enables interface **1/1/2** and assigns an IPv6 address to it. (By default, all interfaces are layer 3 and disabled.)
- Defines an IP helper address of **FF01::1:1000** on interface **1/1/2**.
- Enables DHCP option 79.

```
switch(config)# dhcpv6-relay
switch(config)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-if)# ipv6 address 2002::22/64
switch(config-if)# ipv6 helper-address unicast 2001::1
switch(config-if)# ipv6 helper-address multicast ff01::1:1000 egress 1/1/1
switch(config-if)# ipv6 helper-address multicast all-dhcp-servers egress
1/1/1
switch(config-if)# exit
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# ipv6 address 2002::21/64
switch(config-if)# ipv6 helper-address unicast 2001::1
switch(config-if)# ipv6 helper-address multicast ff01::1:1000 egress 1/1/1
switch(config-if)# ipv6 helper-address multicast all-dhcp-servers egress
1/1/1
switch(config-if)# exit
switch(config)# end
```

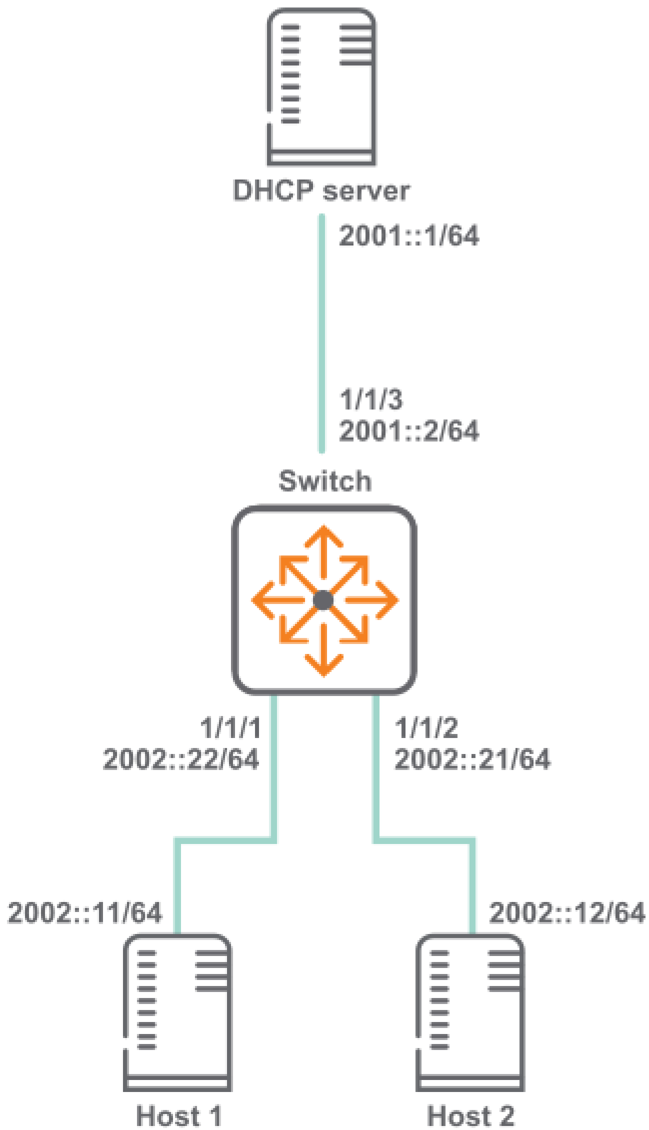
Use Case

This section explain the use cases for DHCPv6 relay agent.

Subtopics

DHCPv6 relay scenario 1

In this scenario, DHCP relay on the switch enables two hosts to obtain their IP addresses from a DHCP server on a different subnet. The physical topology of the network looks like this:



Procedure

1. Enable DHCP relay.

```
switch# config
switch(config)# dhcpv6-relay
```

2. Define an IPv6 helper address.

   - On interface `vlan 10` and `vlan 20`.

```
switch(config)# dhcpv6-relay
switch(config)# interface 1/1/1
switch(config)# no shutdown
switch(config-if)# vlan access 10
```

```
switch(config-if)# exit
switch(config)# interface 1/1/2
switch(config)# no shutdown
switch(config-if)# vlan access 20
switch(config-if)# exit
switch(config)# interface vlan 10
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ipv6 address 2002::22/64
switch(config-if-vlan)# ipv6 helper-address unicast 2001::1
switch(config-if-vlan)# ipv6 helper-address multicast ff01::1:1000
egress vlan30
switch(config-if-vlan)# ipv6 helper-address multicast all-dhcp-
servers egress vlan50
switch(config-if-vlan)# exit
switch(config)# interface vlan 20
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ipv6 address 2002::21/64
switch(config-if-vlan)# ipv6 helper-address unicast 2001::1
switch(config-if-vlan)# ipv6 helper-address multicast ff01::1:1000
egress vlan30
switch(config-if-vlan)# ipv6 helper-address multicast all-dhcp-
servers egress vlan50
switch(config-if-vlan)# exit
switch(config)# end
```

3. Verify DHCP relay configuration.

   - When interface `vlan 10` and `vlan 20` is configured as relay.

```
switch# show dhcpv6-relay
  DHCPv6 Relay Agent : enabled
  Option 79          : disabled
switch# show ipv6 helper-address
IP Helper Addresses
Interface: vlan10
IP Helper Address                                Egress Port
---------------------------------------------- ------------
  all-dhcp-servers                                 vlan50
  ff01::1:1000                                     vlan30
  2001::1                                          -
Interface: vlan20
IP Helper Address                                Egress Port
-----------------------------------------------------------
  all-dhcp-servers                                 vlan50
```

```
ff01::1:1000                                    vlan30
2001::1                          -
```

- When interface `1/1/1` and `1/1/2` is configured as relay.

```
switch# show dhcpv6-relay
DHCPv6 Relay Agent : enabled
Option 79 : disabled
switch# show ipv6 helper-address
Interface: 1/1/1
IPv6 Helper Address                             Egress Port
---------------------------------------------------------
all-dhcp-servers                                1/1/3
ff01::1:1000                                    1/1/3
2001::1                                         -
Interface: 1/1/2
IPv6 Helper Address                             Egress Port
--------------------------------------------------------
all-dhcp-servers                                1/1/3
ff01::1:1000                                    1/1/3
2001::1                                         -
```

DHCP relay (IPv6) commands

Select a command from the list in the left navigation menu.

Subtopics

`dhcpv6-relay`

Syntax

```
dhcpv6-relay [l2vpn-clients | source-interface]
no dhcpv6-relay [l2vpn-clients | source-interface]
```

Description

Enables DHCPv6 relay support. DHCPv6 relay is disabled by default. DHCP relay is not supported on the management interface. Best practices is to disable this configuration on all the VXLAN tunnel endpoints (VTEPs), to avoid forwarding duplicate DHCPv6 requests to the server.

The **no** form of this command disables DHCP relay support.

DHCPv6 Relay requires that you configure the egress interface using the **ipv6 helper-address** command. The egress interface of a VTEP is used as an underlay, so a DHCPv6 Relay Multicast ipv6 address is not supported in a VXLAN topology.

Parameter	Description
`l2vpn-clients`	Enables packets from l2vpn clients to be forwarded to configur ed servers. Enabled by default.
`source-interface`	Enables DHCPv6 relay to use the configured source interface.

Usage

In Asymmetric/Symmetric Integrated Routing and Bridging (IRB) VXLAN deployments with a VLAN extension in subset of VTEPs , client DHCPv6 broadcast requests are received by all the VTEPS where a client VLAN is configured. A DHCPv6 relay agent on those VTEPs forward DHCPv6 packets to configured DHCPv6 server(s). As DHCPv6 requests are forwarded by multiple DHCPv6 relay agents, the DHCPv6 server receives duplicate copies of the same packet. When this configuration is disabled, the DHCPv6 relay agent on VTEPs ignores DHCPv6 request packets that are received from client MACs addresses learned via EVPN.

Examples

Enables DHCPv6 relay support.

```
switch(config)# dhcpv6-relay
```

Removes DHCPv6 relay support.

```
switch(config)# no dhcpv6-relay
```

Command History

Release	Modification
10.12.1000	**l2vpn - clients** and **source - interface** added.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`dhcpv6-relay option 79`

Syntax

```
dhcpv6-relay option 79
no dhcpv6-relay option 79
```

Description

Enables support for DHCP relay option 79. When enabled, the DHCPv6 relay agent forwards the link-layer address of the client. This option is disabled by default.

The **no** form of this command disables support for DHCP relay option 79.

Examples

Enables DHCP option 79 support.

```
switch(config)# dhcpv6-relay option 79
```

Disables DHCP option 79 support.

```
switch(config)# no dhcpv6-relay option 79
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`dhcpv6-relay vsx active-active`

Syntax

```
dhcpv6-relay vsx active-active
no dhcpv6-relay vsx active-active
```

Description

Enables DHCPv6 relay active-active mode in VSX. When enabled, DHCPv6 relay is active on both VSX Primary and Secondary peers. DHCPv6 client requests are processed on the same VSX peer that receives them. A VSX peer that receives DHCPv6 client request packets from its peer silently ignores them.

By default, this feature is disabled, and DHCPv6 request packets are forwarded to the configured helper addresses only by the VSX primary.

The **no** form of this command disables DHCPv6 relay active-active mode, and DHCPv6 request packets are forwarded to configured helper addresses only by the VSX primary.

Example

Enabling relay active-active mode in VSX.

```
switch(config)# dhcpv6-relay vsx active-active
```

Disabling relay active-active mode in VSX.

```
switch(config)# no dhcpv6-relay vsx active-active
```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
5420	`config`	Administrators or local user group members with execution righ ts for this command.

`diag-dump dhcpv6-relay basic`

Syntax

```
diag-dump dhcpv6-relay basic
```

Description

Dumps DHCPv6 relay configurations for all interfaces.

Examples

This example enables DHCP relay support.

```
switch# diag-dump dhcpv6-relay basic
===========================================================================
[Start] Feature dhcpv6-relay Time : Thu Jun  8 09:55:11 2017
===========================================================================
---------------------------------------------------------------------------
[Start] Daemon hpe-relay
---------------------------------------------------------------------------
DHCP Relay : 1
DHCPv6 Relay Option79 : 0
vsx: Present
vsx_status: Primary
isl_status: Up
VSX Active-Active Mode : Enable

System Mac [02:02:02:02:0a:0a]
Interface vlan10: 1
Intf selected IP -  ifindex - 10 kernel ifindex - 1701::1, 1, egress =
(null), Server vrf = default
  Client Packet Statistics:
  Valid         Dropped      vsx_drops
  -----         -------      ---------
```

```
  0             0             0
  Server Packet Statistics:
  Valid         Dropped       To_Dsnoop
  -----         -------       ---------
  0             0             0
Total VRF source-ip entrie: 0
-------------------------------------------------------------------------
[End] Daemon hpe-relay
-------------------------------------------------------------------------
=========================================================================
[End] Feature dhcpv6-relay
=========================================================================
Diagnostic-dump captured for feature dhcpv6-relay
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`ipv6 helper-address`

Syntax

```
ipv6 helper-address unicast <UNICAST-IPV6-ADDR>
no ipv6 helper-address unicast <UNICAST-IPV6-ADDR>
ipv6 helper-address multicast {all-dhcp-servers | <MULTICAST-IPV6-ADDR>}
egress <PORT-NUM>
no ipv6 helper-address multicast {all-dhcp-servers | <MULTICAST-IPV6-ADDR>}
egress <PORT-NUM>
```

Description

Defines the address of a remote DHCPv6 server or DHCPv6 relay agent. Up to eight addresses can be defined. The DHCPv6 agent forwards DHCPv6 client requests to all defined servers.

Not supported on the OOBM interface.

The **no** form of this command removes an IP helper address.

Parameter	Description
<UNICAST-IPV6-ADDR>	Specifies the unicast helper IP address in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where **x** is a hexadecimal number from 0 to F.
<MULTICAST-IPV6-ADDR>	Specifies the multicast helper IP address in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where **x** is a hexadecimal number from 0 to F.
`all-dhcp-servers`	Specifies all the DHCP server IPv6 addresses for the interface.
`egress` *<PORT-NUM>*	Specifies the port number on which DHCPv6 service requests a re relayed to a multicast destination. The egress port must be di fferent than the one on which the multicast helper address is co nfigured. Format: `member/slot/port`.
`vrf` *<VRF-NAME>*	Specifies the name of the VRF from which the specified protoco l sets its source IP address.

Examples

Defining a multicast IPv6 helper address of **2001:DB8::1** on port **1/1/2**:

```
switch(config-if)# ipv6 helper-address multicast 2001:DB8:0:0:0:0:0:1
egress 1/1/2
```

Removing the IP helper address of **2001:DB8::1** on port **1/1/2**:

```
switch(config-if)# no ipv6 helper-address multicast 2001:DB8:0:0:0:0:0:1
egress 1/1/2
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution rights for this command.

`show dhcpv6-relay`

Syntax

```
show dhcpv6-relay
```

Description

Shows DHCP relay configuration settings.

Example

```
switch(config)# show dhcpv6-relay
  DHCPv6 Relay Agent : enabled
  Option 79          : disabled
  L2vpn-clients      : enabled
  Source-interface   : enabled
  VSX Active-Active Mode : enabled
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show ipv6 helper-address`

Syntax

```
show ipv6 helper-address [interface <INTERFACE-ID>]
```

Description

Shows the helper IP addresses defined for all interfaces or a specific interface.

Parameter	Description
`interface <INTERFACE-ID>`	Specifies an interface. Format: **member/slot/port**.

Examples

```
switch# show ipv6 helper-address
 Interface: 1/1/1
 IPv6 Helper Address                               Egress Port
 ----------------------------------------------- -----------
 2001:db8:0:1::                                    -
 FF01::1:1000                                      1/1/2
 Interface: 1/1/2
 IPv6 Helper Address                               Egress Port
 ---------------------------------------------   -----------
 2001:db8:0:1::                                    -
switch# show ipv6 helper-address interface 1/1/1
 Interface: 1/1/1
 IPv6 Helper Address                               Egress Port
 ----------------------------------------------- -----------
 2001:db8:0:1::                                    -
```

```
 FF01::1:1000                                    1/1/2
switch# show ipv6 helper-address interface vlan20
Interface: vlan20
 IP Helper Address                                Egress Port
 ----------------------------------------------  -----------
  2001::1                                          -
 ff01::1:1000                                     vlan30      default
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

Troubleshooting

One or more DHCP clients not getting IP address

Reachability between relay and server(s)

About this task

Server reachability is required for DHCP Relay function to work.

Procedure

1. Ping to check IP reachability between switch and the server(s).
2. Confirm the ping test is made in the appropriate VRF and with the correct gateway IP. By default, DHCP Relay picks the lowest IP address on the client-connected interface as the gateway IP address. When configured, the bootp-gateway IP address is used as the gateway IP.

Source-ip configuration

About this task

With the source-ip configuration in the VRF, DHCP relay uses it as a gateway IP address to communicate with the DHCP server.

Procedure

1. It requires `dhcp_relay source-interface` configuration for `ip source-interface` relay to work.
2. Ping the server with `source-ip` as the configured `source-interface IP`. The same test is applicable for the Inter-VRF route leak (IVRL) configuration.
3. If the client and server are in different VRFs, test server reachability in server-VRF by using configured source-interface IP.

Option-82 conflict with DHCP snooping and Relay [DHCPv4 Specific]

AOS-CX supports configuring DHCP snooping and DHCP relay on the same VLAN. In cases where DHCP relay is configured to use `source-interface` along with `inter-vrf` server, Option 82 is added to the packet sent to the server.

If Option 82 is enabled for both DHCP snooping and DHCP relay, DHCP snooping gets the priority. In this scenario for DHCP relay to work, disable Option-82 for DHCP snooping.

- Check if the helper-address is configured to the correct VRF where the DHCP server is reachable
- CoPP drop checks—In case of excessive DHCP traffic, CoPP does the rate limit by dropping some of the packets.

Client packets are dropped

Check if client packets are getting dropped by CoPP using `show copp statistics class dhcp -ipv4` or `show copp statistics class dhcp-ipv6` commands.

Server pool configuration and exhaustion

Procedure

1. Check the server side configuration to see if IP lease criteria is being met for the client.
2. Check if the right VRF selection is happening for the lease allocation.

Results

DHCP Smart - Relay

It allows the DHCP relay agent to use non - lower IP addresses from the client - connected interfaces as giaddr and for pool selection when the DHCP server does not reply to the DHCP discover messages with the lowest IP address. More than one IP address and helper address must be configured on the client -

connected interfaces. The server must be configured with the DHCPv4 pools for the client - connected IP addresses

- Server Reachability
  Ensure the ping test is made in the appropriate VRF and with the correct gateway IP (IPs from the client - connected interfaces).

- Bootp gateway IP configuration
  When a bootp gateway IP is configured, it gets the highest priority and is used as a gateway IP.

- Source - ip configuration
  When a source IP is configured, Option 82 needs to be enabled. The Source IP is used as a gateway IP, and Option 82 is used as sub - option 5 contains the IP used for the pool selection.

- Client Cache and timeout
  For debugging, a client cache is built, which contains information on the gateway IP for every client.

  The entry is added to the client cache for the client which does not have an existing entry. The client entries are deleted if the client is idle for 360 seconds. In other cases, a particular client entry might get deleted if it is the oldest entry in time and if the client cache is full. The client cache is deleted completely if the functionality is disabled.

  It uses the lowest IP address as giaddr or for pool selection when all client - connected IPs are exhausted or the client retries after an idle timeout. The client cache is rebuilt upon HA and VSF switchover, VSX - MM (redundancy) switchover, and reboot of the switch.

DHCP server

The dynamic host configuration protocol (DHCP) enables a server to automate the assignment of IP addresses, and other networking settings, to host computers. The DHCP server on the switch provides both IPv4 and IPv6 support and is independently configurable on each VRF.

Subtopics

Protocol and feature details

Key features

- Supports multiple address pools and static address bindings.
- Supports DHCP options, enabling the server to provide additional information about the network when DHCP clients request an address.
- Supports BOOTP to distribute boot image files using an external TFTP server.
- VRF aware, meaning that DHCP client requests received on an interface are processed by the DHCP server instance configured for a VRF. DHCP server responses are forwarded to clients on the VRF.
- Supports external storage of lease information on a remote host. This enables the DHCP server to restore lease information after a reboot or a failure. Lease information is stored in a flat file on the configured external device. It is important that the external device provide persistent external storage to allow restoration of lease information. If external storage is not configured, then after a failure or reboot, all existing lease information is lost.

DHCP relay interoperation

Both DHCP relay and DHCP server can be configured on the same VRF.

DHCP snooping interoperation

DHCP snooping can not be configured with DHCP server.

Supported platform and standards

The following table list the supported platforms for DHCPv4 server and DHCPv6 server.

Table 1. Support platforms for DHCPv4 server and DHCPv6 server

Platform	DHCPv4 server
5420	Yes
6200	Yes

The below limit is based on system stability.

- Single VRF
    - Maximum pools—64
    - Maximum range —64 (Every pool having one range)
    - One static host per pool
    - Maximum clients - 8192 (Every range providing 128 leases)
- Across 32 VRF's
    - Maximum pools - 64 (Every VRF having 2 pools)
    - Maximum range - 64 (Every pool having one range)
    - One static host per pool
    - Maximum clients - 8192 (Every range providing 128 leases)

Configuring a DHCPv4 server on a VRF

Prerequisites

- An enabled layer 3 interface.
- A VRF.
- An external TFTP server to host BOOTP image files (optional).
- An external storage device installed and configured (optional).

Procedure

1. Assign the DHCPv4 server to a VRF with the command **dhcp-server vrf**. This switches to the DHCPv4 server configuration context.
2. If you want the DHCPv4 server to be the sole authority for IP addresses on the VRF, enable authoritative mode with the command **authoritative**.
3. Define an address pool for the VRF with the command **pool**. This switches to the DHCPv4 server pool context. Customize pool settings as follows:
    a. Define the range of addresses in the pool with the command **range**.
    b. Set the lease time for addresses in the pool with the command **lease**.
    c. Set the domain name for the pool with the command **domain-name**.
    d. Define up to four default routers with the command **default-router**.
    e. Define up to four DNS servers with the command **dns-server**.
    f. Create static bindings for specific addresses in the pool with the command **static-bind**.
    g. Configure custom DHCPv4 options for the pool with the command **option**.
    h. Configure NetBIOS support with the commands **netbios-name-server** and **netbios-node-type**.
    i. Configure BOOTP options with the command **bootp**.
    j. Exit the DHCPv4 server pool context with the command **exit**.
4. Enable the DHCP server on the VRF with the command **enable**.
5. Configure support for persistent external storage of DHCP settings with the command **dhcp-server external-storage**.
6. View DHCPv4 server configuration settings with the command **show dhcp-server all-vrfs**.

Example

This example creates the following configuration:

- Configures the DHCPv4 server on VRF **primary-vrf**.
- Enables authoritative mode.
- Defines the pool **primary-pool** with the following settings:
    - Address range: **10.0.0.1** to **10.0.0.100**.
    - Lease time: 12 hours.
    - Domain name: **example.org.in**.
    - Default routers: **10.30.30.1** and **10.30.30.2**.
    - DNS servers: **125.0.0.1** and **125.0.0.2**.
    - Static binding of **10.0.0.11** for MAC address **24:be:05:24:75:73**.
    - DHCP custom option **3** with IP address **10.30.30.3**.
- Enables the DHCPv4 server.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# range 10.0.0.1 10.0.0.100
switch(config-dhcp-server-pool)# lease 12:00:00
switch(config-dhcp-server-pool)# domain-name example.org.in
switch(config-dhcp-server-pool)# default-router ip 10.30.30.1 10.30.30.2
switch(config-dhcp-server-pool)# dns-server 125.0.0.1 125.0.0.2
switch(config-dhcp-server-pool)# static-bind ip 10.0.0.11 mac
24:be:05:24:75:73
switch(config-dhcp-server-pool)# option 3 ip 10.30.30.3
switch(config-dhcp-server-pool)# exit
switch(config-dhcp-server)# enable
```

Configuring the DHCPv6 server on a VRF

Prerequisites

- An enabled layer 3 interface.
- A VRF.
- An external storage device installed and configured (optional).

Procedure

1. Assign the DHCPv6 server to a VRF with the command **dhcpv6-server vrf**. This switches to the DHCPv6 server configuration context.
2. If you want the DHCP server to be the sole authority for IP addresses on the VRF, enable authoritative mode with the command **authoritative**.
3. Define an address pool for the VRF with the command **pool**. This switches to the DHCPv6 server pool context. Customize pool settings as follows:
    a. Define the range of addresses in the pool with the command **range**.
    b. Set the DHCP lease time for addresses in the pool with the command **lease**.
    c. Define up to four DNS servers with the command **dns-server**.
    d. Create static bindings for specific addresses in the pool with the command **static-bind**.
    e. Configure custom DHCP options for the pool with the command **option**.
    f. Exit the DHCP server pool context with the command **exit**.
4. Enable the DHCPv6 server on the VRF with the command **enable**.
5. Configure support for persistent external storage of DHCP settings with the command **dhcpv6-server external-storage**.
6. View DHCPv6 server configuration settings with the command **show dhcpv6-server all-vrfs**.

Example

This example creates the following configuration:

- Configures a DHCPv6 server on VRF **primary-vrf**.
- Enables authoritative mode.
- Defines the pool **primary-pool** with the following settings:
    - Address range: **2001::1** to **2001::100**.
    - Lease time: 12 hours.
    - DNS servers: **2101::14** and **2101::14**.
    - Static binding of **2001::101** for client ID **1:0:a0:24:ab:fb:9c**.
    - DHCP custom option: **22** with IP address **2101::15**.
- Enables the DHCPv6 server.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# pool primary-pool
switch(config-dhcpv6-server-pool)# range 2001::1 2001::100 prefix-len 64
switch(config-dhcpv6-server-pool)# lease 12:00:00
switch(config-dhcpv6-server-pool)# dns-server 2101::13 2101::14
switch(config-dhcpv6-server-pool)# static-bind ipv6 2001::10 client-id
1:0:a0:24:ab:fb:9c
switch(config-dhcpv6-server-pool)# option 22 ipv6 2101::15
switch(config-dhcpv6-server-pool)# exit
switch(config-dhcpv6-server)# enable
```

DHCP server IPv4 commands

Select a command from the list in the left navigation menu.

Subtopics

`authoritative`

Syntax

```
authoritative
no authoritative
```

Description

Configures the DHCPv4 server as authoritative on the current VRF. This means that the server is the sole authority for the network on the VRF. Therefore, if a client requests an IP address lease for which the server has no record, the server responds with DHCPNAK, indicating that the client must no longer use that IP address. If the server is not authoritative, then it will ignore DHCPv4 requests received for unknown leases from unknown hosts.

The **no** form of this command disables authoritative mode on the current VRF.

Example

Configures DHCPv4 server authoritative mode on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# authoritative
```

Removes the DHCPv4 server authoritative mode on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# no authoritative
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcp-server`	Administrators or local user group members with execution righ ts for this command.

bootp

Syntax

```
bootp <REMOTE-URL>
no bootp <REMOTE-URL>
```

Description

Sets the BOOTP options that are returned by the DHCPv4 server for the current pool. BOOTP provides a way to distribute an IP address and boot image file to client stations. The DHCPv4 server returns the IP address and the location of the boot image file, which must be stored on an external TFTP server.

The **no** form of this command disables support for BOOTP.

Parameter	Description
<REMOTE-URL>	Specifies the name and location of a BOOTP file on a TFTP ser ver in the format: `tftp://{<IP> | <HOST>}/<FILE>` • `<IP>` : Specifies the IP address of the TFTP server hosti ng the file in IPv4 format (**x.x.x.x**), where **x** is a decimal nu mber from 0 to 255. You can remove leading zeros. For exa mple, the address `192.169.005.100` becomes `192.168.5.100`. • `<HOST>` : Specifies the fully - qualified domain name of the TFTP server hosting the file. Range: 1 to 64 printable A SCII characters. • `<FILE>` : Specifies the name of the BOOTP file. Range: 1 to 64 printable ASCII characters.

Example

Defines BOOTP support on the DHCPv4 server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# bootp tftp://10.0.0.1/mybootfile
```

Deletes BOOTP support on the DHCPv4 server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# no bootp tftp://10.0.0.1/mybootfile
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcp-server-pool`	Administrators or local user group members with execution righ ts for this command.

`clear dhcp-server leases`

Syntax

```
clear dhcp-server leases [all-vrfs | <IPV4-ADDR> vrf <VRF-NAME>] | vrf <VRF-NAME>]
```

Description

Clears DHCPv4 server lease information. The DHCPv4 server must be disabled before clearing lease information.

Parameter	Description
`all-vrfs`	Clears leases for all VRFs.
<IPV4-ADDR> vrf *<VRF-NAME>*	Clears the lease for a specific client on a specific VRF. Specify th e client address in IPv4 format (**x.x.x.x**), where **x** is a decimal n umber from 0 to 255. You can remove leading zeros. For examp le, the address `192.169.005.100` becomes `192.168.5.100`.
vrf *<VRF-NAME>*	Clears leases for a specific VRF.

Examples

Clearing all DHCPv4 server leases.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# disable
switch(config-dhcp-server)# exit
switch(config)# exit
switch# clear dhcp-server leases
```

Clearing all DHCPv4 server leases for VRF **primary-vrf**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# disable
switch(config-dhcp-server)# exit
switch(config)# exit
switch# clear dhcp-server leases vrf primary-vrf
```

Clear the DHCPv4 server lease for IP address **10.10.10.1** on VRF **primary-vrf**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# disable
switch(config-dhcp-server)# exit
switch(config)# exit
switch# clear dhcp-server leases 10.10.10.1 vrf primary-vrf
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`default-router`

Syntax

```
default-router <IPV4-ADDR-LIST>
no default-router <IPV4-ADDR-LIST>
```

Description

Defines up to four default routers for the current DHCPv4 server pool.

The **no** form of this command removes the specified default routers from the pool.

Parameter	Description
<IPV4-ADDR-LIST>	Specifies the IP addresses of the default routers in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255. You can remove leading zeros. For example, the address `192.169.005.100` becomes `192.168.5.100`. Separate addresses with a space. A maximum of four IP addresses can be defined.

Example

Defines two default routers, **10.0.0.1** and **10.0.0.10**, for the server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# default-router ip 10.0.0.1 10.0.0.10
```

Deletes the default router **10.0.0.1** from the server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# no default-router ip 10.0.0.1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcp-server-pool`	Administrators or local user group members with execution righ ts for this command.

`dhcp-server external-storage`

Syntax

```
dhcp-server external-storage <VOLUME-NAME> file <LEASE-FILENAME> [delay
<DELAY>]
no dhcp-server external-storage <VOLUME-NAME> file <LEASE-FILENAME> [delay
<DELAY>]
```

Description

Configures the external storage file location for DHCPv4 server lease information. This file provides persistent storage, enabling DHCPv4 server settings to be restored when the switch is restarted. Lease information is stored in a flat file on the configured external device.

If external storage is not configured, then after a failure or reboot, all existing lease information is lost.

Lease information is saved to external storage each time the delay timer expires, which by default is every 300 seconds.

Lease information is not restored when issuing the command `dhcp-server enable`.

The **no** form of this command removes external storage support for the DHCPv4 server.

Parameter	Description
`<VOLUME-NAME>`	Specifies the external storage volume name. Range: 1 to 64 prin table ASCII characters.
`file <LEASE-FILENAME>`	Specifies the external storage filename. Range: 1 to 255 printab le ASCII characters.
`delay <DELAY>`	Specifies the interval in seconds between updates to the extern al storage file. Range: 15 to 86400. Default: 300.

Example

Stores the lease file on external storage volume **Storage1** in file **LeaseFile** at an interval of 600 seconds.

```
switch(config)# dhcp-server external-storage Storage1 file LeaseFile delay
600
```

Disables storage of the lease file on external storage volume **Storage1** in file **LeaseFile**.

```
switch(config)# no dhcp-server external-storage Storage1 file LeaseFile
delay 600
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`dhcp-server vrf`

Syntax

```
dhcp-server vrf <VRF-NAME>
no dhcp-server vrf <VRF-NAME>
```

Description

Configures the DHCPv4 server to support a VRF and changes to the `config-dhcp-server` context for that VRF.

The **no** form of this command removes DHCPv4 server support on a VRF.

Parameter	Description
<VRF-NAME>	Name of a VRF.

Example

Configures DHCPv4 server support on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
```

Removes DHCPv4 server support on VRF **primary**.

```
switch(config)# no dhcp-server vrf primary
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`disable`

Syntax

```
disable
```

Description

Disables the DHCPv4 server on the current VRF. The DHCPv4 server is disabled by default when configured on a VRF.

Example

Disables the DHCPv4 server on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcp-server`	Administrators or local user group members with execution righ ts for this command.

`dns-server`

Syntax

```
dns-server <IPV4-ADDR-LIST>
no dns-server <IPV4-ADDR-LIST>
```

Description

Defines up to four DNS servers for the current DHCPv4 server pool.

The **no** form of this command removes the specified DNS servers from the pool.

Parameter	Description
<IPV4-ADDR-LIST>	Specifies the IP addresses of the DNS servers in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255.Separate add resses with a space.

Example

Defines DNS servers for the server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# dns-server 10.0.20.1
```

Deletes a DNS server from the server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# no dns-server 10.0.20.1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcp-server-pool`	Administrators or local user group members with execution righ ts for this command.

`domain-name`

Syntax

```
domain-name  <DOMAIN-NAME>
no domain-name  <DOMAIN-NAME>
```

Description

Defines a domain name for the current DHCPv4 server pool.

The **no** form of this command removes the specified domain name from the pool.

Parameter	Description
<DOMAIN-NAME>	Specifies a domain name. Range: 1 to 255 printable ASCII chara cters.

Example

Defines a domain name for the server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# domain-name example.org.in
```

Deletes a domain name from the server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# no domain-name example.org.in
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcp-server-pool`	Administrators or local user group members with execution righ ts for this command.

`enable`

Syntax

```
enable
```

Description

Enables the DHCPv4 server on the current VRF. The DHCPv4 server is disabled by default when configured on a VRF.

Example

Enables the DHCPv4 server on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcp-server`	Administrators or local user group members with execution righ ts for this command.

`lease`

Syntax

```
lease {<TIME> | infinite}
no lease
```

Description

Sets the length of the DHCPv4 lease time for the current pool. The lease time determines how long an IP address is valid before a DHCPv4 client must request that it be renewed.

The **no** form of this command returns the DHCPv4 lease time to its default value 1 hour.

Parameter	Description
`<TIME>`	Sets the DHCPv4 lease time. Format: DD:HH:MM. Default: 01:0 0:00.
`infinite`	Sets the DHCPv4 lease time to infinite. This means that addres ses do not need to be renewed.

Example

Sets the lease time for DHCPv4 server pool **primary-pool** on VRF **primary** to **12** hours.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# lease 00:12:00
```

Deletes the lease time for DHCPv4 server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# no lease 00:12:00
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcp-server-pool`	Administrators or local user group members with execution righ ts for this command.

netbios-name-server

Syntax

```
netbios-name-server <IPV4-ADDR-LIST>
no netbios-name-server <IPV4-ADDR-LIST>
```

Description

Defines up to four NetBIOS WINS servers for the current DHCPv4 server pool. WINS is used by Microsoft DHCP clients to match host names with IP addresses.

The **no** form of this command removes the specified WINS servers from the pool.

Parameter	Description
<IPV4-ADDR-LIST>	Specifies the IP addresses of NetBIOS (WINS) servers in IPv4 fo rmat (**x.x.x.x**), where **x** is a decimal number from 0 to 255. Sep arate addresses with a space. A maximum of four IP addresses can be defined.

Example

Defines two WINS servers for the server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# netbios-name-server ip 10.0.20.1 10.0.30.10
```

Deletes a WINS server from the server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# no netbios-name-server ip 10.0.20.1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcp-server-pool`	Administrators or local user group members with execution righ ts for this command.

`netbios-node-type`

Syntax

```
netbios-node-type <TYPE>
no netbios-node-type <TYPE>
```

Description

Defines the NetBIOS node type for the current DHCPv4 server pool.

The **no** form of this command removes the NetBIOS node type for the current pool.

Parameter	Description
<TYPE>	Specifies the NetBIOS node type: `broadcast`, `hybrid`, `mixed`, or `peer-to-peer`.

Examples

Defines the NetBIOS node type **broadcast** for the DHCPv4 server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# netbios-node-type broadcast
```

Deletes the NetBIOS node type **broadcast** from the DHCPv4 server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# no netbios-node-type broadcast
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcp-server-pool`	Administrators or local user group members with execution rights for this command.

option

Syntax

```
option <OPTION-NUM> {ascii "<ASCII-STR>" | hex <HEX-STR> | ip <IPV4-ADDR-LIST>}
no option <OPTION-NUM> {ascii <ASCII-STR> | hex <HEX-STR> | ip <IPV4-ADDR-LIST>}
```

Description

Defines custom DHCPv4 options for the current DHCPv4 server pool. DHCPv4 options enable the DHCPv4 server to provide additional information about the network when DHCPv4 clients request an address.

The **no** form of this command removes custom DHCPv4 options from the pool.

Parameter	Description
<OPTION-NUM>	Specifies a DHCPv4 option number. For a list of DHCPv4 option numbers, see **https://www.iana.org/assignments/bootp - dhcp - parameters/bootp - dhcp - parameters.xhtml**. Range: 2 to 254.
ascii *<ASCII-STR>*	Specifies a value for the selected option as an ASCII string. Range: 1 to 255 ASCII characters. **NOTE**

Parameter	Description
	If you specify **18** as the **<OPTION - NUM>** parameter, the ASCII string must be enclosed within quotation marks (").
`hex <HEX-STR>`	Specifies a value for the selected option as a hexadecimal string. Range: 1 to 255 hexadecimal characters.
`ip <IPV4-ADDR-LIST>`	Specifies a list of IP addresses in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255. Separate addresses with a space. A maximum of four IP addresses can be defined.

Example

Defines DHCPv4 option **3** for the server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# option 3 ip 192.168.1.1
```

Deletes DHCPv4 option **3** for the server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# no option 3 ip 192.168.1.1
```

Defines DHCPv4 option 18 for the server pool **mgmt-test** on VRF **mgmt**.

```
switch(config)# dhcp-server vrf mgmt
switch(config-dhcp-server)# pool mgmt-test
switch(config-dhcp-server-pool)# option 18 ascii "aswed"
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcp-server-pool`	Administrators or local user group members with execution righ ts for this command.

`pool`

Syntax

```
pool <POOL-NAME>
no pool <POOL-NAME>
```

Description

Creates a DHCPv4 server pool for the current VRF and switches to the `config-dhcp-server-pool` context for it. Multiple pools, each with a distinct range, can be assigned to a VRF. A maximum of 64 pools (IPv4 and IPv6), 64 address ranges, and 8182 clients are supported on the switch across all VRFs.

The **no** form of this command deletes the specified DHCPv4 server pool.

Parameter	Description
<POOL-NAME>	Specifies the DHCPv4 pool name. A maximum of 64 pools (IPv 4 and IPv6) are supported across VRFs on the switch. Range: 1 to 32 printable ASCII characters. First character must be a letter or number.

Example

Creates the DHCPv4 server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)#
```

Deletes the DHCPv4 server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# no pool primary-pool
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcp-server`	Administrators or local user group members with execution righ ts for this command.

`range`

Syntax

```
range <LOW-IPV4-ADDR> <HIGH-IPV4-ADDR> [prefix-len <MASK>]
no range <LOW-IPV4-ADDR> <HIGH-IPV4-ADDR> [prefix-len <MASK>]
```

Description

Defines the range of IP addresses supported by the current DHCPv4 server pool. A maximum of 64 ranges are supported per switch across all VRFs.

The **no** form of this command deletes the address range for the current pool.

Parameter	Description
<LOW-IPV4-ADDR>	Specifies the lowest IP address in the pool in IPv4 format (**x.x.x .x**), where **x** is a decimal number from 0 to 255.
<HIGH-IPV4-ADDR>	Specifies the highest IP address in the pool in IPv4 format (**x.x. x.x**), where **x** is a decimal number from 0 to 255.

Parameter	Description

`prefix-len <MASK>`	Specifies the number of bits in the address mask in CIDR format (**x**), where **x** is a decimal number from 0 to 32.
	
	NOTE
	When active gateway is configured on the inter face serviced by the pool, you must specify a pr efix length that matches the mask on the IP add ress assigned to the interface. Otherwise, client stations will get a prefix length from active gate way that may not be consistent with the configu red range, and a DHCP error will occur. In the fol lowing example, the DHCP range prefix is set to 16 to match the mask on the IP address assigne d to interface VLAN 2.
	
	`switch(config)# interface vlan 2`
	`switch(config-if-vlan)# ip address 200.1.1.1/16`
	`switch(config-if-vlan)# active-gateway ip 200.1.1.3`
	`  mac 00:aa:aa:aa:aa:aa`
	`switch(config-if-vlan)# exit`
	`switch(config)# dhcp-server vrf primary`
	`switch(config-dhcp-server)# pool primary-pool`
	`switch(config-dhcp-server-pool)# range 192.168.1.1`
	`  192.168.1.100 prefix-len 16`

Examples

Defines the address range **192.168.1.1** to **192.168.1.100** with a mask of **24** bits for the DHCPv4 server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# 192.168.1.1 192.168.1.100 prefix-len 24
```

Deletes the address range **192.168.1.1** to **192.168.1.100** with a mask of **24** bits from the DHCPv4 server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
```

```
switch(config-dhcp-server-pool)# no 192.168.1.1 192.168.1.100 prefix-len 24
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcp-server-pool`	Administrators or local user group members with execution righ ts for this command.

`show dhcp-server`

Syntax

```
show dhcp-server [all-vrfs]
show dhcp-server leases {all-vrfs | vrf <VRF-NAME>}
show dhcp-server pool <POOL-NAME> [vrf <VRF-NAME>]
```

Description

Shows configuration settings for the DHCPv4 server.

Parameter	Description
`all-vrfs`	Shows DHCPv4 server configuration settings for all VRFs.
`leases {all-vrfs	vrf <VRF-NAME>}`	Shows DHCPv4 server lease provided by the server for all VRFs or a specific VRF.
`pool <POOL-NAME> [vrf <VRF-NAME>]`	Shows DHCPv4 server pool configuration settings for all VRFs or a specific VRF.

Examples

Showing all DHCPv4 server configuration settings.

```
switch# show dhcp-server
VRF Name             : default
DHCP Server          : enabled
Operational State    : operational
Authoritative Mode : false
Config_status        : Applied
Pool Name            : test
Lease Duration       : 00:01:00
DHCP dynamic IP allocation
--------------------------
Start-IP-Address   End-IP-Address   Prefix-Length
----------------   --------------   --------------
192.168.1.1        192.168.1.20     24
DHCP Server options
-------------------
Option-Number   Option-Type    Option-Value
-------------   -----------    ------------
6               ip             10.0.0.3 10.0.0.4 10.0.0.5 10.0.0.6
DHCP Server static IP allocation
--------------------------------
IP-Address     Client-Hostname      State            MAC-Address
-----------    ----------------     ------           ------------
10.0.0.3       *                    OPERATIONAL      aa:aa:aa:aa:aa:aa
BOOTP Options
---------------
Boot-File-Name    TFTP-Server-Name    TFTP-Server-Address
--------------    ----------------    ---------------------
boot.txt          *                   10.0.0.10
```

Showing DHCP server configuration settings for VRF **primary-vrf**.

```
switch# show dhcp-server vrf primary-vrf
VRF Name             : primary-vrf
DHCP Server          : disabled
Operational State    : disabled
Authoritative Mode : false
Config_status        : Applied
Pool Name            : test
Lease Duration       : 00:01:00
DHCP dynamic IP allocation
--------------------------
Start-IP-Address   End-IP-Address   Prefix-Length
```

```
-----------------   ---------------   ---------------
10.0.0.1            10.0.0.30         *
192.168.1.1         192.168.1.20      24
192.168.10.30       192.168.10.60     16
DHCP Server options
--------------------
Option-Number    Option-Type     Option-Value
-------------    -----------     ------------
6                ip              10.0.0.3 10.0.0.4 10.0.0.5 10.0.0.6
18               ascii           aswed
DHCP Server static IP allocation
---------------------------------
IP-Address     Client-Hostname    MAC-Address
----------     ---------------    -----------------
10.0.0.1         *                aa:bb:cc:11:12:a4
20.0.0.1         *                11:22:11:22:aa:dd
BOOTP Options
---------------
Boot-File-Name    TFTP-Server-Name     State            TFTP-Server-Address
--------------    ----------------     ------
---------------------
boot.txt           *                   OPERATIONAL      10.0.0.10
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

static-bind

Syntax

```
static-bind {ip <IPV4-ADDR>}|{ mac <MAC-ADDR>} [hostname <HOST>]
no static-bind <IPV4-ADDR-LIST>
```

Description

Creates a static binding that associates an IP address in the current pool with a specific MAC address. This causes the DHCPv4 server to only assign the specified IP address to a client station with the specified MAC address.

The **no** form of this command removes the specified binding.

Parameter	Description
`<IPV4-ADDR>`	Specifies an IP address in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255. The IP address must be within the address range defined for the current pool.
`mac <MAC-ADDR>`	Specifies a client station MAC address (`xx:xx:xx:xx:xx:xx`), where **x** is a hexadecimal number from 0 to F.
`hostname <HOST>`	Specifies the host name of the client station. Range: 1 to 255 printable ASCII characters

Examples

Defines a static address for the server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# static-bind ip 10.0.0.1 mac
24:be:05:24:75:73
```

Deletes a static address from the server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcp-server vrf primary
switch(config-dhcp-server)# pool primary-pool
switch(config-dhcp-server-pool)# no static-bind ip 10.0.0.1 mac
24:be:05:24:75:73
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcp-server-pool`	Administrators or local user group members with execution righ ts for this command.

DHCP server IPv6 commands

Select a command from the list in the left navigation menu.

Subtopics

- **authoritative**
- **clear dhcpv6-server leases**
- **dhcpv6-server external-storage**
- **dhcpv6-server vrf**
- **disable**
- **dns-server**
- **enable**
- **lease**
- **option**
- **pool**
- **range**
- **show dhcpv6-server**
- **static-bind**

`authoritative`

Syntax

```
authoritative
no authoritative
```

Description

Configures the DHCPv6 server as authoritative on the current VRF. This means that the server is the sole authority for the network on the VRF. It responds to client solicit messages with advertise messages having a priority/preference value set to 255 (the maximum), instead of 0 (the minimum). Clients always choose the DHCPv6 server with the highest priority/preference value. If two DHCPv6 servers send an advertise message with the same priority/preference value, then the client picks one and discards the other.

The **no** form of this command disables authoritative mode on the current VRF.

Example

Configures DHCPv6 server authoritative mode on VRF **primary**.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# authoritative
```

Removes DHCPv6 server authoritative mode on VRF **primary**.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# no authoritative
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcpv6-server`	Administrators or local user group members with execution righ ts for this command.

`clear dhcpv6-server leases`

Syntax

```
clear dhcpv6-server leases [all-vrfs | <IPV6-ADDR> vrf <VRF-NAME>] | vrf
<VRF-NAME>]
```

Description

Clears DHCPv6 server lease information. The DHCPv6 server must be disabled before clearing lease information.

Parameter	Description
`all-vrfs`	Clears leases for all VRFs.
`<IPV6-ADDR> vrf <VRF-NAME>`	Clears the lease for a specific client on a specific VRF. Specify the client address in IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address `2222:0000:3333:0000:0000:0000:4444:0055` becomes `2222:0:3333::4444:55`.
`vrf <VRF-NAME>`	Clears leases for a specific VRF.

Examples

Clearing all DHCPv6 server leases.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# disable
switch(config-dhcpv6-server)# exit
switch(config)# exit
switch# clear dhcpv6-server leases
```

Clearing all DHCPv6 server leases for VRF **primary-vrf**.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# disable
switch(config-dhcpv6-server)# exit
switch(config)# exit
switch# clear dhcpv6-server leases vrf primary-vrf
```

Clear the DHCPv6 server lease for IP address **2001::1** on VRF **primary-vrf**.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# disable
```

```
switch(config-dhcpv6-server)# exit
switch(config)# exit
switch# clear dhcpv6-server leases 2001::1 vrf primary-vrf
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`dhcpv6-server external-storage`

Syntax

```
dhcpv6-server external-storage <VOLUME-NAME> file <LEASE-FILENAME> [delay
<DELAY>]
no dhcpv6-server external-storage <VOLUME-NAME> file <LEASE-FILENAME>
[delay <DELAY>]
```

Description

Configures the external storage file location for DHCPv6 server lease information. This file provides persistent storage, enabling DHCPv6 server settings to be restored when the switch is restarted. Lease information is stored in a flat file on the configured external device.

If external storage is not configured, then after a failure or reboot, all existing lease information is lost.

Lease information is saved to external storage each time the delay timer expires, which by default is every 300 seconds.

Lease information is not restored when issuing the command **dhcp-server enable**.

The **no** form of this command removes external storage support for the DHCPv6 server.

Parameter	Description
<VOLUME-NAME>	Specifies the external storage volume name. Range: 1 to 64 printable ASCII characters.
`file <LEASE-FILENAME>`	Specifies the external storage filename. Range: 1 to 255 printable ASCII characters.
`delay <DELAY>`	Specifies the interval in seconds between updates to the external storage file. Range: 15 to 86400. Default: 300.

Example

Stores the lease file on external storage volume **Storage1** in file **LeaseFile** at an interval of 600 seconds.

```
switch(config)# dhcpv6-server external-storage Storage1 file LeaseFile
delay 600
```

Disables storage of the lease file on external storage volume **Storage1** in file **LeaseFile**.

```
switch(config)# no dhcpv6-server external-storage Storage1 file LeaseFile
delay 600
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

`dhcpv6-server vrf`

Syntax

```
dhcpv6-server vrf VRF-NAME
no dhcpv6-server vrf VRF-NAME
```

Description

Configures the DHCPv6 server to support a VRF and changes to the **config-dhcpv6-server** context for that VRF.

The **no** form of this command removes DHCPv6 server support on a VRF.

Parameter	Description
VRF-NAME	Name of a VRF.

Example

Configures DHCPv6 server support on VRF **primary**.

```
switch(config)# dhcpv6-server vrf primary
```

Removes the DHCPv6 server support on VRF **primary**.

```
switch(config)# no dhcpv6-server vrf primary
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420	`config`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority
6200		

`disable`

Syntax

```
disable
```

Description

Disables the DHCPv6 server on the current VRF. The DHCPv6 server is disabled by default when configured on a VRF.

Example

Disables the DHCPv6 server on VRF **primary**.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcpv6-server`	Administrators or local user group members with execution righ ts for this command.

`dns-server`

Syntax

```
dns-server <IPVv6-ADDR-LIST>
no dns-server <IPVv6-ADDR-LIST>
```

Description

Defines up to four DNS servers for the current DHCPv6 server pool.

The **no** form of this command removes the specified DNS servers from the pool.

Parameter	Description
<IPVv6-ADDR-LIST>	Specifies the IP addresses of the DNS servers in IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F. Separate addresses with a space. A maximum of four IP addresses can be defined.

Example

Defines DNS server **2001::13** for the server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# pool primary-pool
switch(config-dhcpv6-server-pool)# dns-server 2001::13
```

Deletes DNS server **2001::13** from the server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# pool primary-pool
switch(config-dhcpv6-server-pool)# no dns-server 2001::13
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	`config-dhcpv6-server-pool`	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority
6200		

`enable`

Syntax

```
enable
```

Description

Enables the DHCPv6 server on the current VRF. The DHCPv6 server is disabled by default when configured on a VRF.

Example

Enables the DHCPv6 server on VRF **primary**.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcpv6-server`	Administrators or local user group members with execution righ ts for this command.

`lease`

Syntax

```
lease {<TIME> | infinite}
no lease
```

Description

Sets the length of the DHCPv6 lease time for the current pool. The lease time determines how long an IP address is valid before a DHCPv6 client must request that it be renewed.

The **no** form of this command returns the DHCPv6 lease time to the default value 1 hour.

Parameter	Description
`<TIME>`	Sets the DHCPv6 lease time. Format: DD:HH:MM. Default: 01:00:00.
`infinite`	Sets the DHCPv6 lease time to infinite. This means that addresses do not need to be renewed.

Example

Sets the lease time for DHCPv6 server pool **primary-pool** on VRF **primary** to **12** hours.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# pool primary-pool
switch(config-dhcpv6-server-pool)# lease 00:12:00
```

Sets the lease time for DHCP server pool **primary-pool** on VRF **primary** to the default value.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# pool primary-pool
switch(config-dhcpv6-server-pool)# no lease 00:12:00
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcpv6-server-pool`	Administrators or local user group members with execution righ ts for this command.

`option`

Syntax

```
option <OPTION-NUM> {ascii <ASCII-STR> | hex <HEX-STR> | ip <IPV6-ADDR-LIST>}
no option <OPTION-NUM> {ascii <ASCII-STR> | hex <HEX-STR> | ip <IPV6-ADDR-LIST>}
```

Description

Defines custom DHCPv6 options for the current DHCPv6 server pool.

The **no** form of this command removes custom DHCPv6 options from the pool.

Parameter	Description
`<OPTION-NUM>`	Specifies a DHCPv6 option number. Range: 2 to 254.
`ascii <ASCII-STR>`	Specifies a value for the selected option as an ASCII string. Ran ge: 1 to 255 ASCII characters.
`hex <HEX-STR>`	Specifies a value for the selected option as a hexadecimal string . Range: 1 to 255 hexadecimal characters.
`ip <IPV6-ADDR-LIST>`	Specifies a list of IP addresses for the option in IPv6 format (**xx xx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexad ecimal number from 0 to F.

Example

Defines DHCPv6 option **22** for the server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# pool primary-pool
```

```
switch(config-dhcpv6-server-pool)# option 22 ipv6 2001::12
```

Deletes DHCPv6 option 22 for the server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# pool primary-pool
switch(config-dhcpv6-server-pool)# no option 22 ipv6 2001::12
```

Defines DHCPv6 option **18** for the server pool **mgmt-test** on VRF **mgmt**.

```
switch(config)# dhcpvv6-server vrf mgmt
switch(config-dhcpv6-server)# pool mgmt-test
switch(config-dhcpv6-server-pool)# option 18 ascii "aswed"
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcpv6-server-pool`	Administrators or local user group members with execution righ ts for this command.

`pool`

Syntax

```
pool <POOL-NAME>
no pool <POOL-NAME>
```

Description

Creates a DHCPv6 server pool for the current VRF and switches to the `config-dhcpv6-server-pool` context for it. Multiple pools, each with a distinct range, can be assigned to a VRF. A maximum of 64 pools (IPv4 and IPv6), 64 address ranges, and 8182 clients are supported on the switch across all VRFs.

The **no** form of this command deletes the specified DHCPv6 server pool.

Parameter	Description
<POOL-NAME>	Specifies the DHCPv6 pool name. A maximum of 64 pools (IPv 4 and IPv6) are supported across VRFs on the switch. Range: 1 to 32 printable ASCII characters. First character must be a letter or number.

Example

Creates the DHCPv6 server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# pool primary-pool
switch(config-dhcpv6-server-pool)#
```

Deletes the DHCPv6 server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# no pool primary-pool
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcpv6-server`	Administrators or local user group members with execution righ ts for this command.

`range`

Syntax

```
range <LOW-IPV6-ADDR> <HIGH-IPV6-ADDR> [prefix-len <MASK>]
no range <LOW-IPV6-ADDR> <HIGH-IPV6-ADDR> [prefix-len <MASK>]
```

Description

Defines the range of IP addresses supported by the current DHCPv6 server pool. A maximum of 64 ranges are supported per switch across all VRFs.

The **no** form of this command deletes the address range for the current pool.

Parameter	Description
`<LOW-IPV6-ADDR>`	Specifies the lowest IP address in the pool in IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F.
`<HIGH-IPV6-ADDR>`	Specifies the highest IP address in the pool in IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F.
`prefix-len <MASK>`	Specifies the number of bits in the address mask in CIDR format (**x**), where **x** is a decimal number from 64 to128.

Example

Defines an address range for the DHCPv6 server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# pool primary-pool
switch(config-dhcpv6-server-pool)# range 2001::1 2001::10 prefix-len 64
```

Deletes an address range for the DHCPv6 server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# pool primary-pool
switch(config-dhcpv6-server-pool)# no range 2001::1 2001::10 prefix-len 64
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcpv6-server-pool`	Administrators or local user group members with execution righ ts for this command.

`show dhcpv6-server`

Syntax

```
show dhcpv6-server [all-vrfs]
show dhcpv6-server leases {all-vrfs | vrf <VRF-NAME>}
show dhcpv6-server pool <POOL-NAME> [vrf <VRF-NAME>]
```

Description

Shows configuration settings for the DHCPv6 server.

Parameter	Description
`all-vrfs`	Shows DHCPv6 server configuration settings for all VRFs.
`leases {all-vrfs | vrf <VRF-NAME>}`	Shows DHCPv6 server lease provided by the server for all VRFs or a specific VRF.
`pool <POOL-NAME> [vrf <VRF-NAME>]`	Shows DHCPv6 server pool configuration settings for all VRFs or a specific VRF.

Examples

Showing all DHCPv6 server configuration settings.

```
switch# show dhcpv6-server
VRF Name              : default
DHCPv6 Server         : enabled
Operational State     : operational
Authoritative Mode    : true
Config_status         : Applied
Pool Name             : test
Lease Duration        : 00:01:00
DHCPV6 dynamic IP allocation
```

```
-----------------------------
Start-IPv6-Address  End-IPv6-Address   Prefix-Length
------------------  ----------------   -------------
2001::2             2001::10           64
DHCPv6 Server options
---------------------
Option-Number     Option-Type    Option-Value
-------------     -----------    ------------
7                 ipv6           2001::15
DHCPv6 Server static IP allocation
-----------------------------------
DHCPv6 Server static host is not configured.
```

Showing DHCPv6 server configuration settings for VRF **primary-vrf**.

```
switch# show dhcpv6-server vrf primary-vrf
VRF Name             : primary-vrf
DHCPv6 Server        : disabled
Operational State    : standby
Authoritative Mode : false
Config_status        : Applied
Pool Name            : test
Lease Duration       : 00:01:00
DHCPV6 dynamic IP allocation
-----------------------------
Start-IPv6-Address  End-IPv6-Address   Prefix-Length
------------------  ----------------   -------------
2000::1             2000::20           *
2001::20            2001::50           *
2001::2             2001::10           64
2010::20            2010::40           *
DHCPv6 Server options
---------------------
Option-Number     Option-Type    Option-Value
-------------     -----------    ------------
7                 ipv6           2001::15
23                ipv6           2001::30
30                ipv6           2001::10
DHCPv6 Server static IP allocation
-----------------------------------
DHCPv6 Server static host is not configured.
Pool Name         : v6test
Lease Duration  : 00:01:00
DHCPv6 dynamic IP allocation
-----------------------------
```

```
Start-IPv6-Address    End-IPv6-Address    Prefix-Length
------------------    ----------------    -------------
2001::1               2001::20            64
2010::10              2010::30            *
2020::20              2020::60            *
DHCPv6 Server options
---------------------
Option-Number    Option-Type    Option-Value
-------------    -----------    -----------------
7                ipv6           2001::20
23               ipv6           2001:0db8:85a3:0000:0000:8a2e:0370:7334
2001:0db8:85a3:0000:0000:8a2e:0370:7335
                                2001:0db8:85a3:0000:0000:8a2e:0370:7336
2001:0db8:85a3:0000:0000:8a2e:0370:7337
DHCPv6 Server static IP allocation
------------------------------------
IPv6-Address    Client-Hostname     State          Client-Id
------------    ---------------     -----------    ---------
2100::4         *                   OPERATIONAL    1:0:a0:24:ab:fb:9c
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`static-bind`

Syntax

```
static-bind ipv6 <IPVv6-ADDR> client-id <ID> [hostname <HOST>]
no static-bind ipv6 <IPVv6-ADDR-LIST>
```

Description

Creates a static binding that associates an IP address in the current pool with a client identifier or DUID. This causes the DHCPv6 server to only assign the specified IP address to a client station with the specified client identifier or DUID.

The **no** form of this command removes the specified static binding from the pool.

Parameter	Description
`<IPV6-ADDR>`	Specifies the IP address to assign in IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F. For example, this address `2222:0000:3333:0000:0000:0000:4444:0055` becomes `2222:0:3333::4444:55`.
`client-id <ID>`	Specifies the client identifier or DUID.
`hostname <HOST>`	Specifies the host name of the client station. Range: 1 to 255 printable ASCII characters

Example

Defines a static address for the DHCPv6 server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# pool primary-pool
switch(config-dhcpv6-server-pool)# static-bind ipv6 2001::10 client-id
1:0:a0:24:ab:fb:9c
```

Deletes a static address from the DHCPv6 server pool **primary-pool** on VRF **primary**.

```
switch(config)# dhcpv6-server vrf primary
switch(config-dhcpv6-server)# pool primary-pool
switch(config-dhcpv6-server-pool)# no static-bind ipv6 2001::10 client-id
1:0:a0:24:ab:fb:9c
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcpv6-server-pool`	Administrators or local user group members with execution righ ts for this command.

Troubleshooting

Client not getting IP address

- CoPP drop

In case of excessive DHCP traffic, CoPP does the rate limit by dropping some of the packets.

Check if client packets are getting dropped by CoPP using `show copp statistics class dhcp-ipv4` or `show copp statistics class dhcp-ipv6` commands.

- Configuration issues
  1. IP range within the pool should match the client connected interface IP subnet.
     ```
     dhcp-server vrf default
         pool pool1
             range 172.16.10.1 172.16.10.20
             exit
         enable
     ```
     Check the scale numbers supported on the platform. For example the maximum number of VRFs, and pools supported on the platform.
  2. Check if the DHCP Server is configured in client VRF and is in enabled state.
  3. Check for server pool exhaustion.
- DHCP Relay co-existence
  From AOS-CX 10.07 onwards, you can configure DHCP Relay and DHCP Server simultaneously. If both are enabled for an earlier release, then either the DHCP relay or the DHCP server will fail to start.

DHCP Server daemon taking up high CPU [`dhcp-server-adapter`]

- Due to architectural limitations, the DHCP Server on boot up the CPU utilization is high for about 2 minutes; this is expected behavior.
- In the VSX setup, when the number of clients in the lease table is more, the server daemon will have high CPU usage while syncing leases between primary and secondary VSX peers. Once syncing is complete, then the CPU utilization comes back to normal.

Check point restore

DHCP-Server configuration changes are not allowed when it is enabled. However, through check-point restore or TFTP configuration download or REST, you can change the configuration in the management VRF. This configuration change gets applied when the server is disabled and enabled.

From 10.10 onwards, a configuration flag icon is displayed in `show dhcp-server` command to indicate if there is any configuration that is not applied.

FAQ

1. What ports use DHCP for switch assignments?

You can set up a DHCP switch with a management VRF port; if the management VRF port is unavailable, use an IN-Band data port.

2. Is DHCP assignment available on boot up from the factory?

Yes, DHCP assignment is available on boot up from the factory. This assists Zero Touch depending on the platform; this can be a management VRF or an IN-Band data port on the default VLAN 1.

3. Is DHCP assignment available on both management VRF and IN-Band data ports simultaneously?

Yes, DHCP assignment is possible on both management VRF and IN Band data ports, but it depends on the platform.

If both ports can get a DHCP address, the DHCP address from the management VRF port is used for Zero Touch Provisioning instead of the one from the IN-Band data port. The data ports will wait 30 seconds before taking ownership of Zero Touch Provisioning (ZTP).

4. Which VLAN is assigned a DHCP address?

The factory default VLAN to which the DHCP address is assigned is VLAN1.

Users can configure the VLAN for DHCP allocation on platforms supporting multiple VLAN configurations. Only a single VLAN can support DHCP assignments.

5. Can DHCP assignments be disabled in an HPE Aruba Networking AOS-CX switch?

Use the **no ip dhcp** command on the required VLAN interface or management VRF port you can disable the DHCP assignments in an HPE Aruba Networking AOS-CX switch.

A static configuration option is available on both ports.

6. Can IP DHCP switch assignment be carried out on a Routed only Port (RoP)?

No, RoP is not supported. Only VLAN and OOBM ports can be allocated an address with DHCP.

7. Can an HPE Aruba Networking AOS-CX switch support DHCP IPv6 Allocation?

Yes, an HPE Aruba Networking AOS-CX switch can support DHCP IPv6 allocation. However, for Zero Touch Provisioning (ZTP), IPv4 is required.

8. Do HPE Aruba Networking AOS-CX switches support DHCP server?

Since the software release of AOS-CX 10.10, all platforms support DHCP server, except for 6100, 6000, and 4100i.

9. Can HPE Aruba Networking AOS-CX switches support DHCP address exclusion from a subnet?

Not formally; however, you can use static bindings to unused MAC addresses as a workaround on a smaller scale.

10. Do all HPE Aruba Networking AOS-CX switches support DHCP helper?

Since the software release of AOS-CX 10.10, all platforms support DHCP helper for IPv4 and IPv6.

11. Can DHCP Server and Relay simultaneously be supported on a switch?

Yes, since the software release AOS-CX 10.08, the platforms that support DHCP Server and Relay both can be set up simultaneously.

12. How many helper addresses can be used?

The Switched Virtual Interface can use up to eight helper addresses for IPv4 and IPv6.

13. Is DHCP Relay supported for multi-netted addresses?

Yes, for both IPv4 and IPv6 DHCP Relay supports multi-netted addresses. The Gateway IP Address (GIADDR) used in the DHCP request will be the lowest IP address configured on the interface.

You can override the lowest IP address using the `bootp-gateway` command. IPV6 does not support bootp.

14. Can a multi-netted address support GIADDR from a different address?

Yes, for IPv4 only. For more information, see DHCP relay agent.

15. Can the binding of DHCP and IP remain after a switch reboot?

Yes, the binding of DHCP and IP remains by using an external storage option `dhcpv4-snooping external-storage`.

16. If multiple 'ip-helper' IPs are configured for DHCP relay on an interface or SVI, which switch will forward the DHCP requests?

In case of multiple 'ip-helper' IPs configured for DHCP relay on an interface or SVI, the client will receive multiple offers from different DHCP servers, and the client will choose which one to accept.

17. What happens when global *dhcp-smart-relay* is enabled in an interface or SVI with multiple IP addresses (multi-netting) that services DHCP clients in different subnets?

On enabling global **dhcp-smart-relay** in an interface or SVI with multiple IP addresses (multi-netting) that services DHCP clients in different subnets, the AOS-CX device will first use the primary interface or SVI IP address as the unicast source to the DHCP server. If no response is received the AOS-CXdevice will use the secondary IPs from the lowest-numbered IP to the highest until a DHCP response is received.

18. What are the troubleshooting commands for DHCP relay agent?

The initial troubleshooting for DHCP relay agent can be done using the following commands:

- `show dhcp-relay`
- `show dhcp-relay bootp-gateway`
- `show ip helper-address`

Next level of troubleshooting can be done by taking packets captured at the DHCP server side and AOS-CX side. For more information about AOS-CX packet capture, see Mirroring chapter of AOS-CX Monitoring Guide.

Perform additional AOS-CX side DHCP debugging using **debug dhcprelay** and **debug dhcpv6relay** commands. For more information, see Debug logging chapter of AOS-CX Diagnostics and Supportability Guide.

DHCP snooping

DHCP is a protocol used by DHCP servers in IP networks to dynamically allocate network configuration data to client devices (DHCP clients). Possible network configuration data includes user IP address, subnet mask, default gateway IP address, DNS server IP address, and lease duration. The DHCP protocol enables DHCP clients to be dynamically configured with such network configuration data without any manual setup process.

DHCP snooping is a security feature that helps avoid problems caused by an unauthorized DHCP server on the network that provides invalid configuration data to DHCP clients. A user without malicious intent may cause this problem by unknowingly adding to the network a switch or other device that includes a DHCP server enabled by default. In some cases, a user with malicious intent adds a DHCP server to the network as part of their Denial of Service or Man in the Middle attack.

DHCP snooping helps prevent such problems by distinguishing between trusted ports connected to legitimate DHCP servers and untrusted ports connected to general users. DHCP packets are forwarded between trusted ports without inspection. DHCP packets received on other switch ports are inspected before being forwarded. DHCP Packets from untrusted sources are dropped.

In support of the separate IP source lockdown feature, DHCP snooping dynamically collects client information (VLAN, IPv4 address, MAC address, interface) and adds it to the switch IP binding database. Alternatively, the IP binding database can be statically updated using the `ipv4 source-binding` or `ipv6 source-binding` commands. Statically configured IP binding information supersedes any dynamically collected information for the same client.

> **NOTE**
>
> DHCP Snooping and DHCP relay can be configured on the same switch.
>
> When DHCP snooping and DHCP relay are both enabled on a VLAN, the following actions occur:
>
> - Received packet: DHCP snooping processes the DHCP packet before (possibly) handing it to DHCP relay.
> - Transmitted packet: DHCP packets sent by DHCP relay are intercepted by DHCP snooping to learn IP bindings.

> **NOTE**
>
> For even more rigorous security that is applied in hardware on a packet-by-packet basis, you can use IP source lockdown feature as described in `IP source lockdown`.

DHCPv6 guard

The DHCPv6 guard feature is an extension of DHCPv6 snooping. When the DHCPv6 snooping feature is configured globally and on the VLAN, the ports are configured as trusted and untrusted ports on the VLAN. DHCPv6 guard enhances this feature by creating a policy and applying it on a port and VLAN. This policy contains multiple attributes which are compared against the packet that is received on trusted ports. If the packet complies with the attributes of the policy, it is forwarded to the destination port; otherwise the packet is dropped.

DHCP server interoperation

DHCP server can not be configured with DHCP snooping.

DHCPv4 snooping conditions for dropping DHCPv4 packets

Applies only to DHCPv4 snooping.

Packet types that are dropped	Conditions for dropping the packets
DHCPOFFER, DHCPACK, DHCPNACK	• A packet from a DHCP server is received on an u ntrusted port.

Packet types that are dropped	Conditions for dropping the packets
	• The switch is configured with a list of authorized DHCP server addresses and a DHCP response received on a trusted port, but the source IP address is not an authorized DHCP server.
DHCPRELEASE, DHCPDECLINE	• A broadcast packet that has a MAC address in the DHCP binding database, but the port in the DHCP binding database does not match the port on which the packet is received.
All DHCP packet types	• A DHCP packet received on an untrusted port in which the DHCP client hardware MAC address does not match the source MAC address in the packet. • A DHCP packet containing DHCP relay information (option 82) is received on an untrusted port.

Subtopics

Protocol details

Procedure

1. When a port is configured as a trusted port, all the dynamic IP binding entries learned on that port will be deleted.
2. When a client is connected on a trusted port, the dynamic IP binding entries will not be learned on the switch, even though the client gets an IP address.
3. If DHCPv4 snooping is enabled on two back-to-back access switches, DHCP packets will be dropped, Since by default option 82 is enabled on DHCPv4 snooping and the default policy is drop. The second

switch with DHCPv4 snooping enabled drops the packets. In this scenario the user should enable DHCPv4 snooping option 82 on one switch, or else you can disable on both.

Supported platform and standards

The following table list the supported platforms for DHCP snooping.

Table 1. Support platforms for DHCP snooping

Platform	DHCP snooping
5420	Yes
6200	Yes

Table 2. Scale

Platform	IP Bindings per port
5420	1024
6200	1024

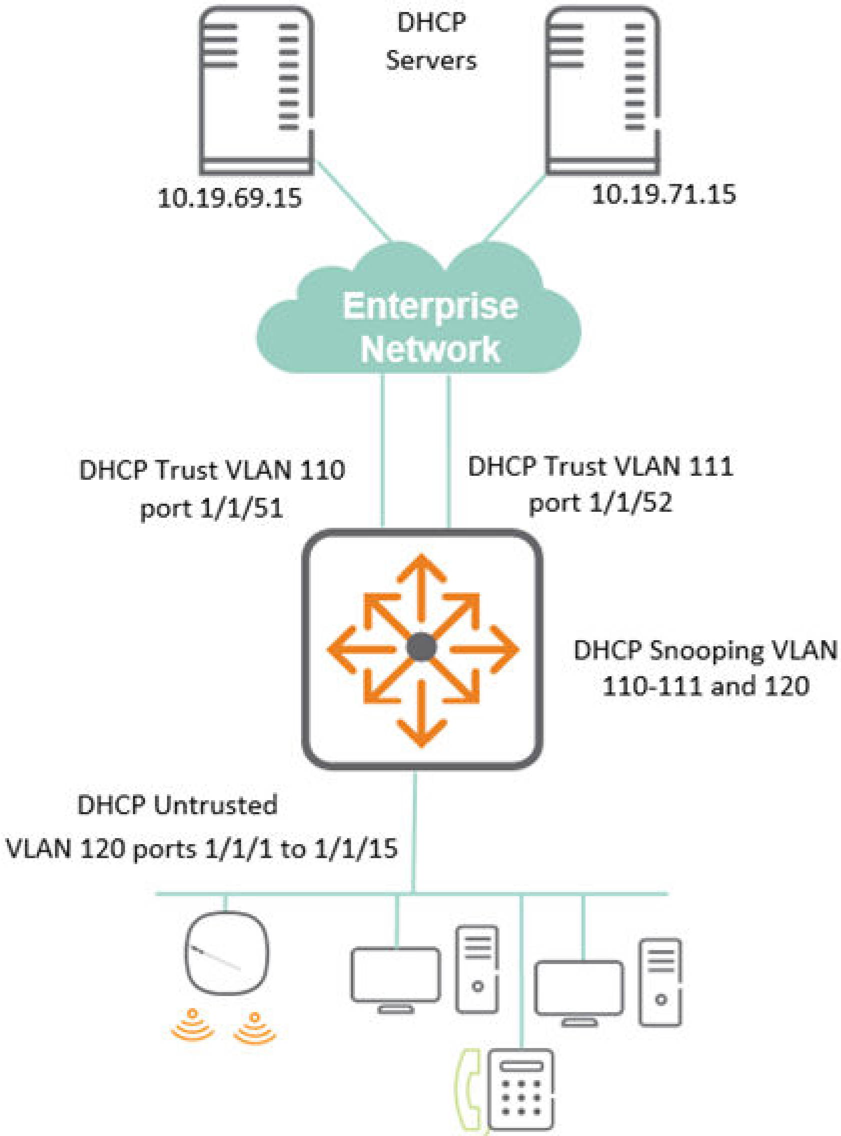
Platform	IP Bindings per port

DHCPv4 Snooping Use case

DHCP Snooping is used to protect from rogue or unwanted attacks from posing DHCP services.

In the following use case:

- Set the switch for DHCPv4 snooping.
- Allow DHCP communication from the uplinks in the Enterprise network port 1/1/51 and 1/1/52.
- For communication to occur, identify the authorized DHCP servers 10.19.69.15 and 10.19.71.15.
- LAN facing user ports 1/1/1 to 1/1/15 is not configured as trusted.



Key configuration portion for above set up is shown with the relevant DHCPv4 snooping parameters:

```
dhcpv4-snooping                                 <--enable DHCPv4 snooping
dhcpv4-snooping authorized-server 10.19.69.15        <--allow authorized
servers
dhcpv4-snooping authorized-server 10.19.71.15
vlan 110
    description uplink
    dhcpv4-snooping                                     <--enable DHCPv4
snooping on required VLANs
vlan 111
    description uplink
    dhcpv4-snooping
vlan 120
    description User
```

```
    dhcpv4-snooping
interface 1/1/1
    no shutdown
    description User
    vlan access 120
...
interface 1/1/15
    no shutdown
    description User
    vlan access 120
interface 1/1/51                      <--Unplinks on 1/1/51 and 52 trusted
    no shutdown
    vlan trunk native 1
    vlan trunk allowed 110
    dhcpv4-snooping trust
interface 1/1/52
    vlan trunk native 1
    vlan trunk allowed 111
    dhcpv4-snooping trust
interface vlan 110
    description uplink
    ip address 172.16.69.1/30
interface vlan 111
    description uplink
    ip address 172.16.71.1/30
interface vlan 120
    description User
    ip address 172.16.10.1/254
    ip helper-address 10.19.69.15
    ip helper-address 10.19.71.15
...
ip route 0.0.0.0/0  172.16.69.2 distance 10
ip route 0.0.0.0/0  172.16.71.2 distance 20
...
```

Using the show dhcpv4-snooping command, you can see some of the following DHCPv4 information:

- Applied VLANs for snooping, 110-112 and 120
- The untrusted policy is set to drop packets
- The list of authorized DHCP servers are set to 10.19.69.15 and 10,19.71.15
- Set the trusted or untrusted ports and any DHCP bindings for those ports. Port 1/1/1 and 1/1/2 offer DHCP.

```
show dhcpv4-snooping
DHCPv4-Snooping Information
  DHCPv4-Snooping         : Yes           Verify MAC Address   : Yes
  Allow Overwrite Binding : No            Enabled VLANs        : 110-111,120
  IP Binding Disabled VLANs :
  Static Attributes : No
  Client Event Logs : No
 Option 82 Configurations
  Untrusted Policy     : drop            Insertion            : Yes
  Option 82 Remote-id  : mac
 External Storage Information
  Volume Name          : --
  File Name            : --
  Inactive Since       : --
  Error                : --
 Flash Storage Information
 File Write Delay : --
 Active Storage : --
 Authorized Server Configurations
  VRF                                         Authorized Servers
------------                        ------------------
 default                    10.19.69.15
 default                    10.19.71.15
 Port Information
                   Max       Static    Dynamic
  Port      Trust  Bindings  Bindings  Bindings
  --------  -----  --------  --------  --------
  1/1/51    Yes    0         0         0
  1/1/52    Yes    0         0         0
  1/1/1     No    1024       0         1
  1/1/2     No    1024       0         1
  !
  !output omitted
  !
  1/1/14    No    1024        0         0
  1/1/15    No    1024        0         0
```

RFC 7610 Compliance for DHCPv6 Guard

DHCPv6 Guard implements **RFC 7610** and enforces all must requirements for filtering rogue DHCPv6 server messages at Layer 2. The following sections describes the compliance and implementation details.

Subtopics

DHCPv6 Guard Extension Header Handling

Extension Header Handling in DHCPv6 Guard

The DHCPv6 Guard implementation processes IPv6 extension headers as follows:

A. Supported Extension Headers (fully parsed and inspected):

- Hop-by-Hop Options Header: DHCPv6 Guard inspects the header chain for DHCPv6 Server packets.
- Destination Options Header: DHCPv6 Guard inspects the header chain for DHCPv6 Server packets.
- Routing Header: DHCPv6 Guard inspects the header chain for DHCPv6 Server packets.
- Authentication Header (AH): Parsed. If DHCPv6 header is found after AH, DHCPv6 Guard policy is enforced.
- Mobility Header: Packets with this header are dropped and not forwarded.

The following table shows the next headers values and the behavior in DHCPv6 Guard:

Extension Header	Next Header Value	Behavior in DHCPv6 Guard
Hop-by-Hop Options	0	Parsed and inspected
Destination Options	60	Parsed and inspected
Routing	43	Parsed and inspected
Authentication Header	51	Parsed and inspected
ESP	50	Not inspected, always allowed
Mobility	135	Parsed and dropped
No Next Header	59	Not inspected, allowed
Unknown/Custom Header	(others)	Forwarded by default

B. Upper-Layer Protocols (not inspected for DHCPv6, always allowed):

Packet / Payload	Next Header Value	Behavior in DHCPv6 Guard
Encapsulating Security Payload (ESP)	50	Treated as an upper-layer protocol. Packets with ESP are not inspected for DHCPv6 and are forwarded as per RFC 7610.
Unsupported or Unrecognized Extension Headers	-	If an unknown or unsupported Next Header value is encountered in the extension header chain, the default behavior is to forward the packet. This is configurable.
No Next Header	59	Not inspected for DHCPv6. Such packets are forwarded as per standard IPv6 behavior.
Custom/Experimental Headers	-	Any custom or experimental extension headers. For example, Next Header values not defined in RFC 8200 or not listed above, are forwarded by default, but can be dropped and logged if the relevant configuration is enabled.

NOTE

The implementation ensures that all supported extension headers are parsed in the order they appear, and the full header chain is inspected to locate DHCPv6 Server packets. Packets are dropped and logged for deviations from the expected header chain or the presence of unsupported or unknown headers only if the relevant configuration is enabled, as per RFC 7610 recommendations.

Client Packet Handling with Extension Headers

DHCPv6 client packets (Solicit, Request, Confirm, Renew, Rebind, Release, Decline, Information-Request) containing IPv6 extension headers are handled based on the **dhcpv6-snooping allow-ipv6-ext-headers** configuration:

- When **allow-ipv6-ext-headers** is **enabled**: Client packets with extension headers are correctly parsed and processed. The DHCPv6 message is correctly located past all extension headers in the chain.
- When **allow-ipv6-ext-headers** is **disabled** (default): Client packets containing valid extension headers are dropped.

DHCPv6 Guard Fragmentation Handling

The DHCPv6 Guard implementation complies with RFC 7610 for fragmented packets. The behavior depends on the fragment type, header completeness, and the **allow-ipv6-ext-headers** configuration.

First Fragment (Fragment Offset = 0)

- **Complete headers + allow-ext-hdrs=yes:** If the first fragment contains the complete IPv6 header chain (including all extension headers, the UDP header, and the DHCPv6 header), the packet is processed by DHCPv6 Guard as per the configured policy and allowed.
- **Complete headers + allow-ext-hdrs=no:** The packet is dropped because extension headers are present (the Fragment header itself is an IPv6 extension header). The **V6_PACKETS_DROP** counter is incremented. No event log is generated.
- **Incomplete UDP/DHCPv6 header + allow-ext-hdrs=yes:** If the first fragment contains the Fragment header and reaches the UDP header (ports 546/547 visible for ASIC classification), but the DHCPv6 header is incomplete, the packet is dropped. An event log is generated and the **V6_PACKETS_DROP** counter is incremented.
- **Incomplete UDP/DHCPv6 header + allow-ext-hdrs=no:** The packet is dropped due to the extension header policy before DHCPv6 header validation. The **V6_PACKETS_DROP** counter is incremented. No event log is generated.

Non-First Fragment (Fragment Offset > 0)

- Non-first fragments do not contain the IPv6 header chain and therefore cannot be classified as DHCPv6 packets by the ASIC. These fragments are **hardware forwarded** regardless of the **allow-ipv6-ext-headers** configuration and are never inspected by DHCPv6 Guard.

Summary:

Only first fragments where the full IPv6 header chain and the DHCPv6 header are present and complete are subject to DHCPv6 Guard inspection. First fragments with incomplete headers are dropped. Non-first fragments are hardware forwarded.

Fragmentation Behavior Summary

Scenario	allow-ext-hdrs	Result
First fragment with complete headers (ext hdrs + frag)	yes	ALLOW
First fragment with complete headers (ext hdrs + frag)	no	DROP
First fragment with complete headers (frag only)	yes	ALLOW
First fragment with complete headers (frag only)	no	DROP
Non-first fragment (offset > 0)	yes	ALLOW
Non-first fragment (offset > 0)	no	ALLOW

Scenario	allow-ext-hdrs	Result
First fragment + incomplete UDP/DHCPv6 header	yes	DROP
First fragment + incomplete UDP/DHCPv6 header	no	DROP

DHCPv6 Guard Unknown Next Header Handling

If the packet contains an unrecognized Next Header value, the default behavior is to forward the packet. The **dhcpv6-snooping drop-unknown-ipv6-ext-headers** configuration enables dropping and logging of such packets, as per RFC 7610 recommendations.

When **drop-unknown-ipv6-ext-headers** is configured, the behavior depends on whether the ingress port is trusted or untrusted:

- **Untrusted ports:** Packets with unknown Next Header values are **reforwarded** instead of dropped, maintaining backward compatibility.
- **Trusted ports:** Packets with unknown Next Header values are **dropped** and logged, as per RFC 7610 compliance.

Port Type	Drop-unknown-ipv6-ext-headers Configured	Behavior for Unknown NH
Untrusted	Yes	Reforwarded
Untrusted	No (default)	Forwarded
Trusted	Yes	Dropped and logged
Trusted	No (default)	Forwarded

DHCPv6 Guard Statistics and Logging

- **Drop Counters:**

  The `show dhcpv6-snooping statistics` command provides a summary of all DHCPv6 packet processing statistics, including drop counters. The drop counter is incremented whenever a packet is dropped due to validation failure, DHCPv6 Guard policy, or RFC 7610 compliance.

- **Event Logging:**

Whenever a packet is dropped by DHCPv6 Guard (including drops due to RFC 7610 compliance), an event log is generated. These logs include details such as the reason for the drop, source/destination addresses, VLAN, and port.

Example event log:

```
2025-06-24:12:34:56.123456|ipsavd|LOG_ERR|DHCPV6GUARD|DHCPv6 server
packet dropped on port:1/1/21 vlan:100. Reason: first fragment missing
complete IPv6 header chain (RFC 7610 compliance)
```

Refer to the **IPSAV event logs** table in the Supportability section for all possible drop log messages and their descriptions.

> **NOTE**
> Event logs are critical for troubleshooting and auditing. Every drop action (including those triggered by extension header or fragmentation checks) is logged for operator visibility.

DHCPv4 snooping commands

Subtopics

DHCP snooping commands

Select a command from the list in the left navigation menu.

Subtopics

dhcp-snooping static-attributes
dhcp-snooping trust
dhcp-snooping verify mac
show dhcp-snooping
show dhcp-snooping binding
show dhcp-snooping statistics

clear dhcp-snooping binding

Syntax

```
clear dhcp-snooping binding {all | ip <IP-ADDR> vlan <VLAN-ID> | port <PORT-
NUM> | vlan <VLAN-ID>}
```

Description

Clears DHCP snooping binding entries.

Parameter	Description
`all`	Specifies that all DHCP binding information is to be cleared.
`ip <IP-ADDR> vlan <VLAN-ID>`	Specifies the IP address and VLAN for which all DHCP binding i nformation is to be cleared.
`port <PORT-NUM>`	Specifies the port number for which all DHCP binding informati on is to be cleared.
`vlan <VLAN-ID>`	Specifies the VLAN for which all DHCP binding information is to be cleared.

Examples

Clearing all DHCP binding information for IP address 192.168.2.4 and VLAN 5:

```
switch(config)# clear dhcp-snooping binding ip 192.168.2.4 vlan 5
```

Clearing all DHCP binding information for port 1/1/1:

```
switch(config)# clear dhcp-snooping binding port 1/1/1
```

Clearing all DHCP binding information for VLAN 10:

```
switch(config)# clear dhcp-snooping binding vlan 10
```

Clearing all DHCP binding information:

```
switch(config)# clear dhcp-snooping binding all
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snooping**. The **ipv4** parameter is deprecated and replaced with **ip**.
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`clear dhcp-snooping statistics`

Syntax

```
clear dhcp-snooping statistics
```

Description

Clears all DHCP snooping statistics.

Examples

Clear all DHCP snooping statistics:

```
switch# clear dhcp-snooping statistics
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snoo ping**.
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`dhcp-snooping`

Syntax

```
dhcp-snooping
no dhcp-snooping
```

Description

Enables DHCP snooping. DHCP snooping is disabled by default. DHCP snooping is not supported on the management interface.

The **no** form of the command disables DHCP snooping, flushing all the IP bindings learned since DHCP snooping was enabled.

Examples

Enabling DHCP snooping:

```
switch(config)# dhcp-snooping
```

Disabling DHCP snooping:

```
switch(config)# no dhcp-snooping
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snoo ping**.
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`dhcp-snooping (in config-vlan context)`

Syntax

```
dhcp-snooping
no dhcp-snooping
```

Description

Enables DHCP snooping for the specified VLAN in the **config-vlan** context. DHCP snooping is disabled by default for all VLANs.

The no form of the command disables DHCP snooping on the specified VLAN, flushing all the IP bindings learned for this VLAN since DHCP snooping was enabled for this VLAN.

Examples

Enabling DHCP snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# dhcp-snooping
switch(config-vlan-100)# exit
switch(config)#
```

Disabling DHCP snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# no dhcp-snooping
switch(config-vlan-100)# exit
switch(config)#
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snooping**.
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	

Command Information

Platforms	Command context	Authority
5420 6200	`config-vlan`	Administrators or local user group members with execution rights for this command.

`dhcp-snooping allow-overwrite-binding`

Syntax

```
dhcp-snooping allow-overwrite-binding
no dhcp-snooping allow-overwrite-binding
```

Description

Allows binding to be overwritten for the same IP address. When enabled, and a DHCP server offers a host an IP address that is already bound to an existing host in the binding table, the existing binding is overwritten for the new host if the new host is successfully able to acquire the same IP address. This overwriting is disabled by default, causing the DHCP server offers to be dropped.

The **no** form of the command disables DHCP snooping overwrite binding.

Examples

Enabling DHCP snooping overwrite binding:

```
switch(config)# dhcp-snooping allow-overwrite-binding
```

Disabling DHCP snooping overwrite binding:

```
switch(config)# no dhcp-snooping allow-overwrite-binding
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snooping**.
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

`dhcp-snooping authorized-server`

Syntax

```
dhcp-snooping authorized-server <IP-ADDR> [vrf <VRF-NAME>]
no dhcp-snooping authorized-server <IP-ADDR> [vrf <VRF-NAME>]
```

Description

Adds an authorized (trusted) DHCP server to a list of authorized servers for use by DHCP snooping. This command can be issued multiple times, adding a maximum of 20 authorized servers per VRF. By default, with an empty list of authorized servers, all DHCP servers are considered to be trusted for DHCP snooping purposes.

> **NOTE**
>
> The **mgmt** VRF cannot be used with this command.

The no form of this command deletes the specified DHCP server from the authorized list.

Parameter	Description
`<IP-ADDR>`	Specifies the IP address of the trusted DHCP server.
`vrf <VRF-NAME>`	Specifies the VRF name. The name must be **default**.

Usage

For authorized server lookup, the VRF is derived from the Switch Virtual Interface (SVI) configured for the incoming VLAN. If the SVI is not configured, the `default` VRF is assumed.

Examples

Adding DHCP servers 192.168.2.2, 192.168.2.3, and 192.168.2.10 to the authorized server list:

```
switch(config)# dhcp-snooping authorized-server 192.168.2.2
switch(config)# dhcp-snooping authorized-server 192.168.2.3 vrf default
switch(config)# dhcp-snooping authorized-server 192.168.2.10 vrf default
```

Removing DHCP server 192.168.2.3 from the authorized server list:

```
switch(config)# no dhcp-snooping authorized-server 192.168.2.3 vrf default
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snoo ping**.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`dhcp-snooping event-log client`

Syntax

```
dhcp-snooping event-log client
no dhcp-snooping event-log client
```

Description

This command enables or disables dhcp-snooping client level event logs that help with client telemetry on a remote management station such as HPE Aruba Networking Central. By default, client level event logs are disabled. The **no** form of this command disables client-level event logs for DHCP snooping after they are enabled. View these logged DHCP snooping events by issuing the command **show events -c dhcp-snooping**.

> **NOTE**
> For additional information on DHCP-related event logging, please refer to the Event Log Message Reference Guide.

Examples

Enabling DHCP client level event logs:

```
switch(config)# dhcp-snooping event-log client
```

Disabling external storage:

```
switch(config)# no dhcp-snooping event-log client
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snoo ping**.
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`dhcp-snooping external-storage`

Syntax

```
dhcp-snooping external-storage volume <VOL-NAME> file <FILE-NAME>
no dhcp-snooping external-storage volume <VOL-NAME> file <FILE-NAME>
```

Description

Configures external storage to be used for backing up IP bindings (used by DHCP snooping) to a file. When configured, the switch stores all the IP bindings in an external storage file so that they are retained after the switch restarts. When the switch restarts, it reads the IP bindings from the configured external storage file to populate its local cache.

> **NOTE**
>
> When both external storage and flash storage are configured to store DHCP snooping IP bindings, the external storage takes priority, and is used exclusively until it becomes unconfigured, at which time flash storage (if configured) is used. Later, if external storage is configured again, flash storage stops and external storage resumes.

The no form of this command disables the saving of IP bindings in an external storage file.

Parameter	Description
`volume <VOL-NAME>`	Specifies the name of the existing external storage volume where the IP bindings file will be saved. Before running the **dhcp - snooping external - storage volume** command, first create the external storage volume using command **external - storage <VOLUME - NAME>**. See External storage commands in the Command-Line Interface Guide .
`file <FILE-NAME>`	Specifies the file name to use for storing IP bindings. Maximum 255 characters.

Configuring IP bindings storage in file **dsnoop_ipbindings** on existing volume **dhcp_snoop**:

```
switch(config)# dhcp-snooping external-storage volume dhcp_snoop file
dsnoop_ipbindings
```

Disabling external storage:

```
switch(config)# no dhcp-snooping external-storage volume dhcp_snoop
```

Disabling external storage when flash storage is also configured (note the message indicating that flash storage will be used):

```
switch(config)# no dhcp-snooping external-storage volume dhcp_snoop
dhcp-snooping will use flash storage to store IP Binding database
switch(config)#
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snooping**.
10.08	Updated example with flash storage information.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

dhcp-snooping flash-storage

Syntax

```
dhcp-snooping flash-storage [delay <DELAY>]
no dhcp-snooping flash-storage [delay <DELAY>]
```

Description

Configures switch flash storage to be used for backing up client IP bindings (used by DHCP snooping). When flash storage is configured (and external storage is not already configured for this purpose), the switch stores the IP bindings in switch flash storage. When the switch restarts, it reads the IP bindings from the switch flash storage to populate its local cache.

Writing the IP bindings to flash storage only occurs after the configured delay and if there has been a change in client IP bindings. Writing is skipped when client IP bindings have not changed since the previous write.

Omitting **delay <DELAY>** sets the default delay of 900 seconds.

> **CAUTION**
>
> To reduce switch flash aging it is recommended that you use external storage (command **dhcp-snooping external-storage**) to backup DHCP snooping IP bindings. Alternatively, consider configuring flash storage with a substantial delay between writes.

> **NOTE**
>
> When both external storage and flash storage are configured to store DHCP snooping IP bindings, the external storage takes priority, and is used exclusively until it becomes unconfigured, at which time flash storage (if configured) is used. Later, if external storage is configured again, flash storage stops and external storage resumes.

The no form of this command disables the saving of IP bindings in flash storage.

Parameter	Description
`delay <DELAY>`	Specifies the delay in seconds between writes (when necessary) to the flash storage, Default: 900. Range: 300 to 86400.

Examples

Configuring switch flash storage for DHCP snooping IP binding storage with a write delay of 1200 seconds:

```
switch(config)# dhcp-snooping flash-storage delay 1200
Warning: Using flash storage reduces switch lifetime. It is recommended to
use an external-storage.
Do you want to continue  (y/n)? y
switch(config)#
```

Unconfiguring usage of switch flash storage for IP bindings :

```
switch(config)# no dhcp-snooping flash-storage
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snooping**.
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

`dhcp-snooping max-bindings`

Syntax

```
dhcp-snooping max-bindings <MAX-BINDINGS>
no dhcp-snooping max-bindings <MAX-BINDINGS>
```

Description

Sets the maximum number of DHCP bindings allowed on the selected interface. For all interfaces on which this command is not run, the default max binding is the maximum value of the range.

The no form of the command reverts max bindings for the selected interface to its default.

Parameter	Description
<MAX-BINDINGS>	Specifies the maximum number of DHCP bindings. Range 1 to 1024.

Examples

Set the DHCP max bindings to 256 on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# dhcp-snooping max-bindings 256
switch(config-if)# exit
switch(config)#
```

Revert DHCP max bindings to its default on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no dhcp-snooping max-bindings 256
switch(config-if)# exit
switch(config)#
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snooping**.
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution rights for this command.

dhcp-snooping option 82

Syntax

```
dhcp-snooping option 82 [remote-id {mac | subnet-ip | mgmt-ip}][untrusted-
policy {drop | keep | replace}]
no dhcp-snooping option 82 [remote-id {mac | subnet-ip | mgmt-ip}]
[untrusted-policy {drop | keep | replace}]
```

Description

Configures the addition of option 82 DHCP relay information to DHCP client packets that are being forwarded on trusted ports. DHCP relay is enabled by default.

In the switch default state and when this command is entered without parameters (**dhcp-snooping option 82**), this default configuration is used:

```
dhcp-snooping option 82 remote-id mac untrusted-policy drop
```

When **remote-id** is omitted, its default (**mac**) is used. When **untrusted-policy** is omitted, its default (**drop**) is used.

The no form of this command disables DHCP snooping option 82.

Parameter	Description
`remote-id`	Specifies what address to use as the remote ID for the `replace` option of `untrusted-policy`. Specify one of these address types:
`mac`	The default. Uses the switch MAC address as the remote ID.
`subnet-ip`	Uses the IP address of the client VLAN as the remote ID.
`untrusted-policy`	Specifies what action to take for DHCP packets (with option 82) that are received on untrusted ports. Specify one of these actions:
`drop`	The default. Drop DHCP packets (with option 82) without forwarding them.
`keep`	Forward DHCP packets (with option 82).
`replace`	Replace the option 82 information in the DHCP packets with whatever is set for **remote - id** (one of: **mac**, **subnet - ip**, or **mgmt - ip**) and forward the packets.

Examples

Configuring DHCP snooping option 82 with the keep action:

```
switch(config)# dhcp-snooping option 82 untrusted-policy keep
```

Configuring DHCP snooping option 82 with `mgmt-ip` as the `remote-id` and the `replace` action:

```
switch(config)# dhcp-snooping option 82 remote-id mgmt-ip untrusted-policy
replace
```

Disabling DHCP snooping option 82:

```
switch(config)# no dhcp-snooping option 82 untrusted-policy keep
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snooping**.
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

`dhcp-snooping static-attributes`

Syntax

```
dhcp-snooping static-attributes
no dhcp-snooping static-attributes
```

Description

Enables storage of static attributes provided to the DHCP client by DHCP server during DHCP packet exchange. Disabled by default. When enabled, the following attributes are stored in OVSDB along with the client IP binding entry:

1. Name server IP addresses: DNS server IPs provided by the DHCP server to the client. Maximum: 3 per client.
2. Default gateway IP address: Router IP addresses provided by DHCP server to the client. Maximum: 3 per client.
3. Server IP address: IP address of the DHCP server that leased the IP to the client.

The **no** form of the command disables storing of client static attributes. After disabling, existing client static attributes will be flushed.

Examples

Enabling the storage of DHCP snooping static attributes:

```
switch(config)# dhcp-snooping static-attributes
```

Disabling the storage of DHCP snooping static attributes:

```
switch(config)# no dhcp-snooping static-attributes
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snooping**.
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

dhcp-snooping trust

Syntax

```
dhcp-snooping trust
no dhcp-snooping trust
```

Description

Enables DHCP snooping trust on the selected port. Only server packets received on trusted ports are forwarded. All the ports are untrusted by default.

The **no** form of the command disables DHCP snooping trust on the selected port.

Examples

Enabling DHCP snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# dhcp-snooping trust
switch(config-if)# exit
switch(config)#
```

Disabling DHCP snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# no dhcp-snooping trust
switch(config-if)# exit
switch(config)#
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snoo ping**.
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`dhcp-snooping verify mac`

Syntax

```
dhcp-snooping verify mac
no dhcp-snooping verify mac
```

Description

This command enables verification of the hardware address field in DHCP client packets. When enabled, the DHCP client hardware address field and the source MAC address must be the same for packets received on untrusted ports or else the packet is dropped. This DHCP snooping MAC verification is enabled by default.

The **no** form of the command disables DHCP snooping MAC verification.

Examples

Enabling DHCP snooping MAC verification:

```
switch(config)# dhcp-snooping verify mac
```

Disabling DHCP snooping MAC verification:

```
switch(config)# no dhcp-snooping verify mac
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snoo ping**.
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.

Release	Modification
10.07 or earlier	

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`show dhcp-snooping`

Syntax

```
show dhcp-snooping
```

Description

Shows the DHCP snooping configuration.

Examples

Showing the DHCP snooping configuration:

```
switch# show dhcp-snooping
dhcp-snooping Information
dhcp-snooping                  : Yes              Verify MAC Address    : Yes
Allow Overwrite Binding        : No               Enabled VLANs         :
1-100
Static Attributes              : Yes
Client Event Logs              : Yes
Option 82 Configurations
Untrusted Policy               : replace           Insertion             : Yes
Option 82 Remote-id            : mac
External Storage Information
Volume Name     : ipbinding
File Name       : ipv4Bindings
Inactive Since : 01:23:20 09/10/2021
Error           : File Write Failure
Flash Storage Information
```

```
File Write Delay : 300 seconds
Active Storage    : External
Authorized Server Configurations
VRF                             Authorized Servers
------------                  ----------------------
default                        1.1.10.3
default                        10.10.10.1
default                        10.10.10.56
default                        200.10.10.3
green                          1.1.10.3
green                          1.10.10.3
green                          10.10.100.3
red                            192.168.122.53
red                            192.168.122.121
Port Information
                      Max         Static      Dynamic
Port        Trust     Bindings    Bindings    Bindings
--------    -----      --------   --------      --------
1/1/2       Yes        5000        50             0
1/1/3       Yes        8192        0              0
1/1/5       Yes        8192        0              22
1/1/16      No         100         0              0
10/10/10    No         8100        320            200
lag120      No         512         0              0
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snooping**.
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.08	Updated example with flash storage information.
10.07 or earlier	

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show dhcp-snooping binding`

Syntax

```
show dhcp-snooping binding
```

Description

Shows the DHCP snooping binding configuration.

Parameter	Description
`detail`	Shows detailed information for active IP bindings on the system.

Examples

Showing the DHCP snooping binding configuration:

```
switch(config)# show dhcp-snooping binding
  MacAddress          IP                VLAN  Interface  Time-Left
  -----------------  ----------------  ----  ---------  ---------
  aa:b1:c1:dd:ee:ff  10.2.3.4          1     1/1/2      582
  aa:b2:c2:dd:ee:ff  10.2.3.5          1     1/1/2      584
```

Showing detailed information for active IP bindings:

```
switch(config)# show dhcp-snooping binding detail
VLAN Id : 2, MAC : 00:50:56:96:74:46
  IP               Interface  Time-Left
  ---------------  ---------  -------------------
  100.1.2.100      1/1/23     194
  Static Attributes:
  Default Router  : 100.1.2.1, 192.1.1.1, 1.1.1.2
```

```
  Server IP        : 10.1.84.2
  Name Servers     : 192.1.1.2, 2.2.2.2, 1.1.1.1
VLAN Id : 3, MAC : 00:50:56:96:e5:8e
  IP               Interface  Time-Left
  ---------------  ---------  --------------------
  100.1.3.100      2/1/22     145
  Static Attributes:
  Default Router   : 100.1.3.1, 192.1.1.1, 1.1.1.2
  Server IP        : 10.1.84.2
  Name Servers     : 192.1.1.2, 2.2.2.2, 1.1.1.1
VLAN Id : 3, MAC : 00:11:01:00:00:03
  IP               Interface  Time-Left
  ---------------  ---------  --------------------
  100.1.3.99       2/1/24     137
  Static Attributes:
  Default Router   : 100.1.3.1, 192.1.1.1, 1.1.1.2
  Server IP        : 10.1.84.2
  Name Servers     :192.168.0.1, 192.168.1.1, 192.168.2.1
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snoo ping**.
10.10	Detail parameter added.
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show dhcp-snooping statistics`

Syntax

```
show dhcp-snooping statistics
```

Description

Shows the DHCP snooping statistics.

Examples

Showing the DHCP snooping statistics:

```
switch(config)# show dhcp-snooping statistics
 Packet-Type  Action   Reason                         Count
 -----------  -------  -----------------------------  ---------
 server       forward  from trusted port              5425
 client       forward  to trusted port                3895
 server       drop     received on untrusted port     117
 server       drop     unauthorized server            214
 client       drop     destination on untrusted port  78
 client       drop     untrusted option 82 field      85
 client       drop     bad DHCP release request       0
 client       drop     failed verify MAC check        5
 client       drop     failed on max-binding limit    15
```

Command History

Release	Modification
10.14	The **dhcpv4 - snooping** keyword is deprecated and replaced with **dhcp - snooping**.
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

DHCPv6 snooping commands

Subtopics

- [DHCPv6 snooping commands]

DHCPv6 snooping commands

Select a command from the list in the left navigation menu.

Subtopics

- clear dhcpv6-snooping binding
- clear dhcpv6-snooping guard-policy statistics
- clear dhcpv6-snooping statistics
- dhcpv6-snooping
- dhcpv6-snooping allow-ipv6-ext-headers
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- dhcpv6-snooping (in config-vlan context)
- dhcpv6-snooping authorized-server
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show dhcpv6-snooping guard-policy vlan
show dhcpv6-snooping
show dhcpv6-snooping binding
show dhcpv6-snooping statistics

clear dhcpv6-snooping binding

Syntax

```
clear dhcpv6-snooping binding {all | ip <IPV6-ADDR> vlan <VLAN-ID> |
interface <IFNAME> | vlan <VLAN-ID>}
```

Description

Clears DHCPv6 snooping binding entries.

Parameter	Description
`all`	Specifies that all DHCPv6 binding information is to be cleared.
`ip <IPV6-ADDR> vlan <VLAN-ID>`	Specifies the IPv6 address and VLAN for which all DHCPv6 bin ding information is to be cleared.
`interface <IFNAME>`	Specifies the interface for which all DHCPv6 binding informatio n is to be cleared.
`vlan <VLAN-ID>`	Specifies the VLAN for which all DHCPv6 binding information is to be cleared. Range: 1 to 4094.

Examples

Clearing all DHCPv6 binding information for 5000::1 vlan 1:

```
switch(config)# clear dhcpv6-snooping binding ip 5000::1 vlan 1
```

Clearing all DHCPv6 binding information for interface 1/1/10:

```
switch(config)# clear dhcpv6-snooping binding interface 1/1/10
```

Clearing all DHCPv6 binding information for VLAN 10:

```
switch(config)# clear dhcpv6-snooping binding vlan 10
```

Clearing all DHCPv6 binding information:

```
switch(config)# clear dhcpv6-snooping binding all
```

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`clear dhcpv6-snooping guard-policy statistics`

Syntax

```
clear dhcpv6-snooping guard-policy statistics [vlan <VLAN-ID> | interface
<INTERFACE-NAME>]
```

Description

Clears all DHCPv6 snooping guard policy statistics from the specified VLAN or interface.

Parameter	Description
`<VLAN-ID>`	Specifies the VLAN ID. Range: 1 - 4094.
	Specifies the interface name.

Parameter	Description
<INTERFACE-NAME>	

Examples

Clearing all DHCPv6 snooping guard policy statistics from VLAN 100:

```
switch# clear dhcpv6-snooping guard-policy statistics vlan 100
```

Clearing all DHCPv6 snooping guard policy statistics from interface 1/1/10:

```
switch# clear dhcpv6-snooping guard-policy statistics interface 1/1/10
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`clear dhcpv6-snooping statistics`

Syntax

```
clear dhcpv6-snooping statistics
```

Description

Clears all DHCPv6 snooping statistics.

Examples

Clear all DHCPv6 snooping statistics:

```
switch# clear dhcpv6-snooping statistics
```

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

dhcpv6-snooping

Syntax

```
dhcpv6-snooping
no dhcpv6-snooping
```

Description

Enables DHCPv6 snooping. DHCPv6 snooping is disabled by default. DHCPv6 snooping is not supported on the management interface.

The no form of the command disables DHCPv6 snooping, flushing all the IP bindings learned since DHCPv6 snooping was enabled.

Examples

Enabling DHCPv6 snooping:

```
switch(config)# dhcpv6-snooping
```

Disabling DHCPv6 snooping:

```
switch(config)# no dhcpv6-snooping
```

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

dhcpv6-snooping allow-ipv6-ext-headers

Syntax

```
[no] dhcpv6-snooping allow-ipv6-ext-headers
```

Description

This command enables or disables forwarding of DHCPv6 packets containing any IPv6 extension headers. By default, such packets are **parsed and dropped** for DHCPv6 Guard policy (RFC 7610 compliance). When enabled, these packets are **forwarded** instead of dropped.

Examples

Enabling forwarding:

```
switch# configure terminal
switch(config)# dhcpv6-snooping allow-ipv6-ext-headers
```

Disabling forwarding (default as per RFC 7610):

```
switch# configure terminal
switch(config)# no dhcpv6-snooping allow-ipv6-ext-headers
```

Command History

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution righ ts for this command.

dhcpv6-snooping guard-policy

Syntax

```
dhcpv6-snooping guard-policy <POLICY-NAME>
no dhcpv6-snooping guard-policy <POLICY-NAME>
```

Description

Configures a DHCPv6 snooping guard policy with the given name and enters the guard policy configuration context.

The no form of the command disables the specified guard policy.

Parameter	Description
<POLICY-NAME>	Specifies the name of the DHCPv6 snooping guard policy. Maxi mum length: 64.

Examples

Creating the DHCPv6 snooping guard policy name pol1:

```
switch(config)# dhcpv6-snooping guard-policy pol1
switch(config-guard-policy-pol1)#
```

Deleting the DHCPv6 snooping guard policy named pol1:

```
switch(config)# no dhcpv6-snooping guard-policy pol1
```

Creating the DHCPv6 snooping guard policy name pol1 on interface 1/1/1:

> **NOTE**
>
> The DHCPv6 snooping guard policy applied on the port takes priority over the policy applied over VLAN.

```
switch(config)# interface 1/1/1
switch(config-if)# dhcpv6-snooping guard-policy pol1
```

Creating the DHCPv6 snooping guard policy name pol1 on a VLAN:

```
switch(config)# vlan 100
switch(config-vlan-100)# dhcpv6-snooping guard-policy pol1
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority

5420
6200
	`config`
	`config-if`
	`config-dhcpv6-guard-policy`
	`config-vlan-<VLAN-ID>`
		Administrators or local user group members with execution rights for this command.

`dhcpv6-snooping (in config-vlan context)`

Syntax

```
dhcpv6-snooping
no dhcpv6-snooping
```

Description

Enables DHCPv6 snooping in the `config-vlan` context. DHCPv6 snooping is disabled by default for all VLANs.

The no form of the command disables DHCPv6 snooping on the specified VLAN, flushing all the IPv6 bindings learned for this VLAN since DHCPv6 snooping was enabled for this VLAN.

Examples

Enabling DHCPv6 snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# dhcpv6-snooping
switch(config-vlan-100)# exit
switch(config)#
```

Disabling DHCPv6 snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# no dhcpv6-snooping
switch(config-vlan-100)# exit
switch(config)#
```

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	

Command Information

Platforms	Command context	Authority
5420 6200	`config-vlan`	Administrators or local user group members with execution righ ts for this command.

dhcpv6-snooping authorized-server

Syntax

```
dhcpv6-snooping authorized-server <IPV6-ADDR> [vrf <VRF-NAME>]
no dhcpv6-snooping authorized-server <IPV6-ADDR> [vrf <VRF-NAME>]
```

Description

Adds an authorized (trusted) DHCPv6 server to a list of authorized servers for use by DHCPv6 snooping. This command can be issued multiple times, adding a maximum of 20 authorized servers per VRF. By default, with an empty list of authorized servers, all DHCPv6 servers are considered to be trusted for DHCPv6 snooping purposes.

> **NOTE**
>
> The `mgmt` VRF cannot be used with this command.

> **NOTE**
>
> Configure the link local IPv6 address instead of global IPv6 address of the DHCPv6 server as the authorized-server. For example:
>
> ```
> switch(config)# dhcpv6-snooping authorized-server
> fe80::2ca4:fa40:d4cd:bc2f
> ```

The no form of this command deletes the specified DHCPv6 server from the authorized list.

Parameter	Description
`<IPV6-ADDR>`	Specifies the IPv6 address of the trusted DHCPv6 server.
`vrf <VRF-NAME>`	Specifies the VRF name. The name must be `default`.

Usage

For authorized server lookup, the VRF is derived from the Switch Virtual Interface (SVI) configured for the incoming VLAN. If the SVI is not configured, the `default` VRF is assumed.

Examples

Adding DHCP servers ABCD:5ACD::2000, and ABCD:5ACD::2010 to the authorized server list:

```
switch(config)# dhcpv6-snooping authorized-server ABCD:5ACD::2000 vrf
default
switch(config)# dhcpv6-snooping authorized-server ABCD:5ACD::2010 vrf
default
```

Removing DHCP server ABCD:5ACD::2000 from the authorized server list:

```
switch(config)# no dhcpv6-snooping authorized-server ABCD:5ACD::2000 vrf
default
```

Command History

Release	Modification
10.07 or earlier	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

dhcpv6-snooping drop-unknown-ipv6-ext-headers

Syntax

```
[no] dhcpv6-snooping drop-unknown-ipv6-ext-headers
```

Description

This command enables or disables dropping of DHCPv6 packets containing **unknown or unsupported IPv6 extension headers** (i.e., Next Header values not defined in RFC 8200 or not supported by the implementation). By default, such packets are forwarded. When this command is configured, these packets are **dropped** and logged (RFC 7610 compliance).

Examples

Dropping packets with unknown IPv6 extension headers:

```
switch# configure terminal
switch(config)# dhcpv6-snooping drop-unknown-ipv6-ext-headers
```

Forwarding packets with unknown IPv6 extension headers (default):

```
switch# configure terminal
switch(config)# no dhcpv6-snooping drop-unknown-ipv6-ext-headers
```

Command History

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution righ ts for this command.

dhcpv6-snooping event-log client

Syntax

```
dhcpv6-snooping event-log client
no dhcpv6-snooping event-log client
```

Description

This command enables or disables DHCPv6 snooping client level event logs that help with client telemetry on a remote management station such as HPE Aruba Networking Central. By default, client level event logs are disabled. The **no** form of this command disables client-level event logs for DHCPv6 snooping after they are enabled. View these logged DHCPv6 snooping events by issuing the command `show events -c dhcpv6-snooping`.

> **NOTE**
>
> For additional information on DHCP-related event logging, please refer to the Event Log Message Reference Guide.

Examples

Enabling DHCPv6 client level event logs:

```
switch(config)# # dhcpv6-snooping event-log client
```

Disabling external storage:

```
witch(config)# # no dhcpv6-snooping event-log client
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`dhcpv6-snooping external-storage`

Syntax

```
dhcpv6-snooping external-storage volume <VOL-NAME> file <FILE-NAME>
no dhcpv6-snooping external-storage volume <VOL-NAME> file <FILE-NAME>
```

Description

Configures external storage to be used for backing up IPv6 bindings (used by DHCPv6 snooping) to a file. When configured, the switch stores all the IP bindings in an external storage file so that they are retained after the switch restarts. When the switch restarts, it reads the IPv6 bindings from the configured external storage file to populate its local cache.

> **NOTE**
>
> When both external storage and flash storage are configured to store DHCP snooping IP bindings, the external storage takes priority, and is used exclusively until it becomes unconfigured, at which time flash storage (if configured) is used. Later, if external storage is configured again, flash storage stops and external storage resumes.

The no form of this command disables the saving of IPv6 bindings in an external storage file.

Parameter	Description
`volume` *<VOL-NAME>*	Specifies the name of the existing external storage volume where the IPv6 bindings file will be saved. Before running the `dhcpv6-snooping external-storage volume` command, first create the external storage volume using command `external-storage <VOLUME-NAME>`. See External storage commands in the Command-Line Interface Guide .
`file` *<FILE-NAME>*	Specifies the file name to use for storing IPv6 bindings. Maximum 255 characters.

Examples

Configuring IPv6 bindings storage in file `ipv6Bindings` on existing volume `dhcp_snoop`:

```
switch(config)# dhcpv6-snooping external-storage volume dhcp_snoop file
ipv6Bindings
```

Disabling external storage:

```
switch(config)# no dhcpv6-snooping external-storage volume dhcp_snoop
```

Disabling external storage when flash storage is also configured (note the message indicating that flash storage will be used):

```
switch(config)#  no dhcpv6-snooping external-storage volume dhcp_snoop
DHCPv6-Snooping will use flash storage to store IP Binding database
switch(config)#
```

Command History

Release	Modification
10.08	Updated example with flash storage information.
10.07 or earlier	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`dhcpv6-snooping flash-storage`

Syntax

```
dhcpv6-snooping flash-storage [delay <DELAY>]
no dhcpv6-snooping flash-storage [delay <DELAY>]
```

Description

Configures switch flash storage to be used for backing up client IP bindings (used by DHCPv6 snooping). When flash storage is configured (and external storage is not already configured for this purpose), the switch stores the IP bindings in switch flash storage. When the switch restarts, it reads the IP bindings from the switch flash storage to populate its local cache.

Writing the IP bindings to flash storage only occurs after the configured delay and if there has been a change in client IP bindings. Writing is skipped when client IP bindings have not changed since the previous write.

Omitting `delay <DELAY>` sets the default delay of 900 seconds.

> **CAUTION**
>
> To reduce switch flash aging it is recommended that you use external storage (command `dhcpv6-snooping external-storage`) to backup DHCP snooping IP bindings. Alternatively, consider configuring flash storage with a substantial delay between writes.

> **NOTE**
>
> When both external storage and flash storage are configured to store DHCP snooping IP bindings, the external storage takes priority, and is used exclusively until it becomes unconfigured, at which time flash storage (if configured) is used. Later, if external storage is configured again, flash storage stops and external storage resumes.

The no form of this command disables the saving of IP bindings in flash storage.

Parameter	Description
`delay <DELAY>`	Specifies the delay in seconds between writes (when necessary) to the flash storage, Default: 900. Range: 300 to 86400.

Examples

Configuring switch flash storage for DHCP snooping IP binding storage with a write delay of 1200 seconds:

```
switch(config)# dhcpv6-snooping flash-storage delay 1200
Warning: Using flash storage reduces switch lifetime. It is recommended to
use an external-storage.
Do you want to continue  (y/n)? y
switch(config)#
```

Unconfiguring usage of switch flash storage for IP bindings :

```
switch(config)# no dhcpv6-snooping flash-storage
```

Command History

Release	Modification
10.07	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`dhcpv6-snooping max-bindings`

Syntax

```
dhcpv6-snooping max-bindings <MAX-BINDINGS>
no dhcpv6-snooping max-bindings <MAX-BINDINGS>
```

Description

Sets the maximum number of DHCPv6 bindings allowed on the selected interface. For all interfaces on which this command is not run, the default max binding is the maximum value of the range.

The no form of the command reverts max bindings for the selected interface to its default.

Parameter	Description
<MAX-BINDINGS>	Specifies the maximum number of DHCP bindings. Range 1 to 1024.

Examples

Set the DHCPv6 max bindings to 256 on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# dhcpv6-snooping max-bindings 256
switch(config-if)# exit
switch(config)#
```

Revert DHCPv6 max bindings to its default on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no dhcpv6-snooping max-bindings 256
switch(config-if)# exit
switch(config)#
```

Command History

Release	Modification
10.07	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`dhcpv6-snooping trust`

Syntax

```
dhcpv6-snooping trust
no dhcpv6-snooping trust
```

Description

Enables DHCPv6 snooping trust on the selected interface. Only server packets received on trusted interfaces are forwarded. All the interfaces are untrusted by default.

The no form of the command disables DHCPv6 snooping trust on the selected interface.

```
config-if
```

Examples

Enabling DHCPv6 snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# dhcpv6-snooping trust
switch(config-if)# exit
switch(config)#
```

Disabling DHCPv6 snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# no dhcpv6-snooping trust
switch(config-if)# exit
switch(config)#
```

Command History

Release	Modification
10.07	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`match client prefix-list`

Syntax

```
match client prefix-list <PREFIX-LIST-NAME>
no match client prefix-list <PREFIX-LIST-NAME>
```

Description

Configures a prefix-list for the DHCPv6 snooping guard policy enabling the policy to allow the assigned IPv6 addresses within a specific prefix range.

The **no** form of the command removes a prefix list from the DHCPv6 snooping guard policy.

Parameter	Description
<PREFIX-LIST-NAME>	Specifies the name of the IPv6 prefix list.

Examples

Adding a prefix list named pref1 to the pol1 DHCPv6 snooping guard policy:

```
switch(config)# ipv6 prefix-list pref1 permit 2001:db8::/64 le 128
switch(config)# dhcpv6-snooping guard-policy pol1
switch(config-dhcpv6-guard-policy)# match client prefix-list pref1
```

Deleting the prefix list named prf1 from the pol1 DHCPv6 snooping guard policy:

```
switch(config)# dhcpv6-snooping guard-policy pol1
switch(config-dhcpv6-guard-policy)# no match client prefix-list <ipv6-
prefix-list-name>
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcpv6-guard-policy`	Administrators or local user group members with execution righ ts for this command.

`match server access-list`

Syntax

```
match server access-list <ACL-NAME>
no match server access-list <ACL-NAME>
```

Description

Configures an access list to a DHCPv6 snooping guard policy, enabling the DHCPv6 snooping guard policy to allow or deny the specific DHCP server to assign an IPv6 address. If no filters are applied, DHCP server traffic from any source IP address is allowed in the trusted port.

The **no** form of the command removes the specified access list from the DHCPv6 snooping guard policy.

Parameter	Description
<ACL-NAME>	Specifies the name of the IPv6 access list to be matched.

Examples

Creating an access-list acl1 on DHCPv6 snooping guard policy pol1 :

```
switch(config)# dhcpv6-snooping guard-policy pol1
switch(config-dhcpv6-guard-policy)# match server access-list acl1
```

Deleting the access list acl1 from the DHCPv6 snooping guard policy pol1:

```
switch(config)# dhcpv6-snooping guard-policy pol1
switch(config-dhcpv6-guard-policy)# no match server access-list acl1
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcpv6-guard-policy`	Administrators or local user group members with execution righ ts for this command.

`preference`

Syntax

```
preference [minimum | maximum ] <VALUE>
no preference
```

Description

Enables a DHCPv6 snooping guard policy to allow or deny the DHCPv6 servers in the specified server preference range. If not configured the minimum preference is set to 0 and maximum preference is set to 255.

The **no** form of the command removes the server preference limits on the specified DHCPv6 snooping guard policy.

Parameter	Description
`minimum <VALUE>`	Specifies the minimum value for the server preference range. Range: 1-255.
`maximum <VALUE>`	Specifies the maximum value for the server preference range. Range: 1-255.

Examples

Setting the minimum and maximum server preference range to 6-250 on DHCPv6 snooping guard policy pol1:

```
switch(config)# dhcpv6-snooping guard-policy pol1
switch(config-dhcpv6-guard-policy)# preference min 6
switch(config-dhcpv6-guard-policy)# preference max 250
```

Disabling the server preference range on DHCPv6 snooping guard policy pol1:

```
switch(config)# dhcpv6-snooping guard-policy pol1
switch(config-dhcpv6-guard-policy)# no preference
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-dhcpv6-guard-policy`	Administrators or local user group members with execution righ ts for this command.

show dhcpv6-snooping guard-policy

Syntax

```
show dhcpv6-snooping guard-policy[<POLICY_NAME>]
```

Description

Shows the DHCPv6 snooping guard policy configuration.

Parameter	Description
<POLICY-NAME>	Specifies the DHCPv6 snooping guard policy for which the infor mation is displayed.

Examples

Showing the DHCPv6 snooping guard policy configuration:

```
switch# show dhcpv6-snooping guard-policy
 DHCPv6-Snooping guard-policy Information
    DHCPV6 Guard Policy name     : POL1
    Attached Access List         : ACL1
    Attached Prefix List         : PRF1
    Preference Range             : 0-255
    Applied on VLAN              : 5,7
    Applied on Port
    DHCPV6 Guard Policy name     : POL2
    Attached Access List         : ACL2
    Attached Prefix List         : PRF2
    Preference Range             : 2-20
```

```
    Applied on VLAN
    Applied on Port              : 1/1/1, 1/1/2
    DHCPV6 Guard Policy name     : POL3
    Attached Access List         : ACL3
    Attached Prefix List         : PRF3
    Preference Range             : 3-60
    Applied on VLAN              : 4,6
    Applied on Port
```

Showing the DHCPv6 snooping guard policy configuration for the policy named POLICY_NAME1:

```
switch# show dhcpv6-snooping guard-policy POLICY_NAME1
  DHCPv6-Snooping guard-policy Information
  ========================
    DHCPV6 Guard Policy name     : POLICY_NAME1
    Attached Access List         : ACL1
    Attached Prefix List         : PRF1
    Preference Range             : 0-255
        vsx-sync
    Applied on VLAN              : 5,7
    Applied on Port              : 1/1/1, 1/1/2
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show dhcpv6-snooping guard-policy interface

Syntax

```
show dhcpv6-snooping guard-policy  [interface <INTERFACE-NAME>]
```

Description

Shows the DHCPv6 snooping guard policy configuration and statistics for the specified interface.

Parameter	Description
`<INTERFACE-NAME>`	Specifies the interface name for which the DHCPv6 guard counter information is displayed.

Examples

Showing the DHCPv6 snooping guard policy configuration and statistics for interface 1/1/1:

```
switch# show dhcpv6-snooping guard-policy int 1/1/1
 DHCPv6 Guard Policy  Applied    : pol1
  DHCPv6 Guard Policy Counters
  ==========================
 DHCPv6 Packets Received          : 20
 DHCPv6 Packets Forwarded         : 5
 DHCPv6 Packets Dropped           : 15 [Total]
                                Access list error        [7]
                                Prefix list error        [8]
                                Server preference error  [0]
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show dhcpv6-snooping guard-policy vlan

Syntax

```
show dhcpv6-snooping guard-policy [vlan <VLAN-ID>]
```

Description

Shows the DHCPv6 snooping guard policy configuration and statistics for the specified VLAN.

Parameter	Description
<VLAN-ID>	Specifies the VLAN ID for which the DHCPv6 guard counter in formation is displayed.

Examples

Showing the DHCPv6 snooping guard policy configuration and statistics for VLAN 100:

```
switch# show dhcpv6-snooping guard-policy vlan 2
 DHCPv6 Guard Policy  Applied    : pol1
  DHCPv6 Guard Policy Counters
  ==========================
 DHCPv6 Packets Received         : 20
 DHCPv6 Packets Forwarded        : 5
 DHCPv6 Packets Dropped          : 15 [Total]
                                Access list error        [0]
                                Prefix list error        [8]
                                Server preference error  [7]
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show dhcpv6-snooping`

Syntax

```
show dhcpv6-snooping
```

Description

Shows the DHCPv6 snooping configuration.

Examples

Showing the DHCPv6 snooping configuration:

```
switch# show  dhcpv6-snooping
 DHCPv6-Snooping Information
  DHCPv6-Snooping      : Yes   Enabled VLANs        : 1,5,7,100-110
  Trusted Port Bindings Enabled VLANs :
  Client Event Logs                   : Yes
External Storage Information
   Volume Name           : dhcp_snoop
   File Name             : ip_binding
   Inactive Since        : 01:23:20 09/10/2021
   Error                 : Failed to write external storage
Flash Storage Information
     File Write Delay       : 300 seconds
     Active Storage         : External
Authorized Server Configurations
VRF                                          Authorized Servers
------------                                ------------------
default
2001:0db8:85a3:0000:0000:8a2e:0370:7334
default                                      2002::2
default                                      2004::1
```

```
red                                                2002::1
red                                                2002::2
red                                                2002::9
green                                              5000::1
green                                              5000::2
green                                              5000::3
green                                              5000::7
green                                              5000::8
Port Information
                         Max       Static      Dynamic
Port           Trust     Bindings  Bindings    Bindings
--------       -----     --------   --------  --------
1/1/2          Yes       0         0         0
1/1/3          Yes       0         3         0
1/1/5          Yes       0         22        0
1/1/16         No        256       0         20
10/10/10       No        256       12        7
lag120         No        256       3         0
```

Command History

Release	Modification
10.08	Updated example with flash storage information.
10.07 or earlier	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show dhcpv6-snooping binding`

Syntax

```
show dhcpv6-snooping binding
```

Description

Shows the DHCPv6 snooping binding configuration.

Examples

Showing the DHCPv6 snooping binding configuration:

```
switch# show dhcpv6-snooping binding
 IP Binding Information
 ======================
 MAC-ADDRESS        IPV6-ADDRESS                                 VLAN
INTERFACE   TIME-LEFT
 ----------------  ----------------------------------------  ----
---------  ----------
 00:50:56:96:e4:cf aaaa:bbbb:cccc:dddd:eeee:1234:5678:abcd      1
1/1/1         584
 00:50:56:96:04:4d 1000::3                                    134
1/1/2         435
 00:50:56:96:d8:3d 2000:1000::4                              2002
lag123       21234
```

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

show dhcpv6-snooping statistics

Syntax

```
show dhcpv6-snooping statistics
```

Description

Shows the DHCPv6 snooping statistics.

Examples

Showing the DHCPv6 snooping statistics:

```
switch(config)# show dhcpv6-snooping statistics
 Packet-Type  Action   Reason                         Count
 -----------  -------  -----------------------------  ---------
 server       forward  from trusted port              12
 client       forward  to trusted port                20
 server       drop     received on untrusted port     5
 server       drop     unauthorized server            4
 client       drop     destination on untrusted port  2
 client       drop     bad DHCP release request       5
 server       drop     relay reply on untrusted port  2
 client       drop     failed on max-binding limit    5
```

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Troubleshooting

DHCP client not receiving IP

1. Trusted port configuration
   Check if the server reachable port is configured as trusted.

2. Validate Packet drop counters
   Use `show dhcpv4-snooping statistics` and `show dhcpv6-snooping statistics` commands to view the packet drop counters. This helps to identify a possible reason for the termination of DHCP packet exchange between client and server. A few possible reasons are listed below:

   - Maximum binding limit reached

     - System-wide limit—The maximum binding limit is shared between DHCPv4-Snooping, DHCPv6-Snooping, ND-Snooping, and IP-Source-Bindings. On reaching the limit, DHCP client requests get dropped. To check the limit, use the `show capacities ip-bindings` command.

     - Per port limit: This is a per port configurable value for a protocol. On reaching the limit, the subsequent client requests are dropped.

   - Conflicts with IP-Source-Bindings—User-configured IP-Source-Bindings are given priority over dynamically learned bindings. Validate if the dynamic client is conflicting with any of the existing IP-Source-Binding.

   - Authorized server check failures—Authorized server configuration is global. When one or more authorized servers are set up, all server replies should come from one of the authorized server IP addresses.

   - CoPP packet drops—If the rate of the DHCP packets entering is higher than the configured value, excessive packets are dropped by CoPP. Check CoPP statistics for the DHCP class and configure the value accordingly. Higher CoPP values could cause other protocols to misbehave depending on the platform.

3. Server configuration and reachability
   Validate the following to ensure that the DHCP Server is reachable and configured correctly.

   a. Check that the server is reachable over one of the trusted ports.

   b. Check the server logs to understand the DHCP exchange status between client and server.

   c. Check if DHCP Server pools are exhausted.

4. Option-82 configuration issues (Specific to DHCP Snooping IPv4)

   a. This configuration usually arises when DHCP Snooping and Relay are configured on the same VLAN. Option 82 at DHCP Snooping IPv4 should be disabled when configured for Inter-VRF DHCP-Relay. Use `no dhcpv4-snooping option 82` command to disable the functionality.

   b. Option 82 is enabled by default, and the drop policy is set. If client packets are received with option 82, then they are dropped with this configuration.

5. Server and client are connected to same the VLAN

   a. Check if the server is reachable on the same VLAN as the client.

   b. If the server is on a different L3 network, configure DHCP-Relay to forward the packets between the client and server.

6. Authorized server configurations—When configured, the server assigning IP address should be one of the IP addresses from the authorized server list.
   In DHCP snooping IPv6, if DHCP Relay is between a test device and a server, the authorised server configuration should include a link to the local IP address of the relay interface. This is because, while forwarding the server's response to the client, it will stamp a link to the local IP address as the source IP of the packet.

IP-Binding mismatch in VSX topologies

1. Both primary and secondary VSX peers should have the same clock value. If not, there is a possibility of inconsistent ip-bindings that could result in ip-binding sync in a loop. There is no straight-forward way to check the inconsistencies but by looking at the number of bindings and debug log analysis

2. Ensure the ISL state is up and operational.

DHCP client IP traffic is dropped at data path

1. This can happen when IP-Source-Lockdown is used along with DHCP snooping. Ensure IP-Bindings are learned for clients that are experiencing traffic failure issues

2. Check the hardware programming state of the IP-Binding entry by IP-Source-Lockdown. Use `show ipv4 source-lockdown bindings` or `show ipv6 source-lockdown bindings` commands.

DHCP Options

A number of deployments provide access to WAN or public network only through a proxy server for the network security. This section provides information about the various methods of HTTP proxy configuration on the switch. The HTTP proxy configured is used to connect to HPE Aruba Networking Central as per the zero-touch provisioning requirement. When HTTP proxy location and VRF are configured, they override any existing HTTP proxy location and VRF.

HTTP proxy location can be configured using the CLI or REST interface or auto-configured through the DHCPv4/DHCPv6 server connected to the switch.

There are four sources for HTTP proxy location:

- User configured HTTP proxy via CLI or REST interface.
- DHCPv4 options received via the management/OOBM port
- DHCP options received via VLAN 1 on supported switch platforms.
- DHCPv6 options received via management/OOBM port

Operational configuration for HTTP proxy location is determined by the source with the highest priority. The source priority is as follows:

1. User configured.
2. DHCPv4 options received via the management/OOBM port.
3. DHCP options received via VLAN 1.
4. DHCPv6 options received via the management/OOBM port

NOTE

- DHCPv6 options support FQDN and IPv6 addresses, while DHCPv4 options support FQDN and IPv4 addresses.
- DHCPv6 support is available only on the management/OOBM interface and provides the lowest priority in the arbitration hierarchy.

An **HTTP proxy** FQDN or IP address can be sent with DHCPv4 option 43 sub-option 148 (which supports an FQDN or IPv4 address) or DHCPv6 option 17 sub-option 148 (which supports an FQDN or IPv6 address).

An **HPE Aruba Networking Central** FQDN or IP address can be sent using DHCPv4 option 43 sub-option 146 (which supports an FQDN or IPv4 address) or DHCPv6 option 17 sub-option 146 (which supports an FQDN or IPv6 address).

When the switch uses zero-touch provisioning with HPE Aruba Networking Central, the HPE Aruba Networking Central location, shared secret and alternate location sent using DHCPv6 option 17 suboption 146 is in the format **<group-details>,<aruba-central-fqdn-or-ipv6>,<shared-secret>**, or for dual cluster support, **<group-details>,<aruba-central-fqdn-or-ipv6>,<shared-secret>,<alt-aruba-central-fqdn-or-ipv6>**.

Subtopics

[DHCP options commands]

DHCP options commands

Select a command from the list in the left navigation menu.

Subtopics

[http-proxy]

`http-proxy`

Syntax

```
http-proxy {<FQDN |{IPV4-ADDR|IPV6-ADDR[:PORT]}} [vrf <VRF-NAME>]
no http-proxy [<FQDN | {IPV4-ADDR|IPV6-ADDR[:PORT]}] [vrf <VRF-NAME>]
```

Description

Specifies HTTP proxy location and VRF.

When HTTP proxy location and VRF are configured on the switch, it overrides any existing HTTP proxy location and VRF as this has the highest priority over the values obtained from other sources.

Following locations can be used for the HTTP proxy location:

- A fully qualified domain name (FQDN).
- An IPv4 address with colon separated port number
- An IPv6 address with colon separated port number

> **NOTE**
> When configuring an IPv6 address with a port number, the address must be specified inside square brackets. An example - [2001:0db8:85a3:0000:0000:8a2e:0370:7334]:8080.

If the command is entered without the VRF parameter, then the VRF used will be 'default' VRF.

The **no** form of this command removes a specified HTTP proxy location.

Parameter	Description
<FQDN>	Specifies FQDN for HTTP proxy location.
<IPV4-ADDR>	Specifies IPV4 address for HTTP proxy location.
<IPV6-ADDR>	Specifies IPV6 address for HTTP proxy location.
<VRF-NAME>	Specifies VRF for HTTP proxy.

NOTE

A FQDN or IPV4 address is optional in the **no** form of the command.

Examples

Specifying a FQDN for HTTP proxy location and **MGMT** VRF with an IPv6 address:

```
switch(config)# http-proxy http-proxy.example.com vrf mgmt
switch(config)# http-proxy [2000::100]:8080 vrf mgmt
```

Specifying a FQDN for HTTP proxy location and **MGMT** VRF with an IPv4 address:

```
switch(config)# http-proxy http-proxy.example.com vrf mgmt
switch(config)# http-proxy 192.168.1.3:8080 vrf mgmt
```

Removing HTTP proxy location

```
switch(config)# no http-proxy
```

Command History

Release	Modification
10.13.1000	Command updated to reflect OTP scenario.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

ND snooping

Subtopics

Overview

ND snooping is used in Layer 2 switching networks and prevents ND attacks. ND snooping drops invalid ND packets, and together with DIPLDv6 (Dynamic IP Lockdown for IPv6), blocks data traffic from invalid hosts. ND snooping learns the source MAC addresses, source IPv6 addresses, input interfaces, and VLANs of incoming ND messages and data packets to build IP binding entries.

> **NOTE**
> When DHCPv6 snooping and ND snooping are both enabled, and DHCPv6 clients request and IPv6 address, entries are added to the DHCPv6 snooping table and DHCPv6 snooping takes priority over ND snooping.

ND snooping drops ND packets as follows:

- If the Ethernet source MAC address is mismatched with the address contained in the ICMPv6 Target link layer address field of the ND packet.
- If the global IPv6 address in the source address field is mismatched with the ND snooping prefix filter table.
- If the global IPv6 address or the link-local IPv6 address in the source IP address field is mismatched with the ND snooping binding table.

ND snooping drops RA and RR packets on untrusted ports. To block only RA packets on VLANs with ND snooping enabled, use `nd-snooping ra-drop`. RA (Router Advertisement) drop is disabled by default on VLANs. When enabled (with `nd-snooping ra-drop`), ND snooping blocks RA packets on both trusted and untrusted ports. When RA drop is disabled, ND snooping allows RA packets on trusted ports and blocks them on untrusted ports.

When RA guard policy is enabled (with `ipv6 nd-snooping ra-guard policy`), RA packets received on trusted ports are validated against a set of parameters configured on the policy and assigned to a port or VLAN.

NOTE

For more information on RA guard including RA guard policies, see this related video on the HPE Aruba Networking **AirHeads Broadcasting Channel**.

Dynamic IPv6 lockdown is performed for ND snooping entries. Based on the DAD NS received from the hosts by the switch, ND snooping entries are programmed into the IP binding table and the hardware (as allowed). And ND Binding table entries are added when NA packets are received from hosts. Therefore, data packets from invalid hosts and transit traffic are blocked.

NOTE

Statically-configured IP binding information supersedes any information collected dynamically by ND snooping for the same client.

RFC 7113 Compliance for RA Guard

RA Guard implements **RFC 7113**, which addresses evasion techniques using IPv6 extension headers and fragmentation. It is possible to allow IPv6 Header Chain Parsing. This option is disabled by default, see **nd-snooping allow-ipv6-ext-header** command for more details. IPv6 Header range is also supported.

> **NOTE**
>
> This feature is currently supported in the following Switch series:
>
> - 5420
> - 6000
> - 6100
> - 6200
> - 6300/6300M/6300F
> - 6400
> - 8100
> - 8325H
> - 8360
> - 8400

1. Source Address and Hop Limit Checks

- Source Address: RA Guard passes all RA and RR packets if the IPv6 source address is not link-local (fe80::/10), as required by RFC 7113 Section 3, Rule 1.
- Hop Limit: RA Guard passes all RA and RR packets if the IPv6 Hop Limit is not 255, as required by RFC 7113 Section 3, Rule 2.

These checks are already supported in the current implementation.

2. IPv6 Header Chain Parsing (Extension Headers)

RA Guard parses the entire IPv6 header chain for supported extension headers to identify RA and RR messages, as required by RFC 7113 Section 3, Rule 3.

Supported Extension Headers:

- Hop-by-Hop Options Header (Next Header = 0)
- Destination Options Header (Next Header = 60)

- Routing Header (Next Header = 43)
- Authentication Header (AH, Next Header = 51)
- Mobility Header (Next Header = 135)

Extension Headers which are forwarded:

Packets containing the following extension headers are forwarded as upper-layer protocols and are not inspected for RA/RR messages:

- **Encapsulating Security Payload (ESP, Next Header = 50)**: Treated as an upper-layer protocol and not inspected for RA/RR messages.
- **Unknown or Custom Extension Headers**: By default, RA/RR packets containing unknown or custom extension headers (i.e., Next Header values not defined in RFC 8200 or not listed above) are forwarded. Optionally, you can configure the system to drop and log these packets.

ASIC Pipeline Limits for IPv6 Extension Headers:

The ASIC pipeline has the following constraints when processing IPv6 extension headers for ICMPv6 (RA/RR) packets:

Parameter	Limit	Description
Maximum Extension Headers	8	Maximum number of IPv6 extension headers that can be parsed by the ASIC.
Maximum Extension Header Length	200 bytes	Maximum total length of all extension headers combined.

NOTE

The specific byte limits were determined through empirical testing. The 200-byte limit for ICMPv6 packets is slightly higher than DHCPv6 (192 bytes) because the ASIC pipeline does not need to parse the L4 header for ICMPv6 classification—ICMPv6 type extraction uses a different ASIC mechanism that does not require the L4_VALID flag. This difference is visible in the platform TCAM rule implementation (**cpurx_iptcam_protocol.c**) where DHCPv6 rules require **MATCH_L4_VALID_TRUE** while ICMPv6 rules do not. Packets with extension headers exceeding this limit may not be correctly classified by TCAM rules.

Example: A packet with 7 standard extension headers (7 × 8 bytes = 56 bytes) plus one larger extension header of up to 144 bytes (56 + 144 = 200 bytes total) can be successfully parsed for RA Guard classification.

Host/Client ND Packet Handling with Extension Headers

ND packets from hosts/clients (NS, NA, RS) containing IPv6 extension headers are handled based on the nd**nd-snooping allow-ipv6-ext-headers** configuration:

- When **allow-ipv6-ext-headers** is **enabled**: Client ND packets with extension headers are correctly parsed and processed. The ICMPv6 ND message is correctly located past all extension headers in the chain.
- When **allow-ipv6-ext-headers** is **disabled** (default): Client ND packets containing valid extension headers are dropped.

3. Fragmentation handling (RFC 7113 Section 3, Rule 4)

- If the first fragment contains the ICMPv6 header type field but the RA payload is truncated or incomplete, the packet is dropped by RA Guard. An event log is generated and the **NDSNOOP_DROP** counter is incremented.
- If the first fragment does not contain the ICMPv6 header (i.e., the ICMPv6 type is not visible), the packet cannot be classified as RA/RR. Such packets are **forwarded** and not inspected by RA Guard.
- Non-first fragments (Fragment Offset > 0) do not contain the IPv6 header chain and cannot be classified as RA/RR. These fragments are **forwarded** regardless of configuration and are not inspected by RA Guard.

> **NOTE**
>
> **Note:** RA Guard does not inspect the following fragmentation scenarios: (1) First fragments where the ICMPv6 header is beyond the parser depth and cannot be classified as RA/RR — these are forwarded without inspection. (2) Non-first fragments (Fragment Offset > 0) — these are always forwarded without inspection.

Subtopics

Examples

Case 1: RA with Destination Options header, fragmented

Original packet structure (before fragmentation):

IPv6 Header NH=60	Destination Options Header NH=58	ICMPv6 RA

Packet structure after fragmentation:

First fragment:

IPv6 Header NH=44	Fragment Header NH=60	Destination Options Header NH=58

Second fragment:

IPv6 Header NH=44	Fragment Header NH=60	Destination Options Header	ICMPv6 RA

In this case, the first fragment does not contain the ICMPv6 RA data, so it cannot be classified as an RA packet. Both the first fragment and the second fragment (non-first, offset > 0) are **forwarded** and not inspected by RA Guard.

Case 2: RA with two Destination Options headers, fragmented

Original packet structure (before fragmentation):

IPv6 Header NH=60	Destination Options Header NH=60	Destination Options Header NH=58	ICMPv6 RA

Packet structure after fragmentation:

First fragment:

IPv6 Header NH=44	Fragment Header NH=60	Destination Options Header NH=60

Second fragment:

IPv6 Header NH=44	Fragment Header NH=60	Destination Options Header	Destination Options Header NH=60	ICMPv6 RA

In this case, the first fragment contains only one of the two Destination Options headers and does not include the ICMPv6 RA data. The packet cannot be classified as an RA, so both fragments are **forwarded** and not inspected by RA Guard.

Case 3: RA with Fragment header only (ICMPv6 type visible but truncated RA payload)

Original packet structure (before fragmentation):

IPv6 Header NH=58	ICMPv6 RA

Packet structure after fragmentation:

First fragment:

IPv6 Header NH=44	Fragment Header NH=58	ICMPv6 RA (trunc) type=134, but...

The first fragment contains the Fragment header (NH=58) and the beginning of the ICMPv6 header (type field = 134, identifying it as an RA), but the RA payload is truncated.

Second fragment:

IPv6 Header NH=44	Fragment Header offset > 0	RA payload (cont)

Since the ICMPv6 type (134 = RA) is visible in the first fragment, the packet is classified as an RA. RA Guard validates the RA payload and finds it truncated, so the first fragment is **dropped** with an event log and the **NDSNOOP_DROP** counter is incremented. The second fragment (non-first, offset > 0) is **forwarded** as it cannot be classified as RA/RR. Since the first fragment is dropped, the receiving host cannot reassemble the RA packet.

Fragmentation Behavior Summary

Scenario	allow-ext-hdrs	Result	Event Log	Counter	Notes
First fragment with complete headers (ext hdrs + frag)	yes	ALLOW	-	-	Processed by RA Guard per policy.
First fragment with complete headers (ext hdrs + frag)	no	DROP	-	NDSNOOP_DROP	Dropped due to extension header policy.
First fragment with complete headers (frag only)	yes	ALLOW	-	-	First fragment with complete ICMPv6 header - allowing.
First fragment with complete headers (frag only)	no	DROP	-	NDSNOOP_DROP	Dropped due to extension header policy.
Non-first fragment (offset > 0)	any	ALLOW	-	-	Forwarded
First fragment + incomplete ICMPv6 RA header	any	ALLOW	NA	NA	Forwarded (cannot be classified as RA/RR).
First fragment with ICMPv6 but truncated RA payload	any	DROP	yes	NDSNOOP_DROP	RA packet dropped event log.

ND snooping commands

Select a command from the list in the left navigation menu.

Subtopics

`clear nd-snooping binding`

Syntax

```
clear nd-snooping bindings {all | ipv6 <IPV6-ADDR> vlan <VLAN-ID> |
      port <PORT-NUM> | vlan <VLAN-ID>}
```

Description

Clears ND snooping binding entries.

Command context

Parameter	Description
`all`	Specifies that all ND binding information is to be cleared.
`ip <IPV6-ADDR> vlan <VLAN-ID>`	Specifies the IPv6 address and VLAN for which all ND binding i nformation is to be cleared.
`port <PORT-NUM>`	Specifies the port (interface) for which all ND binding informati on is to be cleared.
`vlan <VLAN-ID>`	Specifies the VLAN for which all ND binding information is to be cleared. Range: 1 to 4094.

Examples

Clearing all ND binding information for 5000::1 vlan 1:

```
switch(config)# clear nd-snooping bindings ipv6 5000::1 vlan 1
```

Clearing all ND binding information for port 1/1/10:

```
switch(config)# clear nd-snooping bindings port 1/1/10
```

Clearing all ND binding information for VLAN 10:

```
switch(config)# clear nd-snooping bindings vlan 10
```

Clearing all ND binding information:

```
switch(config)# clear nd-snooping bindings all
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`clear nd-snooping ra-guard-policy statistics`

Syntax

```
clear nd-snooping ra-guard-policy statistics [vlan <VLAN-ID>]|[interface
<IFNAME>]
```

Description

Clear all RA Guard policy statistics from the specified interface or VLAN.

Command context

Parameter	Description
`vlan <VLAN-ID>`	Clear all RA Guard policy information on the specified VLAN
`interface <IFNAME>`	Clear all RA Guard policy information on the specified interface

Examples

Clear all RA Guard policy statistics for VLAN 10:

```
switch# clear nd-snooping ra-guard-policy statistics vlan 10
```

Clear all RA Guard policy statistics for interface 1/1/10

```
switch# clear nd-snooping ra-guard-policy statistics interface 1/1/10
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`clear nd-snooping statistics`

Syntax

```
clear nd-snooping statistics
```

Description

Clears all ND snooping statistics.

Examples

Clear all ND snooping statistics:

```
switch# clear nd-snooping statistics
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`nd-snooping`

Syntax

```
nd-snooping {enable|disable}
no nd-snooping {enable|disable}
```

Description

Enables or disables ND snooping. ND snooping is disabled by default. ND snooping is not supported on the management interface.

Examples

Enabling ND snooping:

```
switch(config)# nd-snooping enable
```

Disabling ND snooping:

```
switch(config)# nd-snooping disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

nd-snooping allow-ipv6-ext-headers

Syntax

```
[no] nd-snooping allow-ipv6-ext-headers
```

Description

This command enables or disables forwarding of RA/RR packets containing any IPv6 extension headers. By default, such packets are **dropped**. When enabled, these packets are **forwarded** instead of dropped (RFC 7113 compliance).

Examples

Enabling forwarding:

```
switch# configure terminal
switch(config)# nd-snooping allow-ipv6-ext-headers
```

Disabling forwarding (default):

```
switch# configure terminal
switch(config)# no nd-snooping allow-ipv6-ext-headers
```

Command History

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
5420	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority
6200		

nd-snooping (in config-vlan context)

Syntax

```
nd-snooping
no nd-snooping
```

Description

Enables ND snooping in the **config-vlan** context. ND snooping is disabled by default for all VLANs.

The no form of the command disables ND snooping on the specified VLAN, flushing all the IPv6 bindings learned for this VLAN since ND snooping was enabled for this VLAN.

Examples

Enabling ND snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# nd-snooping
switch(config-vlan-100)# exit
switch(config)#
```

Disabling ND snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# no nd-snooping
switch(config-vlan-100)# exit
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-vlan`	Administrators or local user group members with execution righ ts for this command.

nd-snooping drop-unknown-ipv6-ext-headers

Syntax

```
[no] nd-snooping drop-unknown-ipv6-ext-headers
```

Description

This command enables or disables dropping of RA/RR packets containing **unknown or unsupported IPv6 extension headers**. This refers to Next Header values not defined in RFC 8200 or not supported by the implementation. These are within the 142-255 range. By default, such packets are **forwarded**. When this command is configured, these packets are **dropped** and logged by RA Guard (RFC 7113 compliance).

Examples

Dropping packets with unknown IPv6 extension headers:

```
switch# configure terminal
switch(config)# nd-snooping drop-unknown-ipv6-ext-headers
```

Forwarding packets with unknown IPv6 extension headers (default):

```
switch# configure terminal
switch(config)# no nd-snooping drop-unknown-ipv6-ext-headers
```

Command History

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution righ ts for this command.

`nd-snooping mac-check`

Syntax

```
nd-snooping mac-check
no nd-snooping mac-check
```

Description

This command enables verification of the hardware address field in ND snooping packets. When enabled, the ICMPv6 target link layer address field and the source MAC address must be the same for packets received on untrusted ports or else the packets are dropped. This ND snooping MAC verification is enabled by default.

The no form of the command disables ND snooping MAC verification.

Examples

Enabling ND snooping MAC verification:

```
switch(config)# nd-snooping mac-check
```

Disabling ND snooping MAC verification:

```
switch(config)# no nd-snooping mac-check
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`nd-snooping prefix-list`

Syntax

```
nd-snooping prefix-list <IPV6-ADDR>
no nd-snooping prefix-list <IPV6-ADDR>
```

Description

Configures the ND snooping prefix list for the selected VLAN and the specified IPv6 address prefix. ND snooping must be enabled both globally and on this VLAN before this prefix list configuration takes effect.

The no form of this command removes the prefix list configuration for the selected VLAN and IPv6 address.

Parameter	Description
<IPV6-ADDR>	Specifies the IPv6 address.

Examples

Configuring ND snooping prefix-list on VLAN 1:

```
switch(config)# vlan 1
switch(config-vlan-1)# nd-snooping prefix-list 2001::1/64
switch(config-vlan-1)# exit
switch(config)#
```

Remove configuration of ND snooping prefix-list on VLAN 100:

```
switch(config)# vlan 1
switch(config-vlan-1)# no nd-snooping prefix-list 2001::1/64
switch(config-vlan-1)# exit
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-vlan-<VLAN-ID>`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`nd-snooping max-bindings`

Syntax

```
nd-snooping max-bindings <MAX-BINDINGS>
no nd-snooping max-bindings
```

Description

Sets the maximum number of ND bindings allowed on the selected interface. For all interfaces on which this command is not run, the default max bindings applies.

The no form of the command reverts max bindings for the selected interface to its default.

Parameter	Description
<MAX-BINDINGS>	Specifies the maximum number of ND bindings. You can use the **show capacities** command to see the maximum available for yo ur switch model.

Examples

Set the ND max bindings to 768 on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# nd-snooping max-bindings 768
```

```
switch(config-if)# exit
switch(config)#
```

Revert ND max bindings to its default on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# no nd-snooping max-bindings
switch(config-if)# exit
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`nd-snooping nd-guard`

Syntax

```
nd-snooping nd-guard
no nd-snooping nd-guard
```

Description

This command enables ND guard on the selected VLAN.

The no form of the command disables ND guard and deletes all the IPv6 bindings learned on the VLAN.

> **NOTE**
> ND snooping must be enabled in both the global context and the config-vlan context before this command can be used.

Examples

Enabling ND snooping ND guard on VLAN 100:

```
switch(config)# nd-snooping enable
switch(config)# vlan 100
switch(config-vlan-100)# nd-snooping nd-guard
switch(config-vlan-100)# exit
switch(config)#
```

Disabling ND snooping ND guard on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# no nd-snooping nd-guard
switch(config-vlan-100)# exit
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-vlan`	Administrators or local user group members with execution righ ts for this command.

`nd-snooping ra-guard`

Syntax

```
nd-snooping ra-guard [log]
no nd-snooping ra-guard
```

Description

This command enables Routing Advertisement (RA) guard on the selected VLAN. When enabled, ingress Routing Advertisement (RA) and Routing Redirect (RR) packets on the selected VLAN are blocked on untrusted ports. The packets are forwarded when received on trusted ports.

The no form of the command disables RA guard on the VLAN.

NOTE

ND snooping must be enabled in both the global context and the config-vlan context before this command can be used.

Parameter	Description
`[log]`	Logs messages along with drop functionality.

Examples

Enabling ND snooping RA guard on VLAN 100:

```
switch(config)# nd-snooping enable
switch(config)# vlan 100
switch(config-vlan-100)# nd-snooping ra-guard
switch(config-vlan-100)# exit
switch(config)#
```

Enabling ND snooping RA guard on VLAN 100 with event logging on dropped packets:

```
switch(config)# nd-snooping enable
switch(config)# vlan 100
switch(config-vlan-100)# nd-snooping ra-guard log
switch(config-vlan-100)# exit
switch(config)#
```

Disabling ND snooping RA guard on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# no nd-snooping ra-guard
switch(config-vlan-100)# exit
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-vlan-<VLAN-ID>`	Administrators or local user group members with execution righ ts for this command.

`nd-snooping ra-drop`

Syntax

```
nd-snooping ra-drop
no nd-snooping ra-drop
```

Description

This command enables Routing Advertisement (RA) drop on the selected VLAN. When enabled, ingress RA packets on the selected VLAN are blocked on both trusted and untrusted ports. When disabled, RA packets are forwarded on the selected VLAN with ND snooping trusted port validation. RA drop is disabled by default.

> **NOTE**
> ND snooping must be enabled in both the config context and the config-vlan context before this command can be used.

The no form of the command disables ND snooping RA drop on the selected VLAN.

Examples

Enabling ND snooping RA drop on VLAN 100:

```
switch(config)# nd-snooping enable vlan 100
switch(config-vlan-100)# nd-snooping ra-drop
switch(config-vlan-100)# exit
switch(config)#
```

Disabling ND snooping RA drop on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# no nd-snooping ra-drop
switch(config-vlan-100)# exit
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-vlan-<VLAN-ID>`	Administrators or local user group members with execution righ ts for this command.

`nd-snooping trust`

Syntax

```
nd-snooping trust
no nd-snooping trust
```

Description

Enables ND snooping trust on the selected interface (port). Only server packets received on trusted ports are forwarded. All the ports are untrusted by default.

The no form of the command disables ND snooping trust on the selected port.

Examples

Enabling ND snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# nd-snooping trust
switch(config-if)# exit
switch(config)#
```

Disabling ND snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# no nd-snooping trust
switch(config-if)# exit
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`show nd-snooping`

Syntax

```
show nd-snooping [vlan <VLAN-ID>]
```

Description

Shows either all ND snooping configuration or the configuration for the specified VLAN.

Parameter	Description
`vlan <VLAN-ID>`	Specifies the VLAN for which the ND configuration is to be sho wn. Range: 1 to 4094.

Examples

(Applies to the HPE Aruba Networking 5420, 6200, 6300, 6400, 8100, and 8360 Switch series.) Showing all ND snooping configuration:

```
switch(config)# show nd-snooping
  ND Snooping Information
  ========================
  ND Snooping                           : Enabled
  ND Snooping Enabled VLANs             : 10
  Trusted Port Bindings Enabled VLANs : 10
```

```
  ND Guard Enabled VLANs              : 10
  RA Guard Enabled VLANs              : 10
  RA Drop Enabled VLANs               :
  MAC Address Check                   : Disabled
  IPv6 Allow Ext Headers              : Disabled
  IPv6 Drop Unknown Ext Headers       : Disabled

  PORT    TRUST  MAX-BINDINGS  CURRENT-BINDINGS
  ------- ------ ------------- -----------------
  1/1/1   Yes
  1/1/2   Yes
  1/1/3   No     100           10
  1/1/4   No     200           10
  1/1/5   No     300           10
```

(Applies to the HPE Aruba Networking 5420, 6200, 6300, 6400, 8100, 8360 Switch series.) Showing ND snooping configuration for VLAN 2:

```
switch(config)# show nd-snooping vlan 2
  ND Snooping Information
  =======================
  ND Snooping             : Enabled
  MAC Address Check       : Disabled
  Trusted Port Bindings : Enabled
  ND Guard                : Enabled
  RA Guard                : Disabled
  RA Drop                 : Disabled
  PORT    TRUST  MAX-BINDINGS  CURRENT-BINDINGS
  ------- ------ ------------- -----------------
  1/1/1   Yes
  1/1/2   Yes
  1/1/3   No     100           10
```

(Applies to the HPE Aruba Networking 8325, 9300/9300S, 10000 Switch series.) Showing all ND snooping configuration:

```
switch(config)# show nd-snooping
  ND Snooping Information
  ========================
  ND Snooping                    : Enabled
  ND Snooping Enabled VLANs      : 10
  RA Guard Enabled VLANs         : 10
  MAC Address Check              : Disabled
  PORT    TRUST
  ------- ------
  1/1/1   Yes
```

```
  1/1/2   Yes
  1/1/3   No
  1/1/4   No
  1/1/5   No
```

(Applies to the HPE Aruba Networking 8325, 9300/9300S, 10000 Switch series.) Showing ND snooping configuration for VLAN 2:

```
switch(config)# show nd-snooping vlan 2
  ND Snooping Information
  =======================
  ND Snooping          : Enabled
  MAC Address Check : Disabled
  RA Guard             : Disabled
  PORT    TRUST
  ------- ------
  1/1/1   Yes
  1/1/2   Yes
  1/1/3   No
```

Command History

Release	Modification
10.18	Adds ND snooping information about **IPv6 Allow Ext Headers** and IPv6 **Drop Unknown Ext Headers**.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show nd-snooping binding`

Syntax

```
show nd-snooping bindings
```

Description

Shows the ND snooping binding configuration.

Examples

Showing the ND snooping binding configuration:

```
switch# show nd-snooping binding
  PORT     IPV6-ADDRESS                            MAC-ADDRESS        VLAN
TIME-LEFT STATE
  ------- ---------------------------------------- ------------------ -----
--------- ---------
  1/1/1    2001::1                                 00:00:0A:01:02:03  1
600       Valid
  1/1/2    fe80::250:56ff:fe9a:143c                00:00:0B:01:02:03  2
-         Tentative
  1/1/3    2001:1111:2222:3333:4444:5555:6666:7777  00:00:0C:01:02:03  4094
-         Testing
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show nd-snooping prefix-list`

Syntax

```
show nd-snooping prefix-list
```

Description

Shows the ND snooping prefix list information.

Examples

Showing the ND snooping prefix list information:

```
switch# show nd-snooping prefix-list
  VLAN  IPV6-ADDRESS-PREFIX                         SOURCE
  ----- ------------------------------------------- --------
  1     2001::/64                                   Static
  4094  3001::/64                                   Dynamic
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

show nd-snooping statistics

Syntax

```
show nd-snooping statistics
```

Description

Shows the global ND snooping statistics.

Examples

(Applies to the HPE Aruba Networking 5420, 6200, 6300, 6400, 8100 and 8360 Switch series.) Showing global ND snooping statistics:

```
switch(config)# show nd-snooping statistics
  PACKET-TYPE  ACTION   REASON
COUNT
  ------------ -------- ------------------------------------------------
--------
  RA           forward  RA packets received on trusted port              20
  RA           drop     RA packets received on untrusted port            45
  NS           forward  NS packets received on trusted port              52
  NS           forward  NS packets received on untrusted port            95
  NS           drop     NS packets failed MAC check                      14
  NS           drop     NS packets failed Prefix check                   12
  NS           drop     NS packets failed on max-binding limit           0
  NS           drop     NS packets failed ND snooping validation checks  20
  NA           forward  NA packets received on trusted port              17
  NA           forward  NA packets received on untrusted port            30
  NA           drop     NA packets failed Prefix check                   15
  NA           drop     NA packets failed on max-binding limit           2
  NA           drop     NA packets failed ND snooping validation checks  5
  ND           drop     ND packets dropped in total                      259
```

(Applies to the HPE Aruba Networking 8325, 9300/9300S and 10000 Switch series.) Showing global ND snooping statistics:

```
switch(config)# show nd-snooping statistics
  PACKET-TYPE  ACTION   REASON
COUNT
  ------------ -------- ------------------------------------------------
--------
  RA           forward  RA packets received on trusted port              20
  RA           drop     RA packets received on untrusted port            45
  RR           forward  RR packets received on trusted port              20
  RR           drop     RR packets received on untrusted port            45
```

Command History

Release	Modification
10.18	Added information about ND packets.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

RA guard policy commands

Select a command from the list in the left navigation menu.

Subtopics

- **hop limit**
- **ipv6 nd-snooping ra-guard policy**
- **managed-config-flag**
- **match access-list**
- **match prefix-list**
- **nd-snooping allow-ipv6-ext-headers**
- **nd-snooping drop-unknown-ipv6-ext-headers**
- **nd-snooping ra-guard attach-policy**
- **other-config-flag**
- **router-preference**
- **show nd-snooping ra-guard interface**
- **show nd-snooping ra-guard policy**
- **show nd-snooping ra-guard vlan**

`hop limit`

Syntax

```
hop limit [minimum | maximum] <HOP-LIMIT>
no hop limit [minimum | maximum] <HOP-LIMIT>
```

Description

Enables verification of the advertised hop count limit if the RA guard policy is applied on a VLAN or interface. RA packets with the hop limit within the specified minimum and maximum values are processed. If none of the values are specified for hop limit, the default range is 1-255. If hop limit is not enabled, packets are not validated for hop limit.

The **no** form of the command disables the hop limit on the specified RA guard policy.

> **NOTE**
>
> ND snooping must be enabled in both the global context and the config-vlan context before this command can be used.

Parameter	Description
`<HOP-LIMIT>`	Specifies the hop - limit value. Range: 1 - 255.
`minimum`	Specifies the minimum value for the hop - limit range. Default: 1, Range 1 - 255. The range is minimum–255 if only a minimum value is specified .
`maximum`	Specifies the maximum value for the hop - limit range. Default: 255, Range 1 - 255. The range is 1–maximum if only a maximum value is specified.

Examples

Enabling the hop limit on the RA guard policy and adding minimum and maximum values for hop limit on the policy:

```
switch(config)# ipv6 nd-snooping ra-guard policy <POLICY-NAME>
switch(config-raguard-policy)# hop-limit enable
switch(config-raguard-policy)# hop-limit maximum 150
switch(config-raguard-policy)# hop-limit minimum 50
```

Disabling the hop limit on the RA guard policy:

```
switch(config)# ipv6 nd-snooping ra-guard policy <POLICY-NAME>
switch(config-raguard-policy)# no hop-limit enable
```

Removing minimum and maximum values for the hop limit on the RA guard policy:

```
switch(config)# ipv6 nd-snooping ra-guard policy <POLICY-NAME>
switch(config-raguard-policy)# no hop-limit maximum 150
switch(config-raguard-policy)# no hop-limit minimum 50
switch(config-raguard-policy)# no hop-limit maximum
switch(config-raguard-policy)# no hop-limit minimum
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-raguard-policy`	Administrators or local user group members with execution rights for this command.

`ipv6 nd-snooping ra-guard policy`

Syntax

```
ipv6 nd-snooping ra-guard policy <POLICY-NAME>
no ipv6 nd-snooping ra-guard policy <POLICY-NAME>
```

Description

Creates the Router Advertisement (RA) guard policy with the given name and enters the RA guard policy configuration context.

The **no** form of the command removes the specified RA guard policy from the switch.

> **NOTE**
> ND snooping must be enabled in both the global context and the config-vlan context before this command can be used.

Parameter	Description
<POLICY-NAME>	Specifies the name of the RA guard policy. Maximum length: 64.

Examples

Creating the RA guard policy globally with a specified name:

```
switch(config)# ipv6 nd-snooping ra-guard policy <POLICY-NAME>
switch(config-raguard-policy)#
```

Deleting the specified RA guard policy:

```
switch(config)# no ipv6 nd-snooping ra-guard policy <POLICY-NAME>
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`managed-config-flag`

Syntax

```
managed-config-flag [on | off]
no managed-config-flag [on | off]
```

Description

Enables the verification of the advertised manage configuration flag. Verifies that the advertised managed address configuration flag is On or Off based on the configured value.

The **no** form of the command disables the manage configuration flag verification.

> **NOTE**
> ND snooping must be enabled in both the global context and the config-vlan context before this command can be used.

Parameter	Description
`on`	Verifies that the advertised managed address configuration fla g is On.
`off`	Verifies that the advertised managed address configuration fla g is Off.

Examples

Enabling managed configuration flag verification:

```
switch(config)# ipv6 nd-snooping ra-guard policy <POLICY-NAME>
switch(config-raguard-policy)# managed-config-flag off
switch(config-raguard-policy)# managed-config-flag on
```

Disabling managed configuration flag verification:

```
switch(config)# ipv6 nd-snooping ra-guard policy <POLICY-NAME>
switch(config-raguard-policy)# no managed-config-flag
switch(config-raguard-policy)# no managed-config-flag off
switch(config-raguard-policy)# no managed-config-flag on
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-raguard-policy`	Administrators or local user group members with execution righ ts for this command.

`match access-list`

Syntax

```
match access-list <ACL-NAME>
no match access-list <ACL-NAME>
```

Description

Configures the access list to an RA guard policy. The access list has to be created with the desired match criteria before adding it into RA guard policy. Advertised packets are verified for the match criteria when an RA guard policy with matched access list is enabled on a trusted port or VLANs.

The **no** form of the command removes the access list from the RA guard policy.

> **NOTE**
> ND snooping must be enabled in both the global context and the config-vlan context before this command can be used.

Parameter	Description
<ACL-NAME>	Specifies the name of the access list to be matched.

Examples

Adding an access list named Example_ACL to the RA guard policy POL1:

```
switch(config)# ipv6 nd-snooping ra-guard policy POL1
```

```
switch(config-raguard-policy)# match access-list Example_ACL
```

Deleting the access list named Example_ACL from the RA guard policy POL1:

```
switch(config)# ipv6 nd-snooping ra-guard policy POL1
switch(config-raguard-policy)# no match access-list Example_ACL
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-raguard-policy`	Administrators or local user group members with execution righ ts for this command.

`match prefix-list`

Syntax

```
match prefix-list <PREFIX-LIST-NAME>
no match prefix-list <PREFIX-LIST-NAME>
```

Description

Configures a prefix-list for the RA guard policy. Advertised prefixes in RA packets are compared against the configured prefix-list and if there is no match, the RA packets are dropped. If the RA prefix list is not configured, this check is not performed.

The **no** form of the command removes the prefix list from the RA guard policy.

NOTE

ND snooping must be enabled in both the global context and the config-vlan context before this command can be used.

Parameter	Description
<PREFIX-LIST-NAME>	Specifies the name of the prefix list to be matched.

Examples

Adding a prefix list named PREFIX_LIST_EXAMPLE to the POLICY1 RA guard policy:

```
switch(config)# ipv6 nd-snooping ra-guard policy POLICY1
switch(config-raguard-policy)# match prefix-list PREFIX_LIST_EXAMPLE
```

Deleting the prefix list named PREFIX_LIST_EXAMPLE from the POLICY1 RA guard policy:

```
switch(config)# ipv6 nd-snooping ra-guard policy POLICY1
switch(config-raguard-policy)# no match pefix-list PREFIX_LIST_EXAMPLE
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-raguard-policy`	Administrators or local user group members with execution righ ts for this command.

nd-snooping allow-ipv6-ext-headers

Syntax

```
[no] nd-snooping allow-ipv6-ext-headers
```

Description

This command enables or disables forwarding of RA/RR packets containing any IPv6 extension headers. By default, such packets are **dropped**. When enabled, these packets are **forwarded** instead of dropped (RFC 7113 compliance).

Examples

Enabling forwarding:

```
switch# configure terminal
switch(config)# nd-snooping allow-ipv6-ext-headers
```

Disabling forwarding (default):

```
switch# configure terminal
switch(config)# no nd-snooping allow-ipv6-ext-headers
```

Command History

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution righ ts for this command.

nd-snooping drop-unknown-ipv6-ext-headers

Syntax

```
[no] nd-snooping drop-unknown-ipv6-ext-headers
```

Description

This command enables or disables dropping of RA/RR packets containing **unknown or unsupported IPv6 extension headers**. This refers to Next Header values not defined in RFC 8200 or not supported by the implementation. These are within the 142-255 range. By default, such packets are **forwarded**. When this command is configured, these packets are **dropped** and logged by RA Guard (RFC 7113 compliance).

Examples

Dropping packets with unknown IPv6 extension headers:

```
switch# configure terminal
switch(config)# nd-snooping drop-unknown-ipv6-ext-headers
```

Forwarding packets with unknown IPv6 extension headers (default):

```
switch# configure terminal
switch(config)# no nd-snooping drop-unknown-ipv6-ext-headers
```

Command History

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution righ ts for this command.

nd-snooping ra-guard attach-policy

Syntax

```
nd-snooping ra-guard attach-policy <POLICY-NAME>
no nd-snooping ra-guard attach-policy <POLICY-NAME>
```

Description

Applies the created RA guard policy to a specific L2 port or VLAN.

The **no** form of the command detaches the specified RA guard policy from the L2 port or VLAN.

Parameter	Description
<POLICY-NAME>	Specifies the name of the RA guard policy.

Usage

In the interface configuration (config-if) context:

- RA guard must be enabled on member VLANs of the port for which RA packets need to be inspected using the policy.

In the interface configuration (config-if) and VLAN configuration (config-vlan) contexts:

- RA packets received on untrusted ports are dropped without any inspection.
- RA packets received on trusted ports are validated against the policy.
- The applied policy takes effect only if ND snooping is enabled globally and both ND snooping and RA guard are enabled under the VLAN context.

Policy precedence between VLAN and port:

- If the policy is attached to both VLAN and port, the port policy takes precedence over the VLAN policy.
- Only one policy can be attached per VLAN or port.
- If the port belongs to a different VLAN (for example, in the case of a trunk port) the tagged VLAN takes priority. If the packets are untagged, the native VLAN policy takes precedence.

Examples

Attaching the RA guard policy to an L2 port:

```
switch(config)# interface 1/1/10
switch(config-if)# nd-snooping ra-guard attach-policy POLICY_NAME
```

Attempting to attach the RA guard policy to a port where routing is enabled, the policy is not configured, or it is an untrusted port:

(When prompted, enter "Y" to create the policy and attach it to the interface.)

```
switch(config)# interface 1/1/10
```

```
switch(config-if)# nd-snooping ra-guard attach-policy POLICY_NAME
RA Guard policy can't be attached to an interface with routing enabled.
switch(config-if)# no routing
switch(config-if)# nd-snooping trust
switch(config-if)# nd-snooping ra-guard attach-policy POLICY_NAME
switch(config-if)#6300(config-if)# nd-snooping ra-guard attach-policy
POLICY_NOT_CREATED
RA guard policy does not exist.
Do you want to create (y/n)?
switch(config)# interface 1/1/10
switch(config-if)# nd-snooping ra-guard attach-policy AA
RA Guard policy is ineffective, as 1/1/10 is configured as untrusted port.
```

Attaching the RA guard policy to a VLAN:

```
switch(config)# vlan 10
switch(config-vlan-10)# nd-snooping ra-guard attach-policy POLICY_NAME
```

Detaching the RA guard policy:

```
switch(config)# interface 1/1/10
switch(config-if)# no nd-snooping ra-guard attach-policy POLICY_NAME
```

Attempting to detach a RA guard policy which is not applied on the port or VLAN:

```
switch(config)# interface 1/1/10
switch(config-if)# no nd-snooping ra-guard attach-policy POLICY_NAME
RA Guard Policy POLICY_NAME is not applied on this port.
```

Attempting to detach a non-existent RA guard policy:

```
switch(config-if)# no nd-snooping ra-guard attach-policy POLICY_NOT_CREATED
Could not find the policy POLICY_NOT_CREATED.
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-vlan-<VLAN-ID>`	Administrators or local user group members with execution righ ts for this command.

`other-config-flag`

Syntax

```
other-config-flag [on | off]
no other-config-flag [on | off]
```

Description

Enables the verification of the advertised other configuration flag. Verifies that the advertised Other Stateful Configuration flag is On or Off based on the configured value.

The **no** form of the command disables other configuration flag verification.

> **NOTE**
> ND snooping must be enabled in both the global context and the config-vlan context before this command can be used.

Parameter	Description
`on`	Verifies that the advertised Other Stateful Configuration flag is On.
`off`	Verifies that the advertised Other Stateful Configuration flag is Off.

Examples

Enabling other configuration flag verification:

```
switch(config)# ipv6 nd-snooping ra-guard policy <POLICY-NAME>
```

```
switch(config-raguard-policy)# other-config-flag off
switch(config-raguard-policy)# other-config-flag on
```

Disabling other configuration flag verification:

```
switch(config)# ipv6 nd-snooping ra-guard policy <POLICY-NAME>
switch(config-raguard-policy)# no other-config-flag
switch(config-raguard-policy)# no other-config-flag off
switch(config-raguard-policy)# no other-config-flag on
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-raguard-policy`	Administrators or local user group members with execution righ ts for this command.

`router-preference`

Syntax

```
router-preference {high | medium | low}
no router-preference [high | medium | low]
```

Description

Enables the router preference verification on the RA guard policy for advertised packets and processes the packets only if the router preference is lower than the configured value. If the router preference is not configured, this validation is bypassed.

The **no** form of this command disables router preference verification on the RA guard policy.

Parameter	Description
`high`	Sets the maximum router preference to high.
`medium`	Sets the maximum router preference to medium.
`low`	Sets the maximum router preference to low.

Examples

Enabling router preference verification with the maximum router preference set to high:

```
switch(config)# ipv6 nd-snooping ra-guard policy <POLICY-NAME>
switch(config-raguard-policy)# router-preference high
```

Disabling router preference verification:

```
switch(config)# ipv6 nd-snooping ra-guard policy <POLICY-NAME>
switch(config-raguard-policy)# no router-preference
```

Command History

Release	Modification
10.10 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-raguard-policy`	Administrators or local user group members with execution righ ts for this command.

`show nd-snooping ra-guard interface`

Syntax

```
show nd-snooping ra-guard interface <INTERFACE-ID>
```

Description

Shows RA guard counters for the specified interface. Counters are cleared once the RA guard policy is detached from the interface.

Parameter	Description
<INTERFACE-ID>	Specifies the interface for which the RA guard counters are disp layed.

Examples

Showing RA guard counters for interface 1/1/1:

```
switch# show nd-snooping ra-guard interface 1/1/1
  RA Guard Policy Counters
  ========================
  RA Guard Policy Applied        : POLICY_2
  RA Packets Received            : 10
  RA Packets Forwarded           : 5
  RA Packets Dropped             : 5 [Total]
                          reason : Managed flag error      [0]
                                   Other flag error        [0]
                                   Access list error       [0]
                                   Prefix list error       [0]
                                   Router preference error[0]
                                   Hop limit error         [5]
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

show nd-snooping ra-guard policy

Syntax

```
show nd-snooping ra-guard policy  [<POLICY-NAME>]
```

Description

Shows the RA guard policy configuration.

Parameter	Description
<POLICY-NAME>	Specifies the name of the RA guard policy to be displayed.

Examples

Showing RA guard configuration:

```
switch# show nd-snooping ra-guard policy
RA Guard Policy                  Applied Ports                  Applied VLANs
-----------------------------------------------------------------------------
--------------
POLICY_NAME1       1/1/25,1/1/27,1/1/29-1/1/44,1/1/46          10,20,50-100
POLICY_NAME2       1/1/1-1/1/24
switch# show nd-snooping ra-guard policy POLICY_NAME1
RA Guard policy Information
========================
Policy name               : POLICY_NAME1
Policy Applied Ports      : 1/1/25,1/1/27,1/1/29-1/1/44,1/1/46
Policy Applied VLANs      : 10,20,50-100
Hop Limit         : enabled
    minimum               : 50
    maximum               : 150
Managed config flag       : On
Other config flag         : On
Access List               : ACL1
Prefix List               : PREFIX_LIST_NAME
Router Preference         : high
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show nd-snooping ra-guard vlan

Syntax

```
show nd-snooping ra-guard vlan <VLAN-ID>]
```

Description

Shows RA guard counters for the specified VLAN. Counters are cleared once the RA guard policy is detached from the VLAN.

Parameter	Description
<VLAN-ID>	Specifies a VLAN ID for which the RA guard counters are displayed. Range: 1 to 4094.

Examples

Showing RA guard counters for VLAN 2:

```
switch# show nd-snooping ra-guard vlan 2
  RA Guard Policy Counters
  ========================
  RA Guard Policy  Applied       : POLICY_1
  RA Packets Received            : 20
```

```
RA Packets Forwarded           : 5
RA Packets Dropped             : 15 [Total]
                        reason : Managed flag error      [1]
                                 Other flag error        [4]
                                 Access list error       [1]
                                 Prefix list error       [4]
                                 Router preference error[0]
                                 Hop limit error         [5]
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

IPv6 destination guard

Enabling IPv6 destination guard on a switch prevents ND cache depletion issues and helps in minimizing Denial-of-Service (DoS) attacks. When IPv6 destination guard is enabled address resolution is performed only for the destination addresses that are active on the link.

This feature requires the binding table to be populated with the help of DHCPv6 snooping, ND snooping, or static-ip-bindings. Destination guard enables the destination address based filtering of IPv6 traffic and blocks the Neighbor Discovery (ND) protocol resolution for destination addresses that are not found in the binding table.

Subtopics

IPv6 destination guard commands

IPv6 destination guard commands

Subtopics

IPv6 destination guard commands

Select a command from the list in the left navigation menu.

Subtopics

`ipv6 destination guard`

Syntax

```
ipv6 destination-guard
no ipv6 destination-guard
```

Description

Enables IPv6 destination guard on a VLAN.

The **no** form of the command removes the IPv6 destination guard from a VLAN.

> **NOTE**
> To avoid dropping valid packets when destination guard is enabled, it is recommended to configure DHCPv6 snooping and ND snooping to populate the binding database.

Examples

Enabling IPv6 destination guard policy on a VLAN:

```
switch(config)# vlan 10
switch(config-vlan-10)# ipv6 destination-guard
```

Disabling IPv6 destination guard policy on a VLAN:

```
switch(config-vlan-10)# no ipv6 destination-guard
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-vlan-<VLAN-ID>`	Administrators or local user group members with execution righ ts for this command.

`show ipv6 destination-guard`

Syntax

```
show ipv6 destination-guard
```

Description

Shows the ipv6 destination-guard configuration.

Examples

Showing the IPv6 destination-guard configuration:

```
switch# show ipv6 destination-guard
IPv6 Destination-Guard information
Enabled VLANs         : 10,20,31-35
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ipv6 destination-guard statistics vlan`

Syntax

```
show ipv6 destination-guard statistics {vlan <VLAN-ID>}
```

Description

Shows IPv6 destination guard statistics for the specified VLAN.

Command context

Parameter	Description
`vlan <VLAN-ID>`	Specifies the VLAN for which all destination guard statisics are to be displayed. Range: 1 to 4094.

Examples

Showing IPv6 destination-guard statistics for VLAN 10:

```
switch# show ipv6 destination-guard statistics vlan 10
Packets dropped for VLAN 10 : 25467
```

Showing IPv6 destination-guard statistics for all VLANs:

```
switch# show ipv6 destination-guard statistics
Packets dropped for VLAN 10 : 25467
```

```
Packets dropped for VLAN 30 : 434
Packets dropped for VLAN 50 : 8767
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`clear ipv6 destination-guard statistics vlan`

Syntax

```
clear ipv6 destination-guard statistics vlan <VLAN-ID>
```

Description

Clears IPv6 destination guard statistics from the specified VLAN.

Command context

Parameter	Description
`vlan <VLAN-ID>`	Specifies the VLAN for which all destination guard statistics are to be cleared. Range: 1 to 4094.

Examples

Clearing all ipv6 destination-guard statistics for VLAN 10:

```
switch# clear ipv6 destination-guard statistics vlan 10
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Internet Control Message Protocol (ICMP)

The Internet Control Message Protocol (ICMP) is a supporting protocol in the Internet protocol suite. The protocol is used by network devices, including routers, to send error messages and operational information. For example, an ICMP message might indicate that a requested service is not available. Another example of an ICMP message might be that a host or router could not be reached.

Subtopics

ICMP message types

The type field identifies the type of message sent by the host or gateway.

Type	ICMP messages
0	Echo Reply (Ping Reply, used with Type 8, Ping Request)

Type	ICMP messages
3	Destination Unreachable
4	Source Quench
5	Redirect
8	Echo Request (Ping Request, used with Type 0, Ping Reply)
9	Router Advertisement (Used with Type 9)
10	Router Solicitation (Used with Type 10)
11	Time Exceeded
12	Parameter Problem
13	Timestamp Request (Used with Type 14)
14	Timestamp Reply (Used with Type 13)
15	Information Request (obsolete) (Used with Type 16)
16	Information Reply (obsolete) (Used with Type 15)
17	Address Mask Request (Used with Type 17)
18	Address Mask Reply (Used with Type 18)

When ICMP messages are sent

ICMP messages are sent when one or more of the following scenarios occur:

- A datagram cannot reach its destination.
- The gateway does not have the buffering capacity to forward a datagram.
- The gateway can direct the host to send traffic on a shorter route.

ICMP redirect messages

ICMP redirect messages are used by routers to notify the hosts on the data link that a better route is available for a particular destination.

When ICMP redirect messages are sent

The switch is configured to send redirects by default. ICMP redirect messages are sent when one or more of the following scenarios occur:

- The interface on which the packet comes into the router is the same interface on which the packet gets routed out.
- The subnet or network of the source IP address is on the same subnet or network of the next-hop IP address of the routed packet.
- The datagram is not source-routed.
- The destination unicast address is unreachable. In this case, the router generates the ICMP destination unreachable message to inform the source host about the situation.

ICMP commands

Subtopics

ICMP commands

ICMP commands

Select a command from the list in the left navigation menu.

Subtopics

ip icmp redirect
ip icmp throttle
ip icmp unreachable

`ip icmp redirect`

Syntax

```
ip icmp redirect
no ip icmp redirect
```

Description

Enables the sending of ICMPv4 and ICMPv6 redirect messages to the source host. Enabled by default. ICMP redirect and active forwarding are mutually exclusive.

The **no** form of this command disables ICMPv4 and ICMPv6 redirect messages to the source host.

Examples

Enabling ICMP redirect messages:

```
switch(config)# ip icmp redirect
```

Disabling ICMP redirect messages:

```
switch(config)# no ip icmp redirect
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip icmp throttle`

Syntax

```
ip icmp throttle <PACKET-INTERVAL>
no ip icmp throttle [<PACKET-INTERVAL>]
```

Description

Used to configure the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages.

The **no** form of this command disables the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages.

Parameter	Description
`<PACKET-INTERVAL>`	Specifies the ICMPv4/v6 packet interval in seconds. Default: 1 s econd. Range: 1 - 86400.

Examples

Enabling the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages:

```
switch(config)# ip icmp throttle 3000
```

Disabling the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages:

```
switch(config)# no ip icmp throttle
```

Command History

Release	Modification
10.8	Added the optional `<PACKET-INTERVAL>` parameter to the **no** form of th e command.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip icmp unreachable`

Syntax

```
ip icmp unreachable
no ip icmp unreachable
```

Description

Enables the sending of ICMPv4 and ICMPv6 destination unreachable messages on the switch to a source host when a specific host is unreachable. The unreachable host address originates from the failed packed. Default setting.

The **no** form of this command disables the sending of ICMPv4 and ICMPv6 destination unreachable messages from the switch to a source host when a specific host is unreachable. This command does not prevent other hosts from sending an ICMP unreachable message.

Examples

Enabling ICMPv4 and ICMPv6 destination unreachable messages to a source host:

```
switch(config)# ip icmp unreachable
```

Disabling ICMPv4 and ICMPv6 destination unreachable messages to a source host:

```
switch(config)# no ip icmp unreachable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

DNS

The Domain Name System (DNS) is the Internet protocol for mapping a hostname to its IP address. DNS allows users to enter more readily memorable and intuitive hostnames, rather than IP addresses, to identify devices connected to a network. It also allows a host to keep the same hostname even if it changes its IP address.

Hostname resolution can be either static or dynamic.

- In static resolution, a local table is defined on the switch that associates hostnames with their IP addresses. Static tables can be used to speed up the resolution of frequently queried hosts.
- Dynamic resolution requires that the switch query a DNS server located elsewhere on the network. Dynamic name resolution takes more time than static name resolution, but requires far less configuration and management.

Subtopics

DNS client

The DNS client resolves hostnames to IP addresses for protocols that are running on the switch. When the DNS client receives a request to resolve a hostname, it can do so in one of two ways:

- Forward the request to a DNS name server for resolution.
- Reply to the request without using a DNS name server, by resolving the name using a statically defined table of hostnames and their associated IP addresses.

`Configuring the DNS client`

Syntax

Procedure

1. Configure one or more DNS name servers with the command `ip dns server`.
2. To resolve DNS requests by appending a domain name to the requests, either configure a single domain name with the command `ip dns domain-name`, or configure a list of up to six domain names with the command `ip dns domain-list`.

3. To use static name resolution for certain hosts, associate an IP address to a host with the command `ip dns host`.

4. Review your DNS configuration settings with the command `show ip dns`.

Examples

This example creates the following configuration:

- Defines the domain **switch.com** to append to all requests.
- Defines a DNS server with IPv4 address of **1.1.1.1**.
- Defines a static DNS host named **myhost1** with an IPv4 address of **3.3.3.3**.
- DNS client traffic is sent on the default VRF (named **default**).

```
switch(config)# ip dns domain-name switch.com
switch(config)# ip dns server-address 1.1.1.1
switch(config)# ip dns host myhost1 3.3.3.3
switch(config)# exit
switch# show ip dns
VRF Name : default
Mode : static
Domain Name : switch.com
Name Server(s) : 1.1.1.1
Host configuration: Active
Host Name                                                         Address
----------------------------------------------------------------------------
----
myhost1                                                           3.3.3.3
```

Fully Qualified Domain Name Resolver

The FQDN (Fully Qualified Domain Name) resolver is a subsystem within the DNS client that enables dynamic resolution of domain names to their corresponding IP addresses (IPv4 and/or IPv6). The task of name resolutions can be offloaded to **FQDN Resolver** by the applications. **FQDN Resolver** performs the name resolution and updates the status back to the interested applications. Periodic refresh for name resolution will be carried by this service.

With the command line interface it's possible to configure refresh Interval and force refresh for FQDN resolver entries. The following show commands are also available:

Show FQDN Resolver

Field	Description
Total FQDN	Total count of FQDN resolver entries in the database.
Refresh Time.	Refresh interval for FQDN resolver entries in second s.
Source	Source of the resolution requests.
VRF	Virtual Routing and Forwarding instance.
FQDN	Fully Qualified Domain Name being resolved.
Status	Resolution status (resolved, unresolved or stale).
Addr - Family	Address family requested for resolution (IPv4, IPv6, or both).
ip - address	Resolved IP addresses.

Show FQDN Resolver Detail

Field	Description
Source	Source of the resolution requests.
VRF	Virtual Routing and Forwarding instance.
FQDN	Fully Qualified Domain Name being resolved.
Status	Resolution status (resolved, unresolved or stale).
Requested Address Family	Address family requested for resolution (IPv4, IPv6, or both).
IPv4 Addresses	Resolved IPv4 addresses.
IPv6 Addresses	Resolved IPv6 addresses.

Show FQDN Resolver Refresh Interval

Field	Description
Refresh Interval	Current refresh interval for FQDN resolvers in secon ds.

DNS client commands

Subtopics

DNS client commands

Select a command from the list in the left navigation menu.

Subtopics

`ip dns domain-list`

Syntax

```
ip dns domain-list <DOMAIN-NAME> [vrf
                <VRF-NAME>


            ]
no ip dns domain-list <DOMAIN-NAME> [vrf
```

```
                 <VRF-NAME>
              ]
```

Description

Configures one or more domain names that are appended to the DNS request. The DNS client appends each name in succession until the DNS server replies. Domains can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF.

The **no** form of this command removes a domain from the list.

Parameter	Description
`list <DOMAIN-NAME>`	Specifies a domain name. Up to six domains can be added to th e list. Length: 1 to 256 characters.
`vrf <VRF-NAME>`	Specifies a VRF name. Default: default.

Examples

This example defines a list with two entries: **domain1.com** and **domain2.com**.

```
switch(config)# ip dns domain-list domain1.com
switch(config)# ip dns domain-list domain2.com
```

```
switch(config)# ip dns domain-list domain2.com vrf mainvrf
switch(config)# ip dns domain-list domain5.com vrf mainvrf
```

This example removes the entry **domain1.com**.

```
switch(config)# no ip dns domain-list domain1.com
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip dns domain-name`

Syntax

```
ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
no ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
```

Description

Configures a domain name that is appended to the DNS request. The domain can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF. If a domain list is defined with the command `ip dns domain-list`, the domain name defined with this command is ignored.

The **no** form of this command removes the domain name.

Parameter	Description
<DOMAIN-NAME>	Specifies the domain name to append to DNS requests. Length: 1 to 256 characters.
`vrf` *<VRF-NAME>*	Specifies a VRF name. Default: default.

Examples

Setting the default domain name to `domain.com`:

```
switch(config)# ip dns domain-name domain.com
```

Removing the default domain name `domain.com`:

```
switch(config)# no ip dns domain-name domain.com
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip dns host`

Syntax

```
ip dns host <HOST-NAME>
            <IP-ADDR> [ vrf <VRF-NAME> ]
no ip dns host <HOST-NAME>
            <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Associates a static IP address with a hostname. The DNS client returns this IP address instead of querying a DNS server for an IP address for the hostname. Up to six hosts can be defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a static IP address associated with a hostname.

Parameter	Description
`host <HOST-NAME>`	Specifies the name of a host. Length: 1 to 256 characters.
`<IP-ADDR>`	Specifies an IP address in IPv4 format (**x.x.x.x**), where **x** is a de cimal number from 0 to 255, or IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where **x** is a hexa decimal number from 0 to F.
`vrf <VRF-NAME>`	Specifies a VRF name. Default: default.

Examples

This example defines an IPv4 address of **3.3.3.3** for **host1**.

```
switch(config)# ip dns host host1 3.3.3.3
```

This example defines an IPv6 address of **b::5** for **host 1**.

```
switch(config)# ip dns host host1 b::5
```

This example defines removes the entry for **host 1** with address **b::5**.

```
switch(config)# no ip dns host host1 b::5
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip dns server address`

Syntax

```
ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]
no ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Configures the DNS name servers that the DNS client queries to resolve DNS queries. Up to six name servers can be defined. The DNS client queries the servers in the order that they are defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a name server from the list.

Parameter	Description
`<IP-ADDR>`	Specifies an IP address in IPv4 format (**x.x.x.x**), where **x** is a de cimal number from 0 to 255, or IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where **x** is a hexa decimal number from 0 to F.
`vrf <VRF-NAME>`	Specifies a VRF name. Default: default.

Examples

This example defines a name server at **1.1.1.1**.

```
switch(config)# ip dns server-address 1.1.1.1
```

This example defines a name server at **a::1**.

```
switch(config)# ip dns server-address a::1
```

This example removes a name server at **a::1**.

```
switch(config)# no ip dns server-address a::1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip dns fqdn-resolver refresh-interval`

Syntax

```
[no] ip dns fqdn-resolver refresh-interval <3600-86400>
```

Description

Sets the refresh interval for FQDN resolvers. The interval specifies the time in seconds after which the FQDN resolver entries are refreshed. The valid range for the interval is between 3600 seconds (1 hour) to 86400 seconds (1 day). For each FQDN entry refresh timer is applicable from the time entry is installed.

The **no** form of this command removes the refresh interval and sets it back to default value which is 1 day.

Parameter	Description
`<3600-86400>`	Refresh interval in seconds (default: 86400 seconds). Required

Examples

This example sets a custom refresh interval.

```
switch(config)# ip dns fqdn-resolver refresh-interval 6000
switch(config)# ip dns fqdn-resolver refresh-interval 85400
```

Resetting the refresh interval to the default value.

```
switch(config)# no ip dns fqdn-resolver refresh-interval
```

Attempting to set an invalid interval:

```
switch(config)# ip dns fqdn-resolver refresh-interval 20
Invalid interval. Valid range: 3600-86400 seconds.
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip dns fqdn-resolver force-refresh`

Syntax

```
ip dns fqdn-resolver force-refresh { fqdn WORD | source port-access | vrf
WORD | status (resolved | stale | unresolved) }
```

Description

This command initiates a force refresh for FQDN resolver entry/entries. It can refresh all FQDN entries, or entries filtered by FQDN, source, VRF or status. If no FQDN is specified, all entries are refreshed.

Parameter	Description
`fqdn`	Refresh entries associated with specific FQDN. Optional
`source`	Refresh entries based on the source of the resolution request. Optional
`vrf`	Refresh entries associated with a specific VRF. Optional
`status`	Refresh entries associated with a specific resolution status. Optional

Examples

Refreshing all FQDN entries:

```
switch(config)# ip dns fqdn-resolver force-refresh
```

Refreshing a specific FQDN:

```
switch(config)# ip dns fqdn-resolver force-refresh fqdn www.example.com
```

Refreshing all FQDNs from specific source:

```
switch(config)# ip dns fqdn-resolver force-refresh source port-access
```

Refreshing FQDN in a specific VRF:

```
switch(config)# ip dns fqdn-resolver force-refresh fqdn www.example.com vrf
red
```

Refreshing FQDN for a specific source and vrf:

```
switch(config)# ip dns fqdn-resolver force-refresh source classifier vrf red
```

When a FQDN entry is not found:

```
switch(config)# ip dns fqdn-resolver force-refresh fqdn www.nonexistent.com
No FQDN entries matching the given criteria.
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show ip dns`

Syntax

```
show ip dns [vrf <VRF-NAME>]
```

Description

Shows all DNS client configuration settings or the settings for a specific VRF.

Parameter	Description
`vrf <VRF-NAME>`	Specifies the VRF for which to show information. If no VRF is def ined, the default VRF is used.

Usage

DNS client arbitration on the MGMT interface on a MGMT VRF can be updated via three different methods.

1. Using the **domain-name <name>** or **nameservers <servers>** commands in the command-line interface.
2. Using the **ip dns domain-name** <DOMAIN-NAME> **vrf** MGMT or **ip dns server-address <SERVER> vrf MGMT** commands in the command-line interface.
3. Using the **ip dhcp** command in the command-line interface (dynamic enties).

AOS-CX gives the following priority levels to the these three update methods.

- Priority 1 - standalone CLI configuration
- Priority 2 - static ip dns configuration
- Priority 3 - Dynamic config

Examples

These examples define DNS settings and then show how they are displayed with the **show ip dns** command.

```
switch(config)# ip dns domain-name domain.com
switch(config)# ip dns domain-list domain5.com
switch(config)# ip dns domain-list domain8.com
switch(config)# ip dns server-address 4.4.4.4
switch(config)# ip dns server-address 6.6.6.6
switch(config)# ip dns host host3 5.5.5.5
switch(config)# ip dns host host2 2.2.2.2
switch(config)# ip dns host host3 c::12
switch(config)# ip dns domain-name reddomain.com vrf
                   red
                default
switch(config)# ip dns domain-list reddomain5.com vrf
                   red
                default
switch(config)# ip dns domain-list reddomain8.com vrf
                   red
                default
switch(config)# ip dns server-address 4.4.4.5 vrf
                   red
```

```
                default
switch(config)# ip dns server-address 6.6.6.7 vrf
                   red
                default
switch(config)# ip dns host host3 5.5.5.6 vrf
                   red
                default
switch(config)# ip dns host host2 2.2.2.3 vrf
                   red
                default
switch(config)# ip dns host host3 c::13 vrf

                red



switch# show ip dns
VRF Name : default
Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6
Host Name    Address
-------------------------------
host2            2.2.2.2
host3            5.5.5.5
host3            c::12

VRF Name : red
Domain Name : reddomain.com
DNS Domain list : reddomain5.com, reddomain8.com
Name Server(s) : 4.4.4.5, 6.6.6.7
Host Name    Address
-------------------------------
host2            2.2.2.3
host3            5.5.5.6
host3            c::13

switch(config)# ip dns domain-name domain.com vrf red
switch(config)# ip dns domain-list domain5.com vrf red
switch(config)# ip dns domain-list domain8.com vrf red
switch(config)# ip dns server-address 4.4.4.4 vrf red
switch(config)# ip dns server-address 6.6.6.6 vrf red
switch(config)# ip dns host host3 5.5.5.5 vrf red
switch(config)# no ip dns host host2 2.2.2.2 vrf red
```

```
switch(config)# ip dns host host3 c::12 vrf red
switch# show ip dns vrf red
VRF Name : red
Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6
Host Name    Address
-------------------------------
host3            5.5.5.5
host3            c::12
switch(config)# ip dns domain-name domain.com
switch(config)# ip dns domain-list domain5.com
switch(config)# ip dns domain-list domain8.com
switch(config)# ip dns server-address 4.4.4.4
switch(config)# ip dns server-address 6.6.6.6
switch(config)# ip dns host host3 5.5.5.5
switch(config)# ip dns host host3 c::12
switch# show ip dns
VRF Name : default
Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6
Host Name    Address
-------------------------------
host3            5.5.5.5
host3            c::12
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

Show FQDN Resolver

Syntax

```
show ip dns fqdn-resolver
```

Description

This command displays a summary table of all FQDN resolver entries, including the total count of FQDN entries, refresh time, source information, and a tabular display of all entries.

Examples

Show unresolved FQDN entries with ipv4 requested:

```
switch# show ip dns fqdn-resolver
Total FQDN: 1
Refresh Time: 86400 seconds
Source: port-access
VRF        FQDN                           Status        Addr-Family    ip-
address
---------- ------------------------------ ------------- --------------
-------------
default    host1.search.com               unresolved    ipv4           -
```

Show resolved FQDN entries:

```
switch# show ip dns fqdn-resolver
Total FQDN: 2
Refresh Time: 7200 seconds
Source: port-access
VRF           FQDN                           Status        Addr-Family
ip-address
------------- ------------------------------ ------------- --------------
-------------
default       www.example.com                resolved      ipv4
192.168.1.1, 192.168.1.2
default       api.test.com                   resolved      ipv4
10.0.0.1
                                                           ipv6
2001:db8::1, 2001:db8::2
long_vrf_na   a_very_long_fqdn_name          resolved      ipv4
10.0.0.2
```

When no FQDN entries exist:

```
switch(config)# show ip dns fqdn-resolver
No FQDN resolver entries found.
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

Show IP DNS FQDN Resolver Detail

Syntax

```
show ip dns fqdn-resolver detail {fqdn WORD | source (port-access) | vrf
WORD | requested-address-family (ipv4|ipv6|both)}
```

Description

Displays detailed hierarchical information for FQDN resolver entries, including the FQDN, VRF, requested address family, resolved IPv4 and IPv6 addresses, resolution status, and remaining refresh timer. The command supports filtering by FQDN, source, VRF, and address family. If no filters are provided, all entries are displayed.

Examples

Show detailed view with unresolved entry:

```
switch# show ip dns fqdn-resolver detail
-----------------------------------------------------------------------------
----
Source: port-access
-----------------------------------------------------------------------------
----
FQDN: host1.search.com
VRF: default
Requested Address Family: ipv4
Status: unresolved
```

```
----------------------------------------------------------------------------
----
```

Show detailed view with resolved entries:

```
switch# show ip dns fqdn-resolver detail
Source: port-access
  FQDN: www.example.com
  VRF: default
  Status: resolved
  Requested Address Family: both
       IPv4 Addresses:
      192.168.1.1
      192.168.1.2
    IPv6 Addresses:
    2001:db8::1
      2001:db8::2
----------------------------------------------------------------------------
----
  FQDN: www.hpe.com
  VRF: default
  Status: resolved
  Requested Address Family: ipv4
    IPv4 Addresses:
      10.0.0.1
----------------------------------------------------------------------------
----
```

Show detailed view with filter:

```
switch# show ip dns fqdn-resolver detail fqdn www.example.com
Source: port-access
  FQDN: www.example.com
  VRF: default
  Status: resolved
  Requested Address Family: both
  IPv4 Addresses:
    192.168.1.1
    192.168.1.2
   IPv6 Addresses:
    2001:db8::1
    2001:db8::2
----------------------------------------------------------------------------
----
```

Show detailed view when no entries exist:

```
switch# show ip dns fqdn-resolver detail
No FQDN resolver entries found
```

> **NOTE**
>
> The **source** parameter is not supported in 8320, 8400 or 9300 switch series.

> **NOTE**
>
> For more information on features that use this command, refer to the Security Guide for your switch model.

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

Show IP DNS FQDN Resolver Refresh Interval

Syntax

```
show ip dns fqdn-resolver refresh-interval
```

Description

Displays the current refresh interval for FQDN resolvers.

Examples

Show current refresh interval (default):

```
switch# show ip dns fqdn-resolver refresh-interval
Refresh Interval: 86400 seconds
```

Show refresh interval after configuration change:

```
switch(config)# ip dns fqdn-resolver refresh-interval 7200
switch(config)# exit
switch# show ip dns fqdn-resolver refresh-interval
Refresh Interval: 7200 seconds
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

ARP

ARP (Address Resolution Protocol) is used to map the network address assigned to a device to its physical address. For example, on an Ethernet network, ARP maps layer 3 IPv4 network addresses to layer 2 MAC addresses. (ARP does not work with IPv6 addresses. Instead, the Neighbor discovery protocol is used.)

ARP operates at layer 2. ARP requests are broadcast to all devices on the local network segment and are not forwarded by routers. ARP is enabled by default and cannot be disabled.

Proxy ARP

Proxy ARP allows a routing switch to answer ARP requests from devices on one network on behalf of devices on another network. The ARP proxy is aware of the location of the traffic destination, and offers its own MAC address as the final destination.

For example, if Proxy ARP is enabled on a routing switch connected to two subnets (10.10.10.0/24 and 20.20.20.0/24), the routing switch can respond to an ARP request from 10.10.10.69 for the MAC address of the device with IP address 20.20.20.69.

Typically, the host that sent the ARP request then sends its packets to the switch that has the ARP proxy. This switch then forwards the packets to the intended host through a mechanism such as a tunnel.

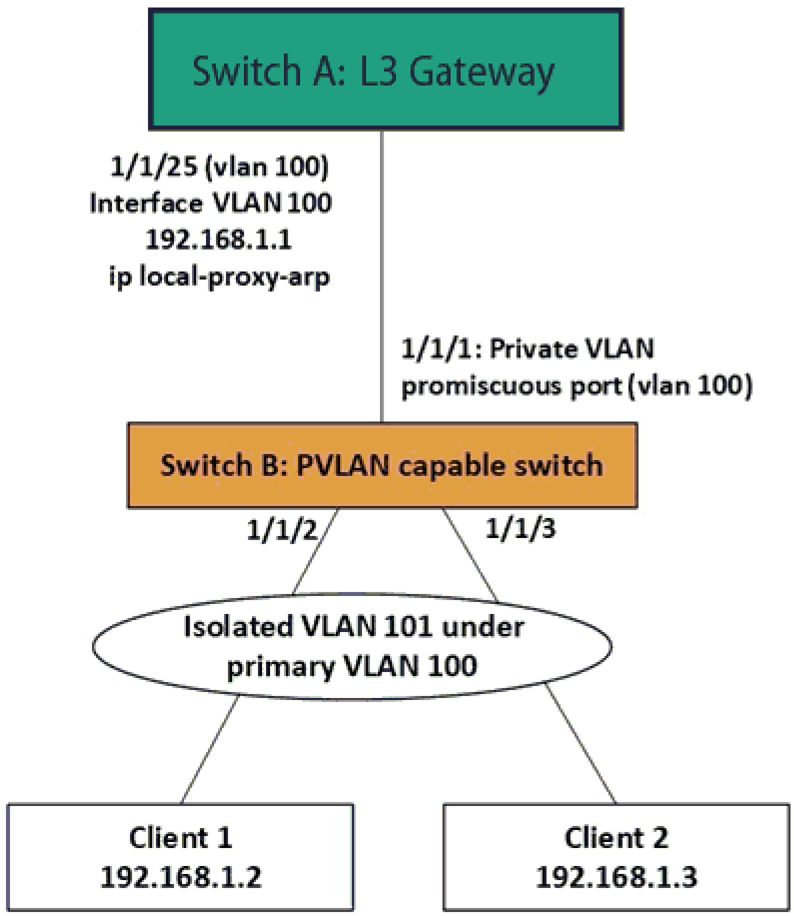
Proxy ARP is supported on L3 physical and VLAN interfaces. It is disabled by default. To enable proxy ARP, routing must be enabled on the interface.

Local proxy ARP

Local proxy ARP is a technique by which a device on a given network answers the ARP queries for a host address that is on the same network. It is primarily used to enable layer 3 communication between hosts within a common subnet that are separated by layer 2 boundaries (Example: PVLAN). Local proxy ARP is supported on L3 physical and VLAN interfaces and is disabled by default. Routing must be enabled on the interface to enable local proxy ARP.

Local proxy ARP can be used along with private VLAN deployments to allow layer 3 communication where clients are isolated at layer 2.

Figure 1. Local Proxy ARP Use Case Example



> **NOTE**
> The local proxy ARP feature is to be configured only if there is a need for the clients to communicate while still maintaining layer 2 isolation.

The following steps configure the local ARP proxy deployment shown in Figure 1, Local Proxy ARP Use Case Example.

Step 1: Configuring private VLAN in switch B

1. Configure the primary and isolated VLAN first

```
PVLAN-Switch(config)# vlan 100
PVLAN-Switch(config-vlan-100)# private-vlan primary
PVLAN-Switch(config-vlan-100)# vlan 101
PVLAN-Switch(config-vlan-101)# private-vlan isolated primary-vlan 100
```

2. Configure the PVLAN secondary ports to which clients are connected

```
PVLAN-Switch(config)# interface 1/1/2,1/1/3
PVLAN-Switch(config-if-<1/1/1,1/1/2>)# vlan trunk allowed 101
PVLAN-Switch(config-if-<1/1/1,1/1/2>)# private-vlan port-type secondary
```

3. Configure the PVLAN promiscuous port on the uplink which connects to the L3 gateway

```
PVLAN-Switch(config)# interface 1/1/1
PVLAN-Switch(config-if)# vlan trunk allowed 100
PVLAN-Switch(config-if)# private-vlan port-type promiscuous
PVLAN-Switch(config-if)# no shutdown
```

For details about private VLAN configuration, refer to the AOS-CX Layer-2 Bridging guide.

Step 2: Configuring L3 gateway in switch A

```
L3-GW(config)# vlan 100
L3-GW(config-vlan-100)# exit
L3-GW(config)# interface 1/1/25
L3-GW(config-if)# vlan trunk allowed 100
L3-GW(config-if)# no shutdown
L3-GW(config-if)# exit
L3-GW(config)# interface vlan 100
L3-GW(config-if-vlan)# ip address 192.168.1.1/24
```

Now the clients will be able to ping the gateway IP; but will not be able to communicate with each other due to the isolation enforced with private VLAN. As expected, the ping from client 1 to client 2 will fail in the above example, although the clients are in the same subnet.

Step 3: Configuring local proxy ARP in switch A

If local proxy ARP is enabled in the L3 gateway, the L3 gateway responds to ARP requests for addresses in the same subnet. In other words, the L3 gateway will be doing proxy for ARP resolution.

```
L3-GW(config)# no ip icmp redirect
L3-GW(config)# interface vlan 100
L3-GW(config-if-vlan)# ip local-proxy-arp
```

With this, the L3 communication will be enabled between client 1 and client 2. Now the ping from client 1 to client 2 will be successful in the above example. The ARP table in client 1 will be populated with the MAC of L3 gateway; not the MAC of client 2. Similar ARP table entry for client 2 as well. In other words, the L3 gateway is doing proxy for ARP resolution for the local network where clients are isolated at layer 2.

Dynamic ARP inspection

Dynamic arp inspection is provided as a mechanism for making ARP more secure.

ARP is used for resolving IP against MAC addresses on a broadcast network segment like the Ethernet and was originally defined by Internet Standard RFC 826. ARP does not support any inherent security mechanism and as such depends on simple datagram exchanges for the resolution, with many of these being broadcast.

Because it is an unreliable and non-secure protocol, ARP is vulnerable to attacks. Some attacks may be targeted toward the networks whereas other attacks may be targeted toward the switch itself. The attacks primarily intend to create denial of service (DoS) for the other entities present in the network.

Most of the attacks are carried out in one of the following three forms:

- Overwhelming the switch control plane with too many ARP packets.
- Overwhelming the switch control plane with too many unresolved data packets.
- Masquerading as a trusted gateway/server by wrongly advertising ARPs.

Several defense mechanisms can be put in place on a switch to protect against attacks:

- Limit the amount of ARP activity allowed from a host or on a port.
- Ensure that all ARP packets are consistent with one or more binding databases, which can be created through various means.
- Enforce integrity checks on the ARP packets to check against different MAC or IP addresses in the Ethernet or IP header and ARP header.

The following is supported:

- Enabling and disabling of Dynamic ARP Inspection on a VLAN level (it does not have to be SVI).
- Defining the member ports of a VLAN as either trusted or untrusted.
- Only ARP traffic on untrusted ports subjected to checks.
- Routed ports (RoPs) always treated as trusted.
- Listening to the DHCP Bindings table and check every ARP packet to match against the binding.

ARP ACLs are not supported and the DHCP snooping table is the only source of binding.

Subtopics

Configuring proxy ARP

Syntax

Procedure

1. Switch to configuration context with the command `config`.
2. Switch to an interface VLAN with the command `interface vlan`.
3. Enable local proxy ARP with the command `ip proxy-arp`.

Examples

```
switch# config
switch(config)# interface 1/1/2
switch(config-if)# ip proxy-arp
```

.

```
switch# config
switch(config)# interface 1/1/2
switch(config)# routing
switch(config-if)# ip proxy-arp
```

.

This example configures proxy ARP on interface VLAN **30**.

```
switch# config
switch(config)# interface vlan 30
switch(config-vlan-30)# ip proxy-arp
```

Configuring local proxy ARP

Syntax

Procedure

1. Switch to configuration context with the command `config`.
2. Switch to an interface VLAN with the command `interface vlan`.
3. Enable local proxy ARP with the command `ip local-proxy-arp`.

Examples

```
switch# config
switch(config)# interface 1/1/2
switch(config-if)# ip local-proxy-arp
```

```
switch# config
switch(config)# interface 1/1/2
switch(config)# routing
switch(config-if)# ip local-proxy-arp
```

This example configures local proxy ARP on interface VLAN **30**.

```
switch# config
switch(config)# interface vlan 30
switch(config-vlan-30)# ip local-proxy-arp
```

ARP commands

Select a command from the list in the left navigation menu.

Subtopics

- show arp
- show arp inspection interface
- show arp inspection statistics
- show arp inspection vlan
- show arp state
- show arp summary
- show arp timeout
- show arp vrf
- show ipv6 neighbors
- show ipv6 neighbors state
- show tech arp-security

arp inspection

Syntax

```
arp inspection
```

Description

Enables Dynamic ARP inspection on the current VLAN, which means that ARP packets received from untrusted interfaces are discarded if they have an Invalid IP-to-MAC address binding.

The **no** form of this command disables Dynamic ARP Inspection on the VLAN.

Examples

Enabling dynamic ARP inspection:

```
switch# configure terminal
switch(config)# vlan 1
switch(config-vlan)# arp inspection
```

Disabling dynamic ARP inspection:

```
switch# configure terminal
switch(config)# vlan 1
switch(config-vlan)# no arp inspection
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-vlan-<VLAN-ID>`	Administrators or local user group members with execution righ ts for this command.

`arp inspection trust`

Syntax

```
arp inspection trust
no arp inspection trust
```

Description

Configures the interface as a trusted. All interfaces are untrusted by default.

The **no** form of this command returns the interface to the default state (untrusted).

Example

Setting an interface as trusted:

```
switch(config-if)# arp inspection trust
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`arp ip`

Syntax

```
arp ip <IP_ADDR> mac <MAC_ADDR>
no arp ip <IP_ADDR> mac <MAC_ADDR>
```

Description

Specifies a permanent static neighbor entry in the ARP table (for IPv4 neighbors).

The no form of this command deletes a permanent static neighbor entry from the ARP table.

Parameter	Description
`ip <IP-ADDR>`	Specifies the IP address of the neighbor or the virtual IP addres s of the cluster in IP format (**x.x.x.x**), where **x** is a decimal num ber from 0 to 255. . Range: 4096 to 131072. Default: 131072.
`mac <MAC-ADDR>`	Specifies the MAC address of the neighbor or the multicast M AC address in IANA format (**xx:xx:xx:xx:xx:xx**), where **x** is a h exadecimal number from 0 to F. Range: 4096 to 131072. Defaul t: 131072.

Example

Configuring a static ARP entry on a interface VLAN **10**:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# arp ip 2.2.2.2 mac 01:00:5e:00:00:01
```

Removing a static ARP entry on interface VLAN**10**:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# no arp ip 2.2.2.2 mac 01:00:5e:00:00:01
```

Command History

Release	Modification
10.14	Replaced the **ipv4** parameter with the **ip** parameter. The **ipv4** parameter is dep recated.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`arp process-grat-arp`

Syntax

```
arp process-grat-arp
no arp process-grat-arp
```

Description

Enables the processing of gratuitous ARP packets on the individual port or group of L3 ports together.

By default, the gratuitous ARP processing is enabled. When gratuitous ARP (GARP) processing is enabled, a switch that is advertising any changes in its MAC through the GARP will reflect in the neighbor table of the switch. However, the switch will not be able to learn the neighbor through the GARP.This configuration is applicable only on L3 interfaces such as ROPs, subinterfaces, and SVIs.

The **no** form of this command disables the processing of gratuitous ARP packets.

Example

Enabling the processing of gratuitous ARP packets on the interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no shutdown
```

```
switch(config-if)# arp process-grat-arp
```

Enabling the processing of gratuitous ARP packets on interfaces **1/1/1** to **1/1/5**:

```
switch(config)# interface 1/1/1-1/1/5
switch(config-if<1/1/1-1/1/5>)# no shutdown
switch(config-if<1/1/1-1/1/5>)# arp process-grat-arp
```

Enabling the processing of gratuitous ARP packets on sub-interface **1/1/1.10**:

> **NOTE**
>
> Applies only to the HPE Aruba Networking 6300, 6400, 8100, and 8360 Switch Series.

```
switch(config)# interface 1/1/1.10
switch(config-subif)# no shutdown
switch(config-subif)# arp process-grat-arp
```

Disabling the processing of gratuitous ARP packets on VLANs **2** to **100**:

```
switch(config)# interface vlan 2-100
switch(config-if-vlan<2-100>)# no shutdown
switch(config-if-vlan<2-100>)# no arp process-grat-arp
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority

All platforms	`config-if`
	`config-if-vlan`
	`config-subif`
		Administrators or local user group members with execution righ ts for this command.

clear arp

Syntax

```
clear arp
   port <PORT-ID> [ip <A.B.C.D>|all]|[ipv6 <X:X::X:X>|all]vla
   vrf [all-vrfs|{<VRF-NAME> [ip <A.B.C.D>]|[ipv6 <X:X::X:X>]}]
```

Description

Clears IPv4 and IPv6 neighbor entries from the ARP table. If you do not specify any VRF or port parameters, ARP table entries are cleared for the **default** VRF.

Parameter	Description
`port <PORT-ID>`	Specifies a port on the switch. For example: 1/1/1 .
`ip <A.B.C.D>|all]`	(Optional) Include an IP address to clear neighbor entries for th at specific address, or use the **all** parameter to clear entries for all IP addresses.
`ipv6 <X:X::X:X>|all`	(Optional) Include an IPv6 address to clear neighbor entries for that specific address, or use the **all** parameter to clear entries fo r all IPv6 addresses.
`vrf`	Clears IPv4 and IPv6 neighbor entries for the specified VRF or for all VRFs. If no VRF is specified he **default** VRF is cleared.
`all-vrfs`	Clear neighbor entries for all VRFs
`<VRF-NAME>`	Clear neighbor entries for the specified VRF.
`ip <A.B.C.D>`	(Optional) Include an IP address to clear just the neighbor entri es for the specified IP address.
`ipv6 <X:X::X:`	(Optional) Include an IPv6 address to clear the neighbor entries for the specified IPv6 address.

Examples

Clearing all IPv4 and IPv6 neighbor ARP entries for the default VRF:

```
switch# clear arp
```

Clearing all ARP neighbor entries for a port:

```
switch# clear arp 1/1/35
```

```
switch# clear arp vrf all-vrfs
```

```
switch# clear arp vrf RED
```

Command History

Release	Modification
10.14	Replaced the **ipv4** parameter with the **ip** parameter. The **ipv4** parameter is dep recated.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

debug arp-security

Syntax

```
debug arp-security <LOG-CATEGORY> [severity <LEVEL>]
no debug arp-security [<LOG-CATEGORY>] [severity <LEVEL>]
```

Description

Enables ARP security debug logs. If <SEVERITY> is omitted, all severities are logged.

The **no** form of this command disables ARP security debug logs.

Parameter	Description
<LOG-CATEGORY>	Selects the ARP security debug log category. Available categori es are: • **all**: Selects all ARP security debug log categories. • **config**: Selects the ARP security config debug log category. • **inspection**: Selects the ARP security inspection debug log category. • **packet**: Selects the ARP security packet debug log categor y.
`severity` *<LEVEL>*	Specifies how to filter the ARP security debug logging by settin g the minimum severity level for which debug logging will be pe rformed. The selected severity level and all severities above (m ore severe) will be included in the logging. • **emerg**: Sets ARP security debug log filtering to Emergency only. • **alert**: Sets ARP security debug log filtering to Alert and a bove. • **critical**: Sets ARP security debug log filtering to Critical and above. • **error**: Sets ARP security debug log filtering to Error and a bove. • **warning**: Sets ARP security debug log filtering to Warning and above. • **notice**: Sets ARP security debug log filtering to Notice and above. • **info**: Sets ARP security debug log filtering to Info and a bove. • **debug**: Sets ARP security debug log filtering to all severiti es.

Examples

Enable ARP security debug logging for all categories and all severities:

```
switch# debug arp-security all
```

Enable ARP security config debug log for severity level Error and above:

```
switch# debug arp-security config severity error
```

Enable ARP security inspection debug log for severity level Notice and above:

```
switch# debug arp-security inspection severity notice
```

Enable ARP security debug packet for severity level Critical and above:

```
switch# debug arp-security packet severity critical
```

Enable ARP security debug logging for all categories and severity level Alert and above:

```
switch# debug arp-security all severity alert
```

Disable ARP security debug logging:

```
switch# no debug arp-security
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`ip local-proxy-arp`

Syntax

```
ip local-proxy-arp
no ip local-proxy-arp
```

Description

Enables local proxy ARP on the specified interface. Local proxy ARP is supported on Layer 3 physical interfaces and on VLAN interfaces. To enable local proxy ARP on an interface, routing must be enabled on that interface.

The **no ip local-proxy-arp** command disables local proxy ARP on the specified interface.

Parameter	Description
`uplink-passthrough`	(For 10000 Switch series only) Enables local proxy ARP passth rough to uplink members. **NOTE** When enabling **ip local - proxy - arp uplink - passthrough**, **ip local - proxy - arp** will be imp licitly enabled.

Examples

Enabling local proxy ARP on interface **1/1/1**:

```
switch# interface 1/1/1
switch(config-if)# ip local proxy-arp
```

Enabling local proxy ARP on interface VLAN **3**:

```
switch# interface vlan 3
switch(config-if-vlan)# ip local-proxy-arp
```

Disabling local proxy ARP on on interface **1/1/1**.

```
switch# interface 1/1/1
switch(config-if)# no ip local-proxy-arp
```

Command History

Release	Modification
10.16	Support added for the 6200 Switch Series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip proxy-arp`

Syntax

```
ip proxy-arp
no ip proxy-arp
```

Description

Enables proxy ARP for the specified Layer 3 interface. Proxy ARP is supported on Layer 3 physical interfaces, LAG interfaces, and VLAN interfaces. It is disabled by default. To enable proxy ARP on an interface, routing must be enabled on that interface.

The **no** form of this command disables proxy ARP for the specified interface.

Examples

Enabling proxy ARP on interface **1/1/1**:

```
switch# interface 1/1/1
switch(config-if)# ip proxy-arp
```

Enabling proxy ARP on VLAN **3**:

```
switch# interface vlan 3
switch(config-if-vlan)# ip proxy-arp
```

Enabling proxy ARP on a LAG **11**:

```
switch(config)# int lag 11
switch(config-lag-if)# ip proxy-arp
```

Disabling proxy ARP on interface 1/1/1:

```
switch# interface 1/1/1
switch(config-if)# no ip proxy-arp
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority

All platforms	`config-if`
	`config-if-vlan`
	`config-lag-vlan`
		Administrators or local user group members with execution righ ts for this command.

`ipv6 local-proxy-nd`

Syntax

```
ipv6 local-proxy-nd
no ipv6 local-proxy-nd
```

Description

Enables local proxy ND for the specified interface.

The **noipv6 local-proxy-nd** command disables local proxy ND for the specified interface.

Usage

Before enabling local proxy ND, disable Internet Control Message Protocol (ICMP) redirect messages. Disabling the ICMP redirect ensures that hosts don't communicate through other gateways which could possibly have better routes and thereby skip the proxy.

Example

Enable local proxy ND on a L3 physical interface.

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 local-proxy-nd
```

Enable local proxy ND on a VLAN interface.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 local-proxy-nd
```

.

```
switch(config)# interface 1/1/1.10
switch(config-subif)# ipv6 local-proxy-nd
```

Disable local proxy ND for the specified interface.

```
switch(config-if)# no ipv6 local-proxy-nd
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ipv6 neighbor mac`

Syntax

```
ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
no ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
```

Description

Specifies a permanent static neighbor entry in the ARP table (for IPv6 neighbors).

The **no** form of this command deletes a permanent static neighbor entry from the ARP table.

Parameter	Description
<IPV6-ADDR>>	Specifies an IP address in IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.

Parameter	Description
`mac <MAC-ADDR>>`	Specifies the MAC address of the neighbor (`xx:xx:xx:xx:xx:xx`), where **x** is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.

Example

Creates a static ARP entry on interface **1/1/1** .

```
switch(config)# interface 1/1/1

switch(config-if)# arp ipv6 neighbor 2001:0db8:85a3::8a2e:0370:7334 mac
00:50:56:96:df:c8
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`show arp`

Syntax

```
show arp
```

Description

Shows the entries in the ARP (Address Resolution Protocol) table.

Usage

This command displays information about ARP entries, including the IP address, MAC address, port, and state.

When no parameters are specified, the `show arp` command shows all ARP entries for the default VRF (Virtual Router Forwarding) instance.

Examples

```
switch# show arp
IPv4 Address     MAC                Port       Physical Port
State
---------------------------------------------------------------------------
---
192.168.1.2        00:50:56:96:7b:e0  vlan10       1/1/29          stale
192.168.1.3        00:50:56:96:7b:ac  vlan10       1/1/1
reachable
Total Number Of ARP Entries Listed- 2.
---------------------------------------------------------------------------
---
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

`show arp inspection interface`

Syntax

```
show arp inspection interface [<IFNAME>] [vlan <VLAN-ID>]
```

Description

Shows the current configuration of dynamic ARP inspection on an interface.

Parameter	Description
<IFNAME>	Specifies the interface.
<VLAN-ID>	Specifies the VLAN ID. Range: 1 to 4094.

Examples

Showing current configuration of dynamic ARP inspection on all interfaces:

```
switch# show arp inspection interface
---------------------------------------------------------------------------
Interface               Trust-State
---------------------------------------------------------------------------
1/1/1                   Untrusted
---------------------------------------------------------------------------
```

switch# show arp inspection interface vsx-peer

```

---------------------------------------------------------------------------
Interface               Trust-State
---------------------------------------------------------------------------
1/1/1                   Untrusted
lag100                  Trusted
---------------------------------------------------------------------------
```

Showing current configuration of dynamic ARP inspection on a particular interface:

```
switch# show arp inspection interface 1/1/1
---------------------------------------------------------------------------
Interface               Trust-State
---------------------------------------------------------------------------
1/1/1                   Untrusted
---------------------------------------------------------------------------
```

Showing current configuration of dynamic ARP inspection on interface VLAN 2:

```
switch# show arp inspection interface vlan 2
-----------------------------------------------------------------
```

```
Interface            Trust-State
-----------------------------------------------------------------
vlan2                Trusted
-----------------------------------------------------------------
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show arp inspection statistics`

Syntax

```
show arp inspection statistics vlan [<VLAN-ID>]
```

Description

Shows statistics about forwarded and dropped ARP packets. When <VLAN-ID> is not specified, information is shown for all configured VLANs.

Parameter	Description
<VLAN-ID>	Specifies the VLAN ID or range of IDs separated by a dash " `-` ". Range: 1 to 4094.

Examples

Showing ARP packet statistics for a range of VLANs:

```
switch# show arp inspection statistics vlan 1-100
-----------------------------------------------------------------
VLAN   Name               Forwarded           Dropped
-----------------------------------------------------------------
1      DEFAULT_VLAN_1     0                   0
-----------------------------------------------------------------
```

```
switch# show arp inspection statistics vlan vsx-peer
-----------------------------------------------------------------
VLAN   Name               Forwarded           Dropped
-----------------------------------------------------------------
1      DEFAULT_VLAN_1     0                   0
200    VLAN200            0                   0
-----------------------------------------------------------------
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show arp inspection vlan`

Syntax

```
show arp inspection vlan [<VLAN-ID>]
```

Description

Shows the current configuration of dynamic ARP inspection on a VLAN. When <VLAN-ID> is not specified, information is shown for all configured VLANs.

Parameter	Description
<VLAN-ID>	Specifies the VLAN ID or range of IDs separated by a dash "`-`". Range: 1 to 4094.

Examples

Showing dynamic ARP configuration for all VLANs:

```
switch# show arp inspection vlan
-----------------------------------------------------------------
VLAN   Name                 ARP Inspection
-----------------------------------------------------------------
1      DEFAULT_VLAN_1       -
100    VLAN100              -
200    VLAN200              Enabled
-----------------------------------------------------------------
```

Showing dynamic ARP configuration for a particular VLAN:

```
switch# show arp inspection vlan 1
-----------------------------------------------------------------
VLAN   Name                 ARP Inspection
-----------------------------------------------------------------
1      DEFAULT_VLAN_1       -
-----------------------------------------------------------------
```

```
switch# show arp inspection vlan vsx-peer
-----------------------------------------------------------------
VLAN   Name                 ARP Inspection
-----------------------------------------------------------------
1      DEFAULT_VLAN_1       -
-----------------------------------------------------------------
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show arp state`

Syntax

```
show arp state {all | failed | incomplete | permanent | reachable | stale}
```

Description

Shows ARP (Address Resolution Protocol) cache entries that are in the specified state.

Parameter	Description
`all`	Shows the ARP cache entries for all VRF (Virtual Router Forwarding) instances.
`failed`	Shows the ARP cache entries that are in `failed` state. The neighbor might have been deleted.
`incomplete`	Shows the ARP cache entries that are in `incomplete` state. An incomplete state means that address resolution is in progress and the link - layer address of the neighbor has not yet been determined. A solicitation request was sent, and the switch is waiting for a solicitation reply or a timeout.
`permanent`	Shows the ARP cache entries that are in `permanent` state. ARP entries that are in a permanent state can be removed by administrative action only.
`reachable`	Shows the ARP cache entries that are in `reachable` state, meaning that the neighbor is known to have been reachable recently.
`stale`	Shows ARP cache entries that are in `stale` state.

Parameter	Description
	ARP cache entries are in the `stale` state if the elapsed time is in excess of the ARP timeout in seconds since the last positive confirmation that the forwarding path was functioning properly.

Examples

```
switch# show arp state failed
IPv4 Address     MAC                      Port         Physical Port    State
----------------------------------------------------------------------------
192.168.1.4                               vlan10                         failed
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show arp summary`

Syntax

```
show arp summary [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows a summary of the IPv4 and IPv6 neighbor entries on the switch for all VRFs or a specific VRF.

Parameter	Description
`all-vrfs`	Selects all VRFs.

Parameter	Description
`vrf <VRF-NAME>`	Specifies the name of a VRF.

Examples

Showing summary ARP information for all VRFs:

```
switch# show arp summary all-vrfs
ARP Entry's State                : IPv4      IPv6
-------------------------------------------------------
Number of Reachable ARP entries  : 2          0
Number of Stale ARP entries      : 0          0
Number of Failed ARP entries     : 2          2
Number of Incomplete ARP entries : 0          0
Number of Permanent ARP entries  : 0          0
-------------------------------------------------------
Total ARP Entries: 6             : 4          2
-------------------------------------------------------
```

Showing a summary of all IPv4 and IPv6 neighbor entries on the primary and secondary (peer) switches:

```
vsx-primary# show arp summary
ARP Entry's State                 IPv4        IPv6
---------------------------------------------------------
Number of Reachable ARP entries   25858       32231
Number of Stale ARP entries       0           1
Number of Failed ARP entries      0           257
Number of Incomplete ARP entries  0           0
Number of Permanent ARP entries   0           0
---------------------------------------------------------
Total ARP Entries- 58347          25858       32489
vsx-primary# show arp summary vsx-peer
ARP Entry's State                 IPv4        IPv6
---------------------------------------------------------
Number of Reachable ARP entries   25858       32168
Number of Stale ARP entries       0           3
Number of Failed ARP entries      0           317
Number of Incomplete ARP entries  0           0
Number of Permanent ARP entries   0           0
---------------------------------------------------------
Total ARP Entries- 58346          25858       32488
---------------------------------------------------------
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show arp timeout`

Syntax

```
show arp timeout [<INTERFACE>]
```

Description

Shows the age-out period for each ARP (Address Resolution Protocol) entry for a port, LAG, or VLAN interface.

Parameter	Description
<INTERFACE>	Specifies a physical port, VLAN, or LAG on the switch. For physi cal ports, use the format member/slot/port (for example, `1/3/1`).

Examples

```
switch# show arp timeout vlan2
Port              VRF                                Timeout
-----------------------------------------------------------
vlan2             default                            1800
```

Showing ARP timeout information for a port:

```
switch# show arp timeout 1/1/1
ARP Timeout:
------------------
Port             VRF                              Timeout
1/1/1            default                          600
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show arp vrf`

Syntax

```
show arp {all-vrfs | vrf <VRF-NAME>}
```

Description

Shows the ARP table for all VRF instances, or for the named VRF.

Parameter	Description
`all-vrfs`	Specifies all VRFs.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Length: 1 to 32 alphanumeric cha racters.

Examples

Showing ARP entries for VRF **vrf1**.

```
switch# show arp vrf vrf1
IPv4 Address      MAC                  Port          Physical Port
State       VRF
-----------------------------------------------------------------------------
------------------
100.1.250.50      00:50:56:8d:44:13  vlan1001     1/1/2
reachable   vrf1
100.2.250.60      00:50:56:8d:45:63  vlan1002     vxlan1(1920:1680:1:1::2)
permanent   vrf1
Total Number Of ARP Entries Listed: 2.
-----------------------------------------------------------------------------
------------------
```

This example from a different network shows ARP entries for all VRFs.

```
switch# show arp all-vrfs
ARP IPv4 Entries:
--------------------------------------------------------
IPv4 Address     MAC                Port     Physical Port   State       VRF
192.168.120.10   00:50:56:bd:10:be  1/1/32   1/1/32          reachable   red
10.20.30.40      00:50:56:bd:6a:c5  1/1/29   1/1/29          reachable   test
--------------------------------------------------------
Total Number Of ARP Entries Listed: 2.
--------------------------------------------------------
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ipv6 neighbors

Syntax

```
show ipv6 neighbors {all-vrfs | vrf <VRF-NAME>}
```

Description

Shows entries in the ARP table for all IPv6 neighbors for all VRFs or for a specific VRF.

When no parameters are specified, this command shows all ARP entries for the default VRF, and state information for `reachable` and `stale` entries only.

Parameter	Description
`all-vrfs`	Specifies all VRFs.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Length: 1 to 32 alphanumeric characters.

Examples

```
switch# show ipv6 neighbors
IPv6 Entries:
--------------------------------------------------------
IPv6 Address                 MAC                 Port      Physical Port
State
fe80::a21d:48ff:fe8f:2700    a0:1d:48:8f:27:00  vlan2300  1/1/31
reachable
fe80::f603:43ff:fe80:a600    f4:03:43:80:a6:00  vlan2300  1/1/30
reachable
--------------------------------------------------------
Total Number Of IPv6 Neighbors Entries Listed: 2.
--------------------------------------------------------
switch# show ipv6 neighbors vrf vrf1
IPv6 Address                                     MAC
Port          Physical Port   State      VRF
-----------------------------------------------------------------------------
------------------
1000:2:1:1::250:60   00:50:56:8d:45:63  vlan1002
vxlan1(1920:1680:1:1::2)   permanent  vrf1
1000:1:1:1::250:50   00:50:56:8d:44:13  vlan1001
1/1/2                      reachable  vrf1
Total Number Of IPv6 Neighbors Entries Listed: 2.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Administrators or local user group members with execution righ ts for this command.

`show ipv6 neighbors state`

Syntax

```
show ipv6 neighbors state {all | failed | incomplete | permanent |
reachable | stale}
```

Description

Shows all IPv6 neighbor ARP (Address Resolution Protocol) cache entries, or those cache entries that are in the specified state.

Parameter	Description
`all`	Shows all ARP cache entries.
`failed`	Shows ARP cache entries that are in `failed` state. The neig hbor might have been deleted. Set the neighbor to be unreacha ble.
`incomplete`	Shows ARP cache entries that are in `incomplete` state. An incomplete state means that address resolution is in progres s and the link - layer address of the neighbor has not yet been determined. This means that a solicitation request was sent, an d you are waiting for a solicitation reply or a timeout.
`permanent`	Shows ARP cache entries that are in `permanent` state.

Parameter	Description
`reachable`	Shows ARP cache entries that are in `reachable` state, mea ning that the neighbor is known to have been reachable recentl y.
`stale`	Shows ARP cache entries that are in `stale` state. ARP cache entries are in the `stale` state if the elapsed time i s in excess of the ARP timeout in seconds since the last positive confirmation that the forwarding path was functioning properly.

Example

```
switch# show ipv6 neighbors state all
IPv6 Address                  MAC                Port     Physical Port
State
--------------------------------------------------------------------------------
----
100::2                        48:0f:cf:af:f1:cc  lag1     lag1
reachable
300::3                        48:0f:cf:af:33:be  vlan3    1/4/20
reachable
fe80::4a0f:cfff:feaf:f1cc     48:0f:cf:af:f1:cc  lag1     lag1
reachable
200::3                        48:0f:cf:af:33:be  1/4/11   1/4/11
reachable
fe80::4a0f:cfff:feaf:33be     48:0f:cf:af:33:be  vlan3    1/4/20
reachable
Total Number Of IPv6 Neighbors Entries Listed- 5.
--------------------------------------------------------------------------------
-----
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show tech arp-security`

Syntax

```
show tech arp-security
```

Description

Shows the output of these three commands:

- **`show arp inspection statistics vlan`**
- **`show arp inspection vlan`**
- **`show arp inspection interface`**

Examples

Showing the output of the three ARP security show commands:

```
switch(config-if)# show tech arp-security
====================================================
Show Tech executed on Mon Nov 28  09:53:54 2019
====================================================
====================================================
[Begin] Feature arp-security
====================================================
*********************************
Command : show arp inspection statistics vlan
*********************************
-----------------------------------------------------------------
VLAN   Name                 Forwarded            Dropped
-----------------------------------------------------------------
1      DEFAULT_VLAN_1       0                    0
200    VLAN200              0                    0
```

```
---------------------------------------------------------------------
*********************************
Command : show arp inspection vlan
*********************************
---------------------------------------------------------------------
VLAN   Name               ARP-Inspection
---------------------------------------------------------------------
1      DEFAULT_VLAN_1     -
200    VLAN200            Enabled
---------------------------------------------------------------------
*********************************
Command : show arp inspection interface
*********************************
-------------------------------------------------------------------------------
Interface           Trust-State
-------------------------------------------------------------------------------
1/1/1               Untrusted
lag100              Trusted
-------------------------------------------------------------------------------
====================================================
[End] Feature arp-security
====================================================
====================================================
Show Tech commands executed successfully
====================================================
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution righ ts for this command.

Support and Other Resources

Access HPE Aruba Networking support and updates, and view warranty and regulatory information

Subtopics

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	**https://www.hpe.com/us/en/networking/hpe - aruba - networking - support - services.html**
AOS - CX Switch Software Documentation Portal	**https://arubanetworking.hpe.com/techdocs/ArubaDocPortal/content/new-portal/aoscx.html**
HPE Aruba Networking Support Portal	**https://networkingsupport.hpe.com/home**
North America telephone	1 - 800 - 943 - 4526 (US & Canada Toll - Free Number) +1 - 650 - 750 - 0350 (Backup—Toll Number)
International telephone	**https://www.hpe.com/psnow/doc/a50011948enw**

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs

- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	**https://developer.arubanetworks.com/hpe - aruba - networking - aoscx/docs/about**
Airheads social forums and Knowledge Base	**https://community.arubanetworks.com/**
AOS - CX Software Technical Update channel on You Tube.	Videos on new features introduced in this release : **https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP - Q_UL3CskS**
HPE Aruba Networking Hardware Documentation and Translations Portal	
HPE Aruba Networking software	**https://networkingsupport.hpe.com/downloads https://networkingsupport.hpe.com/downloads**
Software licensing and Feature Packs	**https://licensemanagement.hpe.com/**
End - of - Life information	**https://networkingsupport.hpe.com/end - of - life**

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at **https://networkingsupport.hpe.com**.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

https://networkingsupport.hpe./notifications/subscriptions (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to **https://www.arubanetworks.com/support-services/product-warranties/**.

Regulatory Information

To view the regulatory information for your product, view the Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products, available at **https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts**

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see **https://www.arubanetworks.com/company/about-us/environmental-citizenship/**.

Documentation Feedback

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